

The Eye

Department Anatomy
Head and Neck (10)

First Part

Dr. Osman Ahmed Osman

Lecture Points

- **The Eye.**
- **The orbital cavity.**
 - ✓ **The Roof.**
 - ✓ **The Floor.**
 - ✓ **Medial Wall.**
 - ✓ **Lateral Wall.**
 - ✓ **Orbital Openings.**
- **Extrinsic Muscles.**

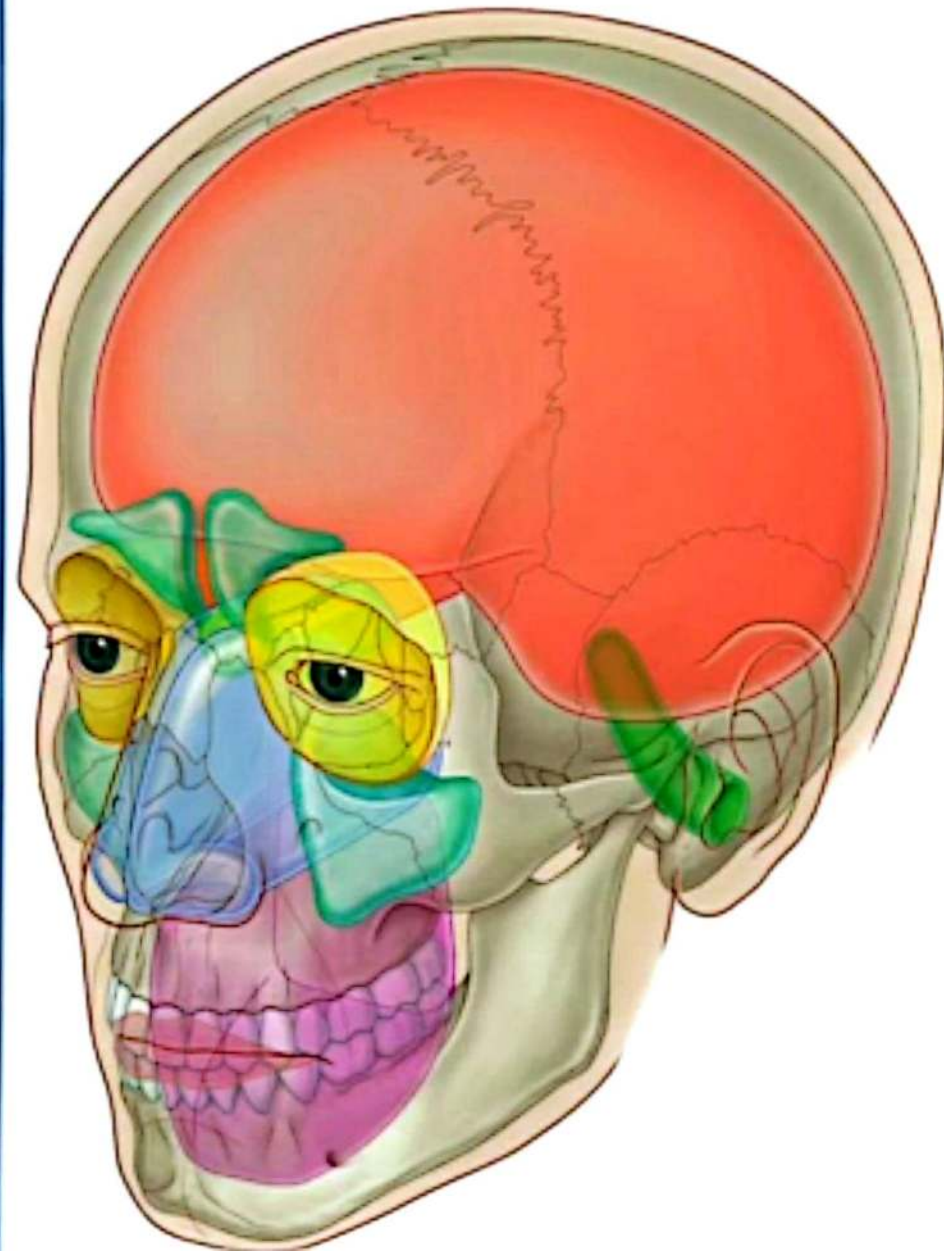
- **Structures (layer) of Eye.**
- **The outer fibrous tunic (the fibrous coat).**
 - ✓ **sclera.**
 - ✓ **cornea.**
- **The middle vascular tunic (the vascular pigmented coat).**
 - ✓ **The choroid.**
 - ✓ **The ciliary body.**
 - ✓ **The iris and Pupil.**
 - **Lens**
- **The inner nervous tunic.**
 - ✓ **The Retina.**
 - ✓ **Optic Disc**

- Contents of the Eyeball
(The refractive media):**
- ✓ **The aqueous humor.**
 - ✓ **The vitreous body.**
 - ✓ **The lens.**
 - ✓ **When you look at any object**

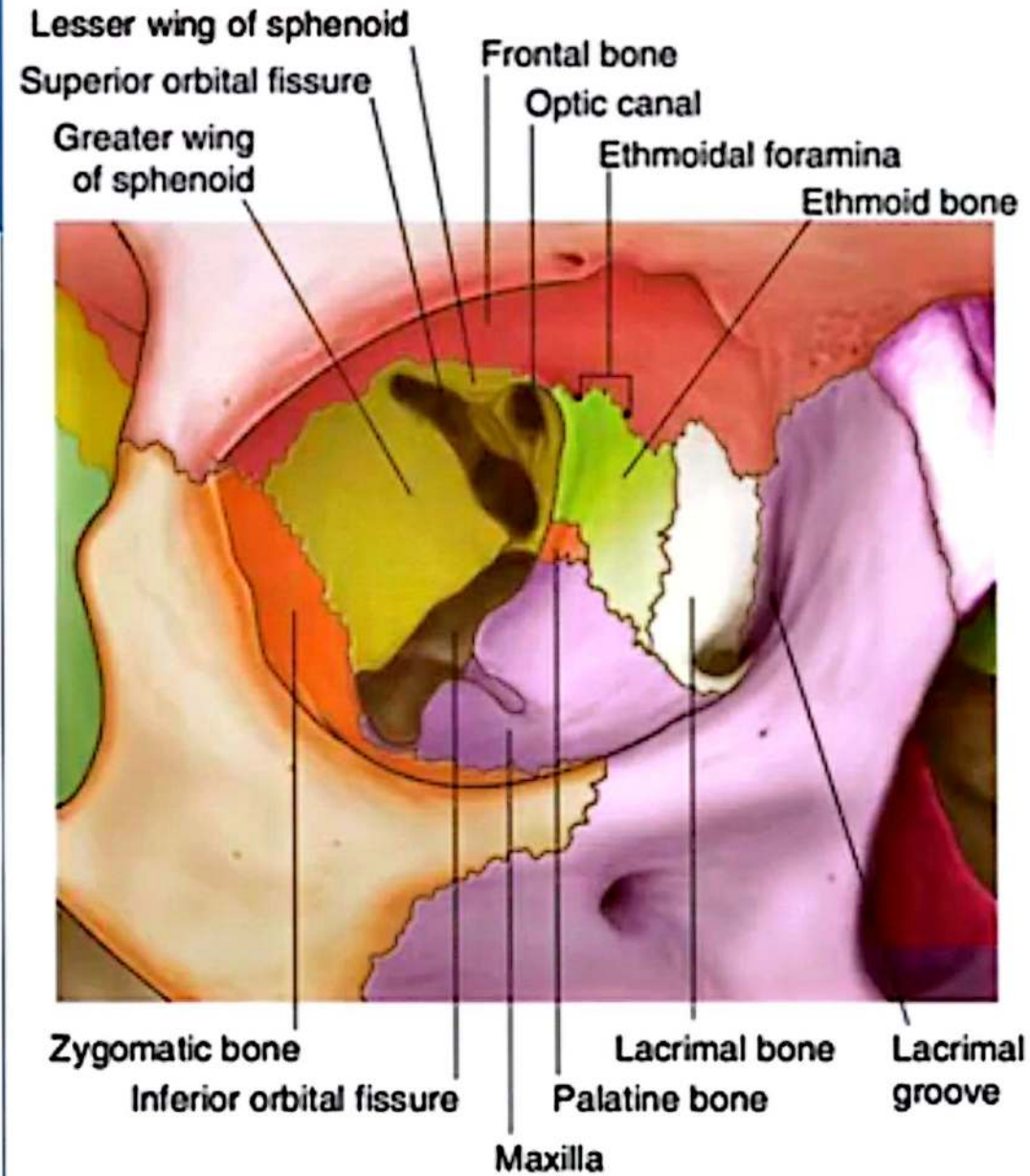
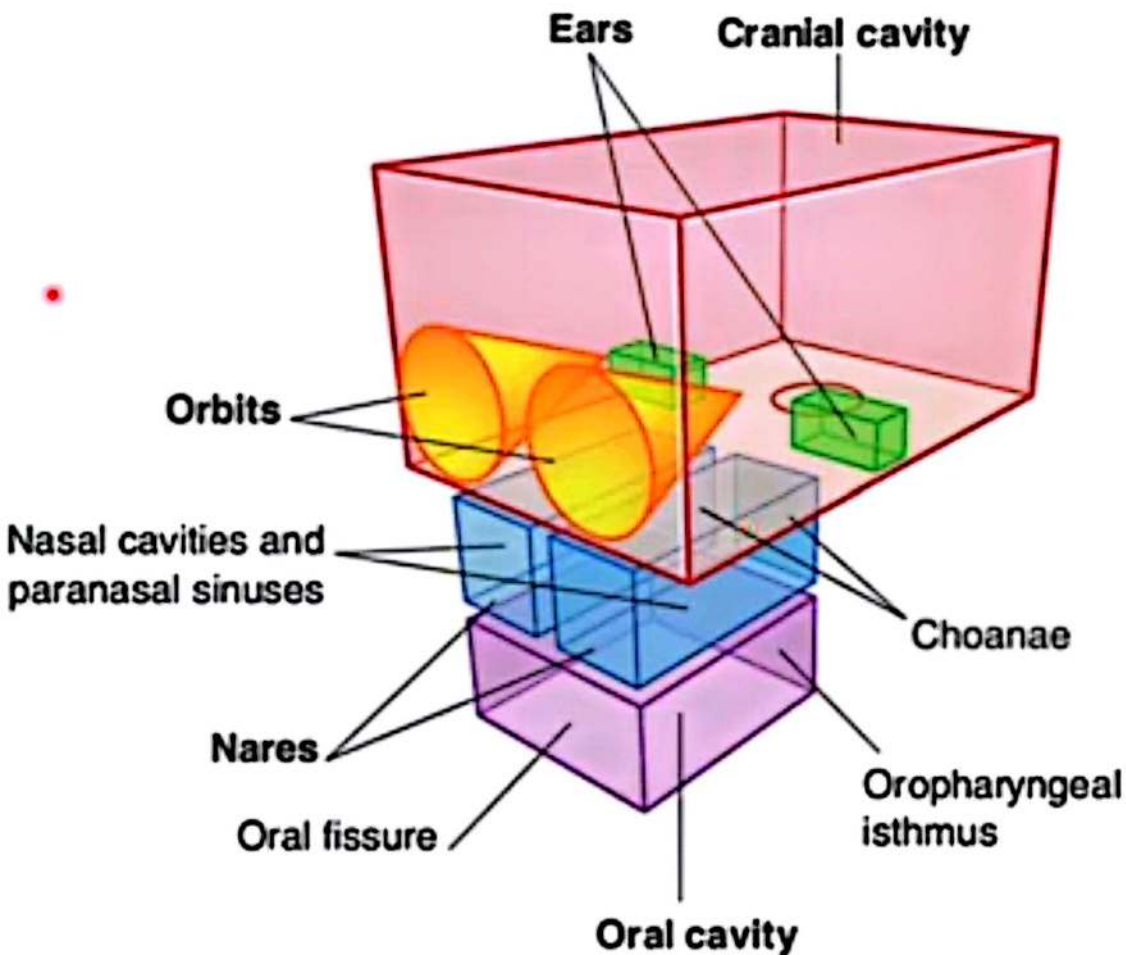
- Accessory Structures**
- ✓ **The Eyelids**
 - ✓ **Lacrimal Apparatus**

The Eye

- The eye allows us to **see** and **interpret** the **shapes, colors** and **dimensions** of objects in the world by processing the **light** they reflect or **send** to the brain (visual area).
- The eye is able to **see** in **bright** or **dim light**, but it **cannot see objects** when **light is absent**.



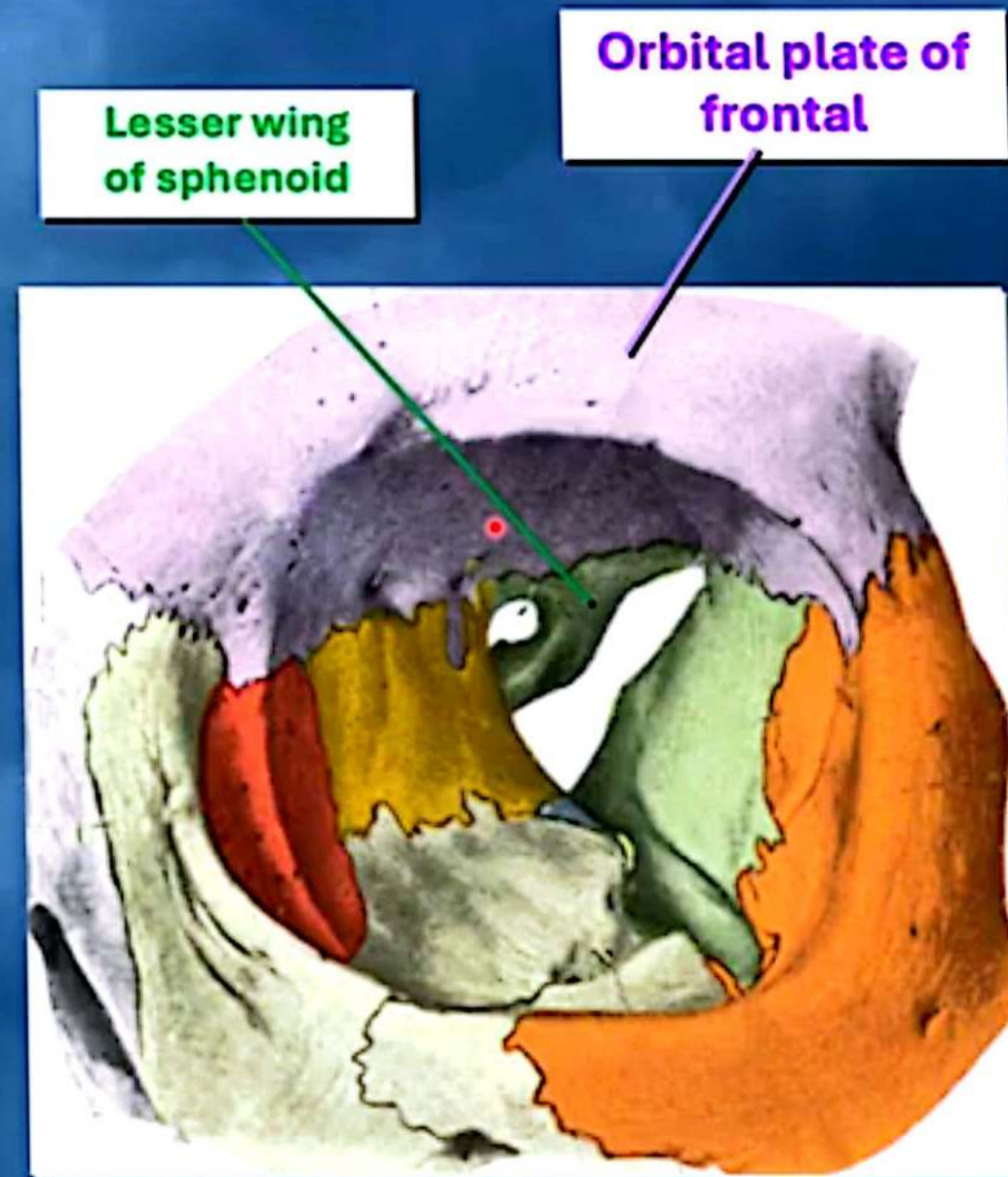
The Eye was found in orbital cavity



The Roof

Roof is formed by:

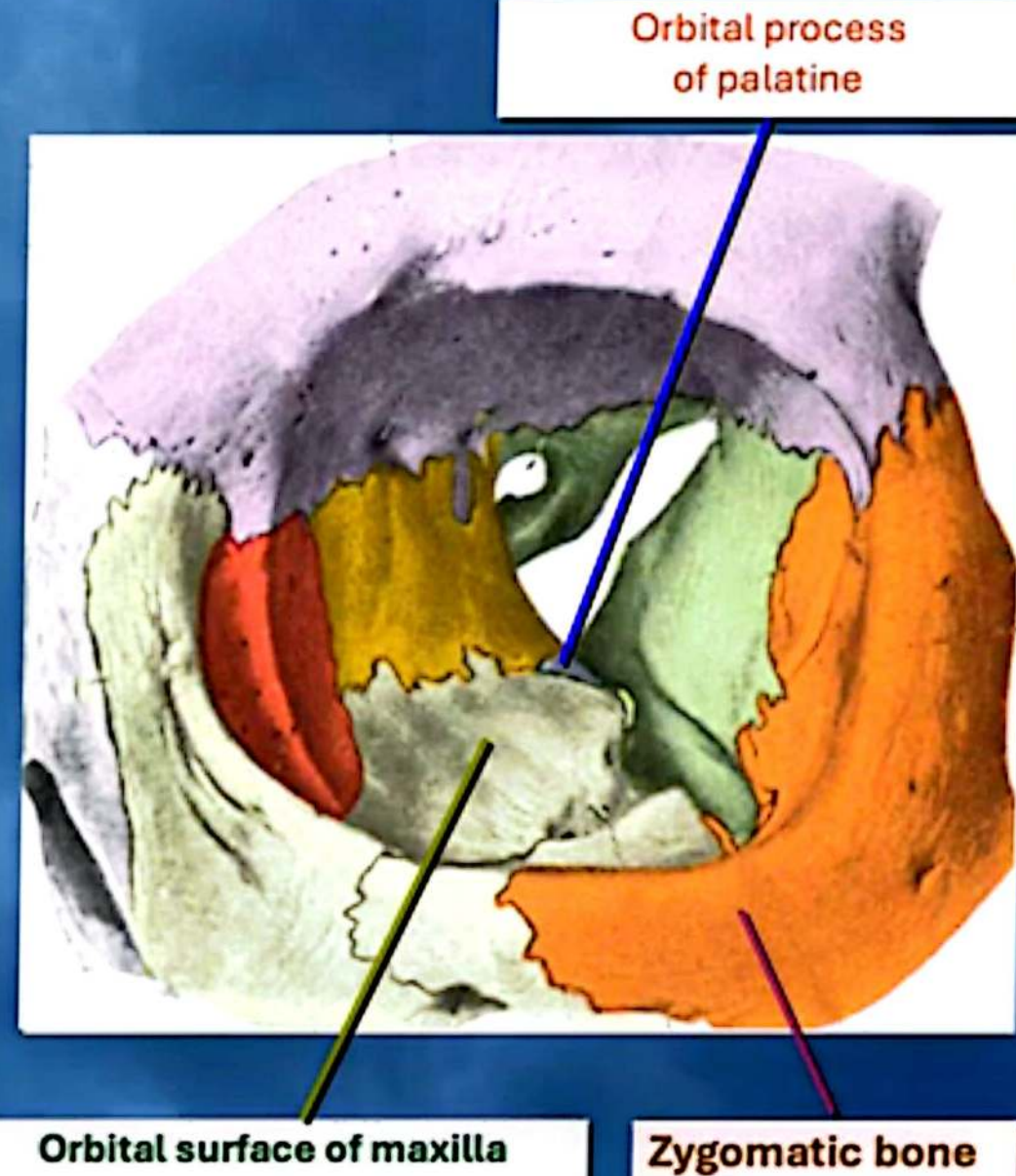
- **Orbital plate of frontal bone.**
- **Lesser wing of sphenoid.**



The Floor

Floor is formed by:

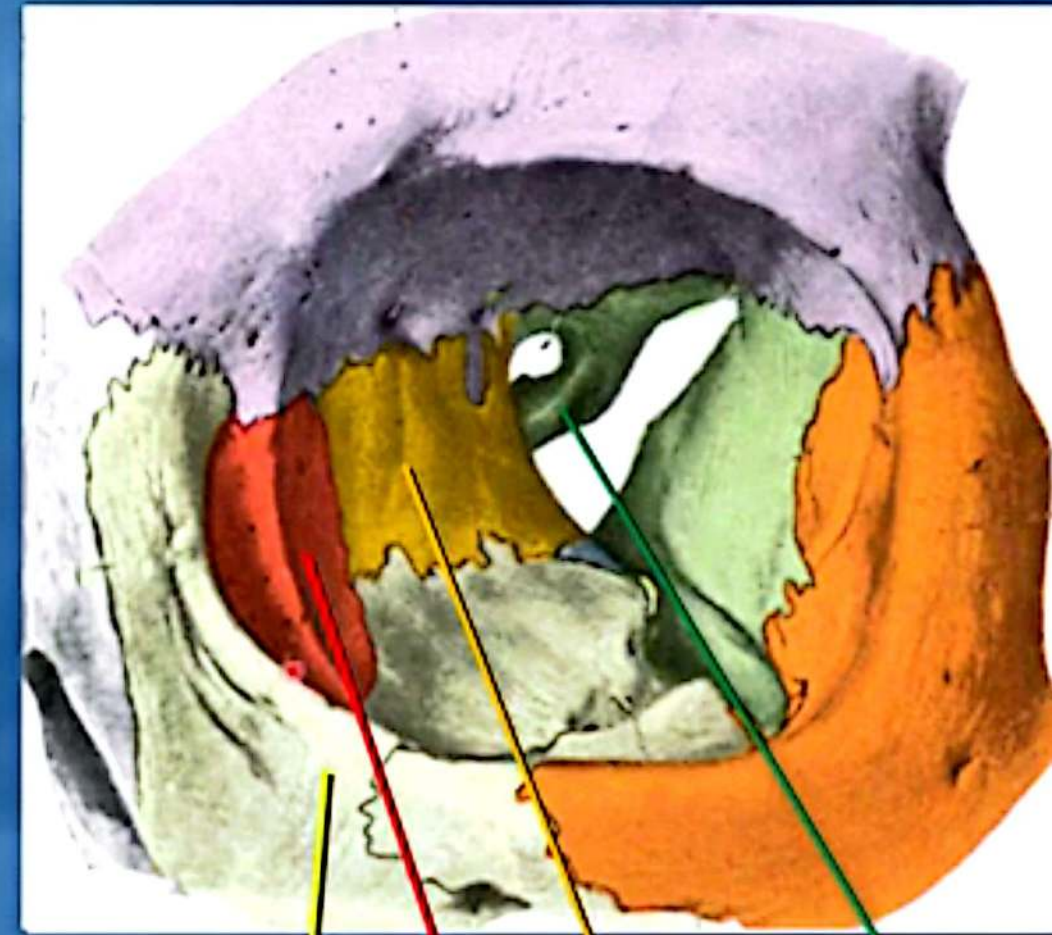
- Maxillary bone.
- Orbital plate and maxillary process of zygomatic bone.
- Orbital process of palatine bone.



Medial Wall

Medial wall is formed by:

- Frontal process of maxillary bone.
- Lacrimal bone.
- Ethmoidal bone.
- Body of sphenoid.



Frontal process of
maxilla

Lacrimal bone

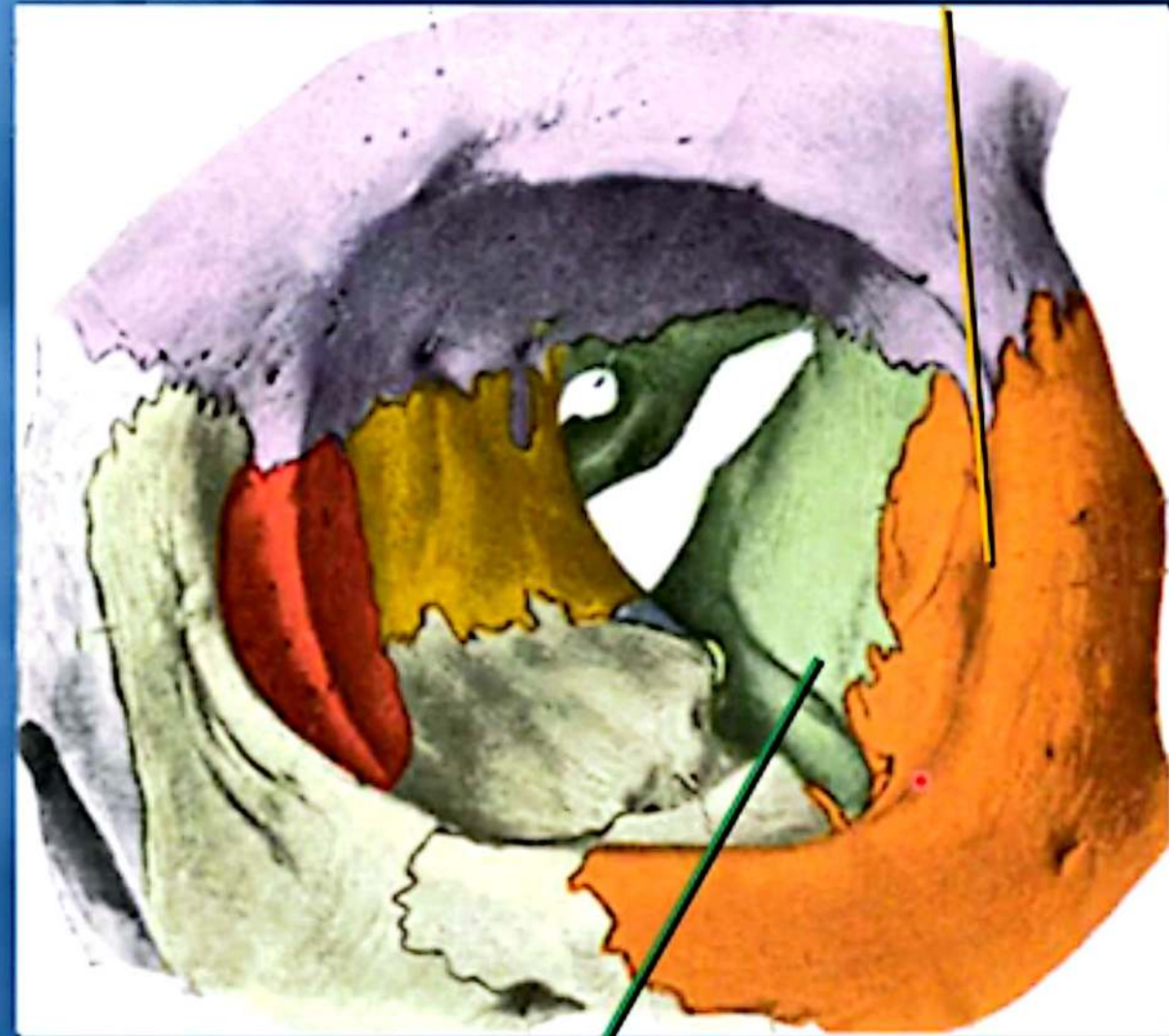
Orbital plate of ethmoid

Body of sphenoid
bone

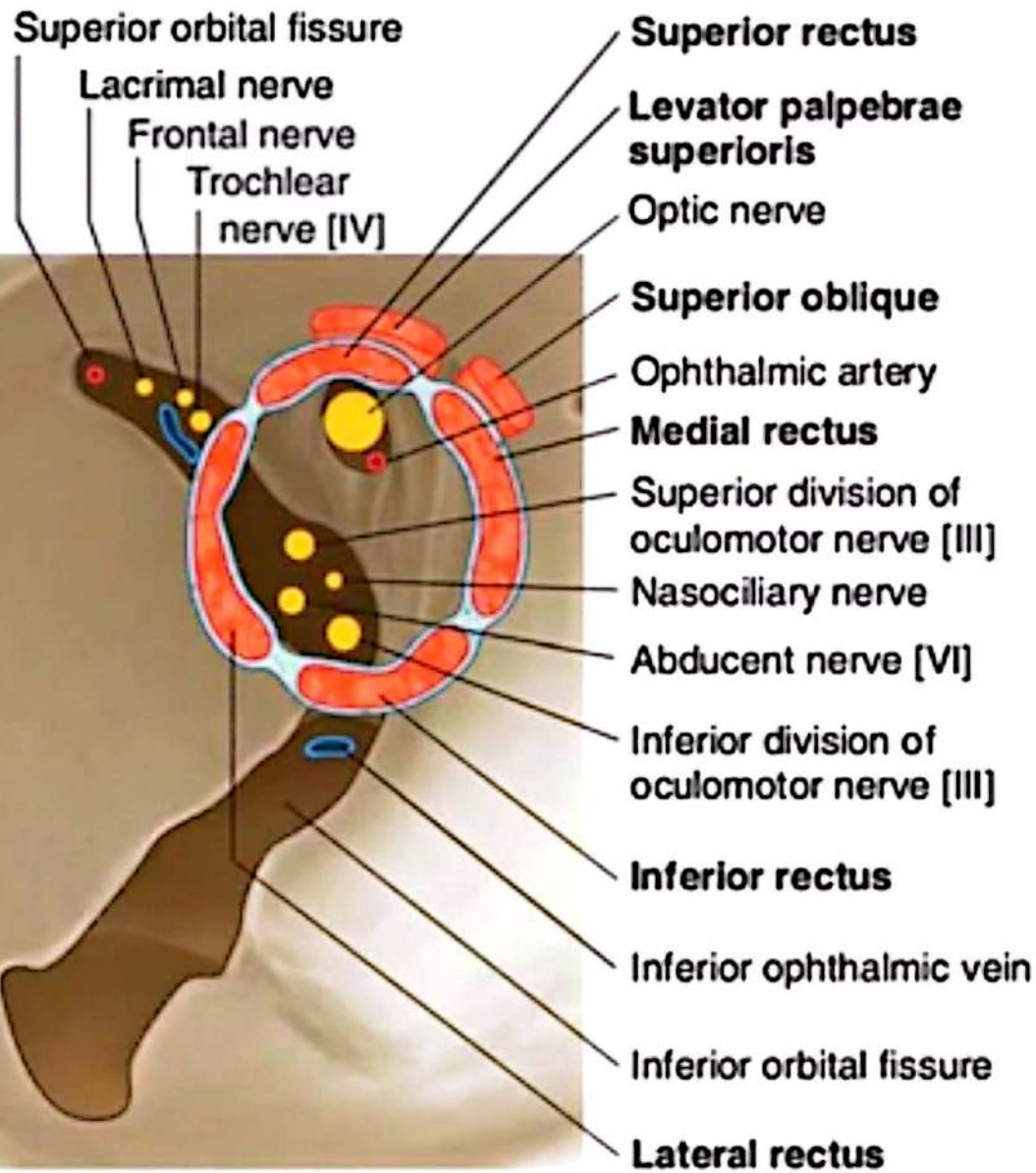
Lateral Wall

Lateral wall is formed by:

- Zygomatic bone.
- Greater wing of sphenoid.

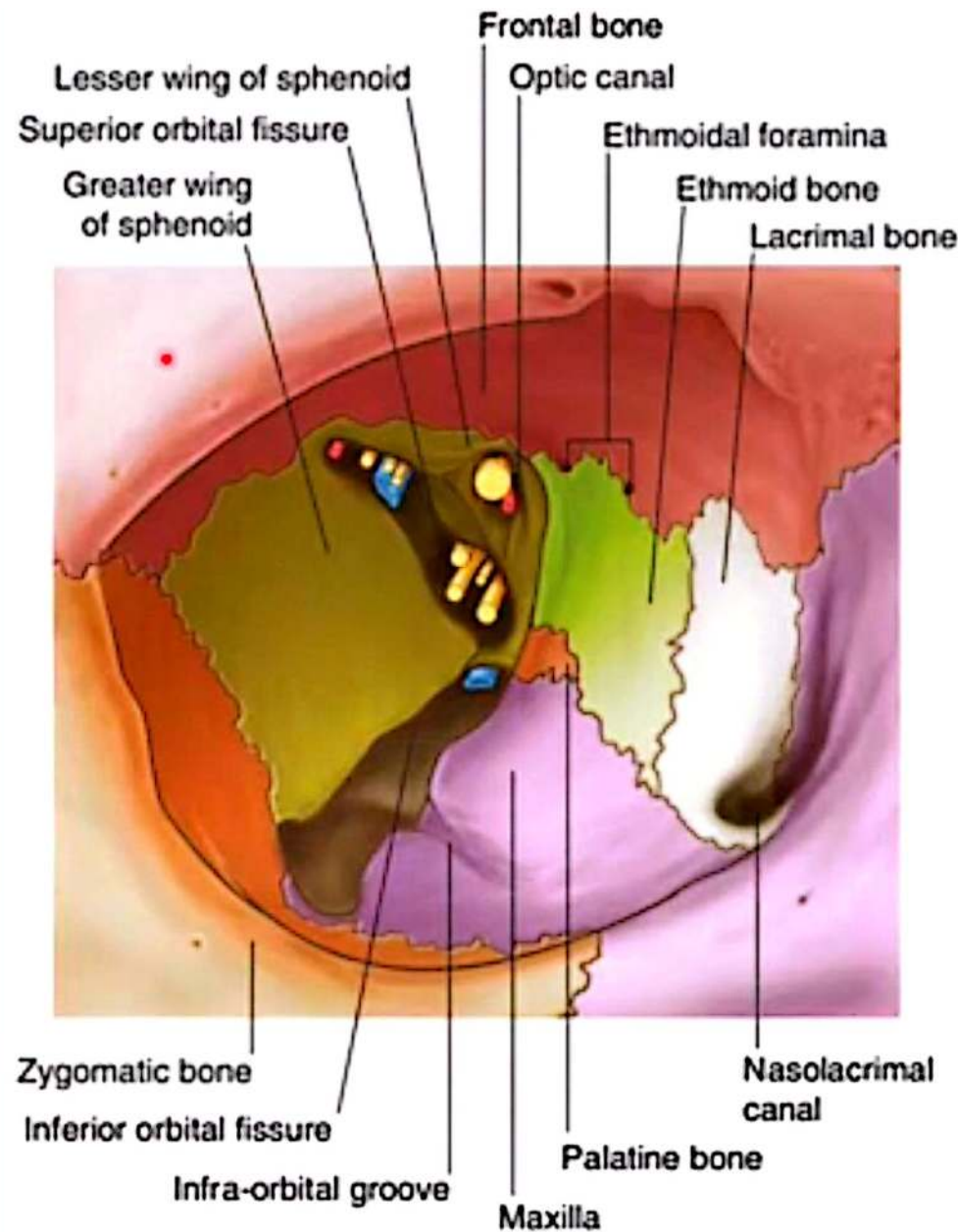


Greater wing of sphenoid bone



Lateral

Medial

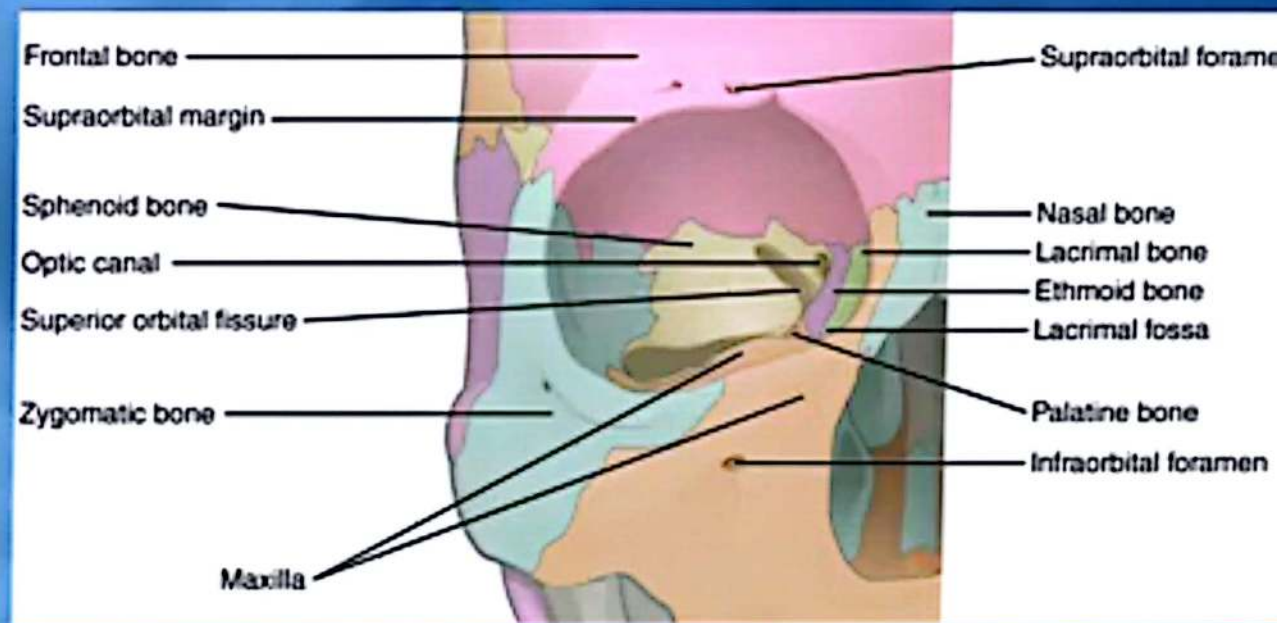


Orbital Openings

➤ **Optic canal optic nerve** with meninges and **ophthalmic artery**.

➤ **Superior orbital fissure:**

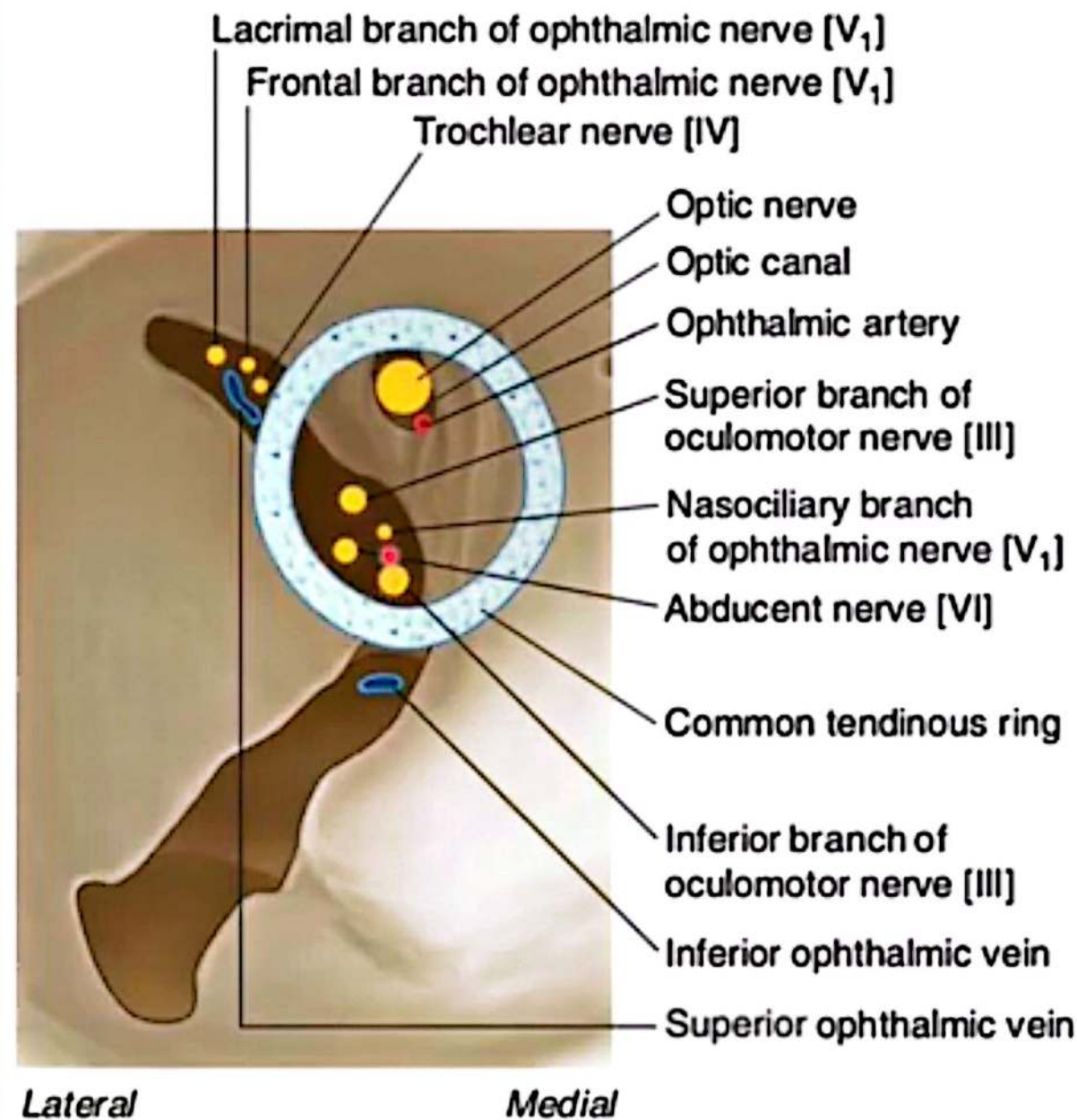
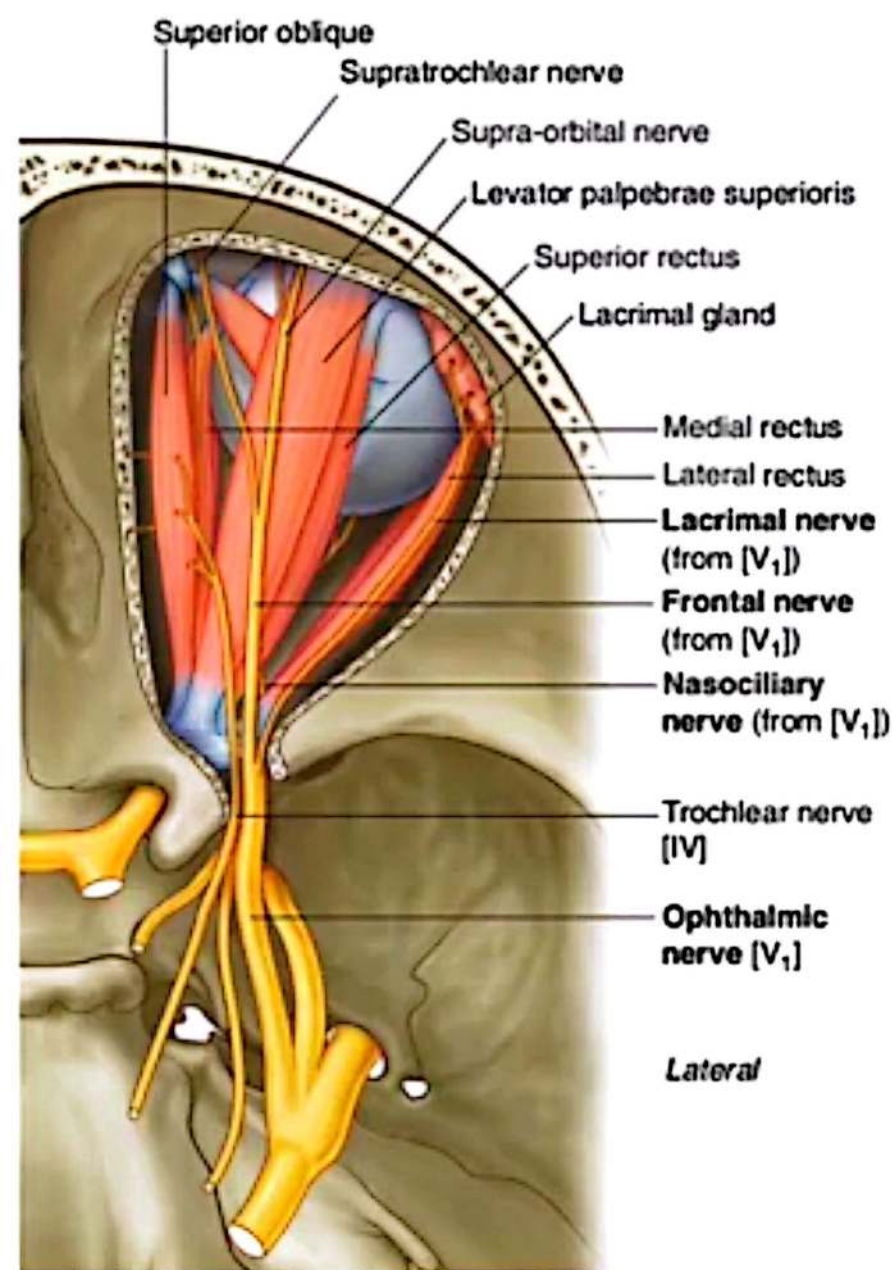
- ✓ Lacrimal nerve.
- ✓ Frontal nerve.
- ✓ Trochlear nerve.
- ✓ Superior and inferior veins.
- ✓ Oculomotor upper division.
- ✓ Nasociliary nerve.
- ✓ Abducent nerve.
- ✓ Oculomotor lower division.



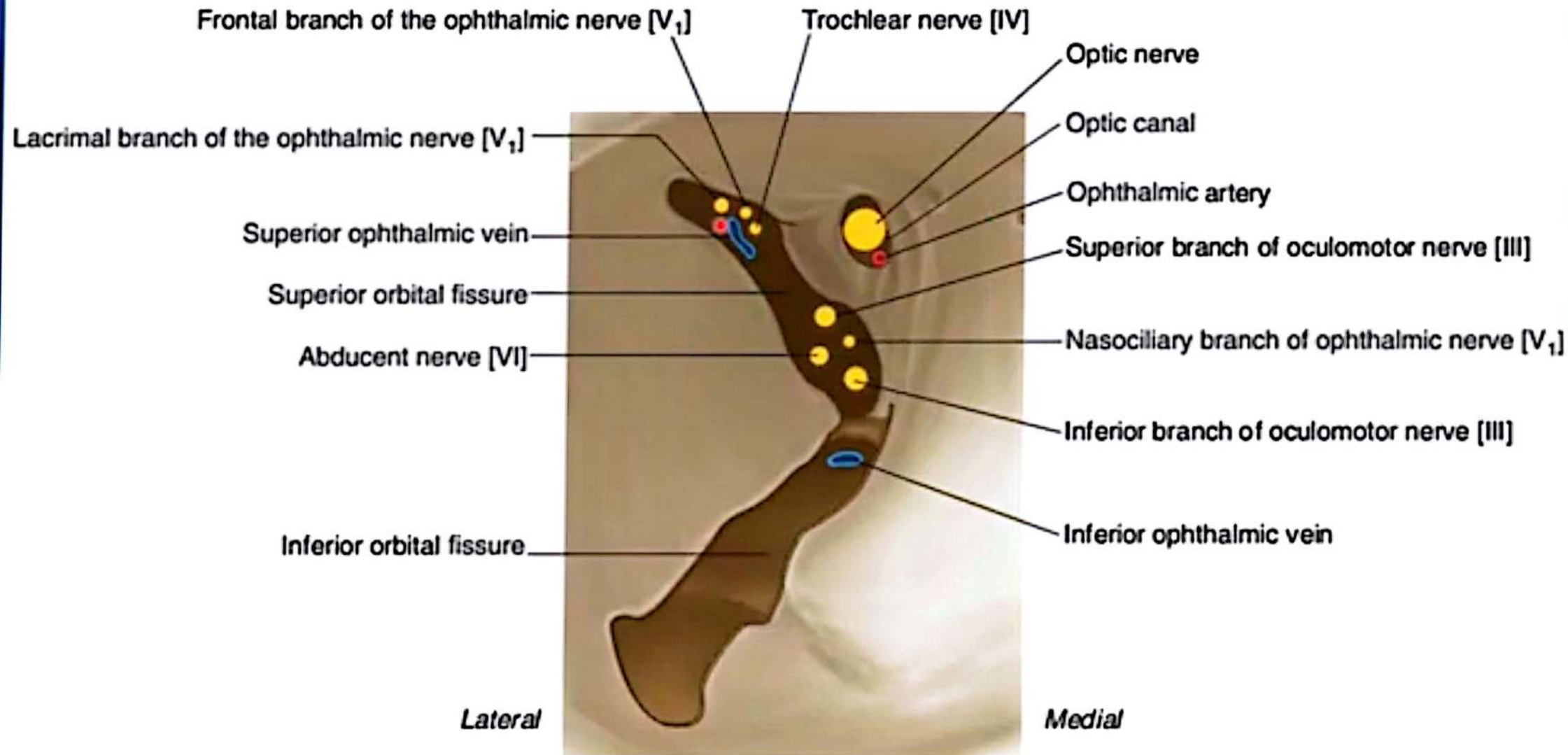
➤ **Inferior orbital fissure-** inferior ophthalmic vein

➤ **Superior orbital notch-** supraorbital nerve and vessels.

➤ **Infra orbital foramen-** infraorbital nerve and artery.

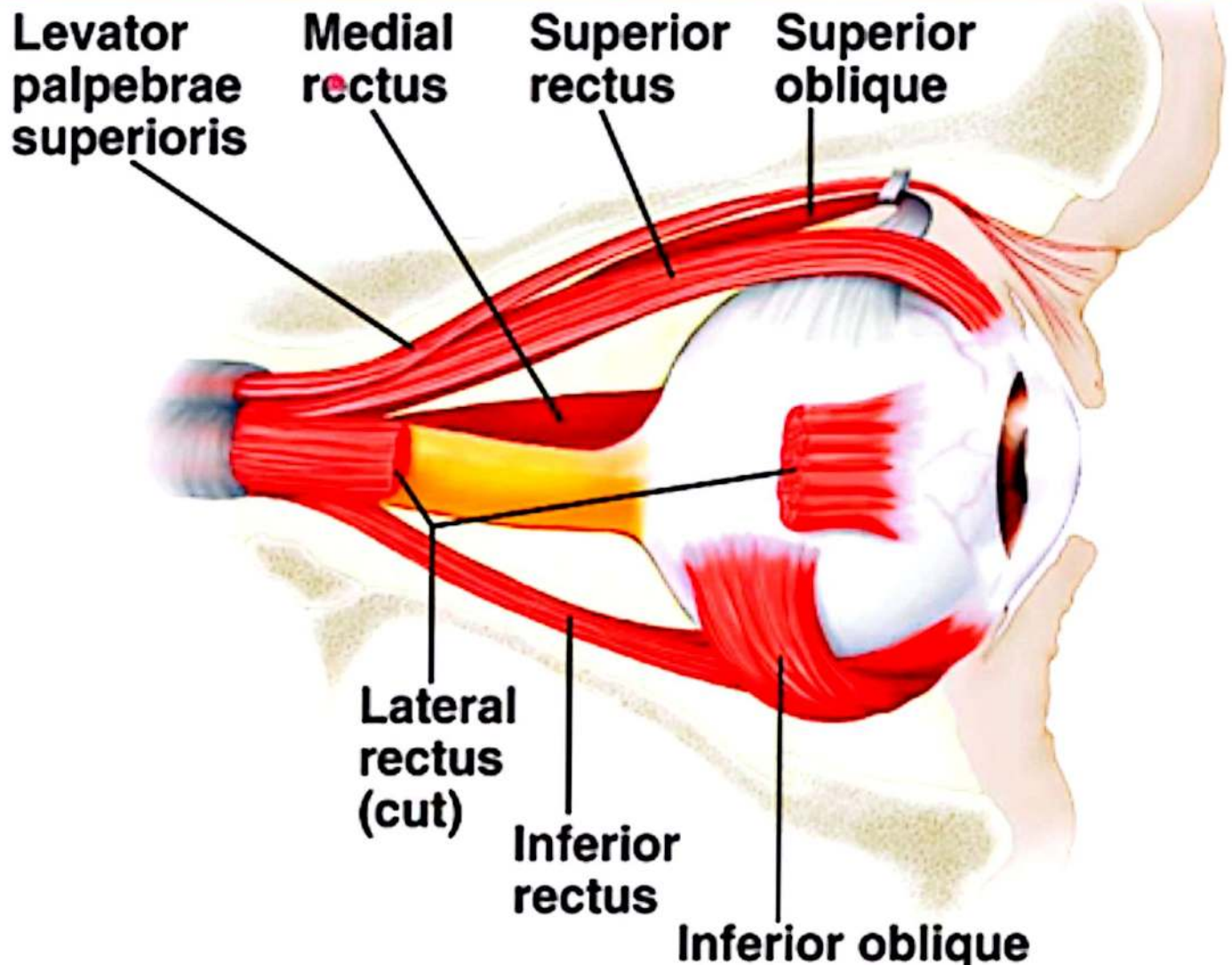


Orbital Openings



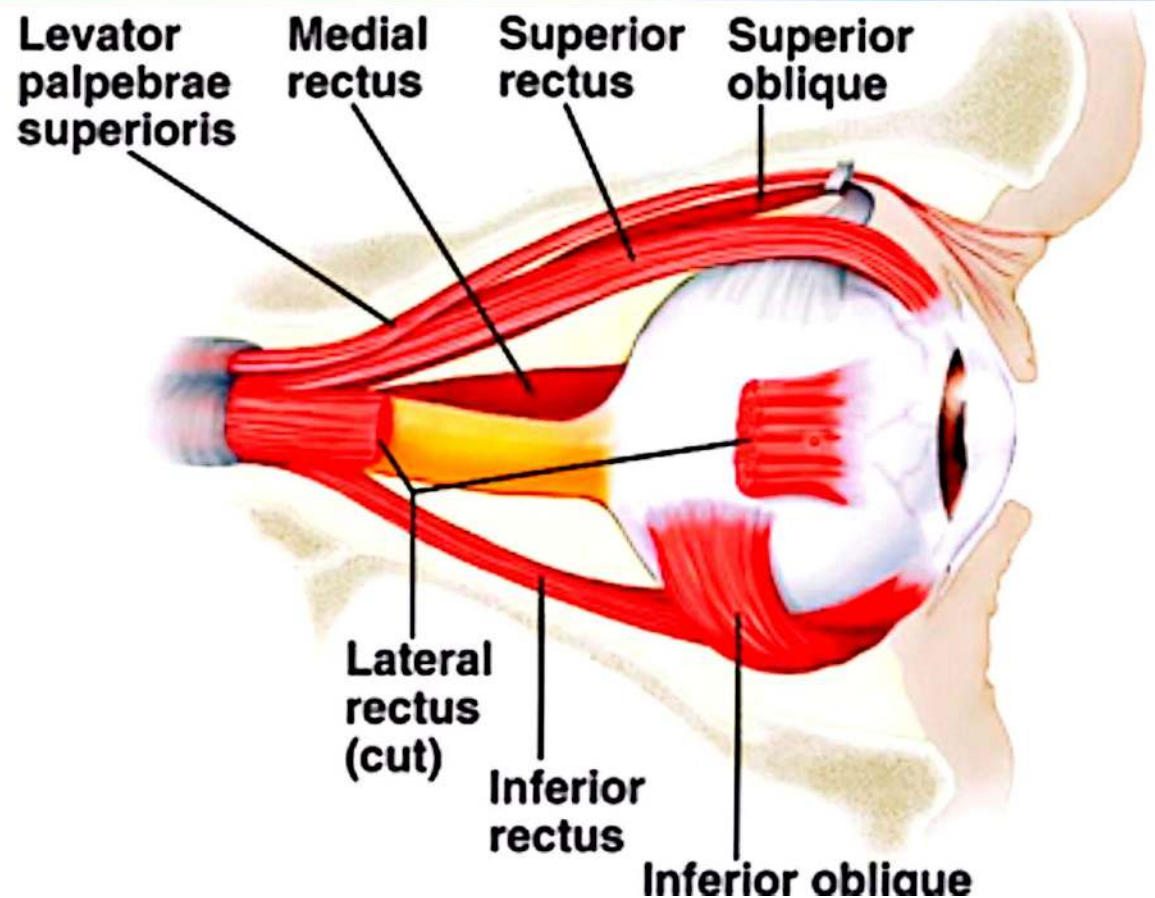
- The **extrinsic muscles** of the eye are **skeletal muscles** that move the eyeball.
- They originate from the **walls of the orbit** and insert upon the tough outer surface of the eyeball.
- There are **six extrinsic muscles**.

Extrinsic Muscles



Extrinsic Muscles

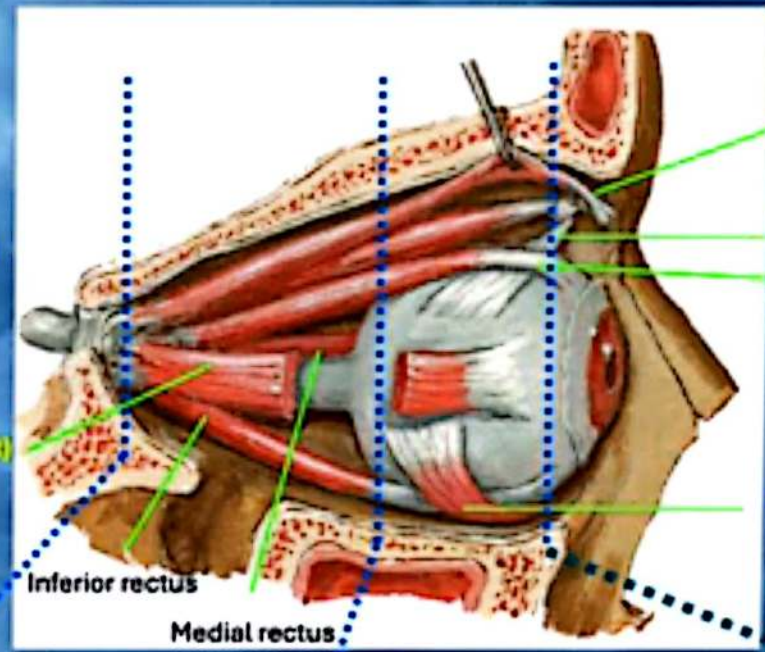
- **Lateral rectus:** moves eye laterally
- **Medial rectus:** moves eye medially
- **Superior rectus:** elevates or rolls eye upward
- **Inferior rectus:** depresses or rolls eye downward
- **Inferior oblique:** elevates and turns eye laterally
- **Superior oblique:** depresses and turns eye laterally



Ocular muscles

Optic nerve

Lateral rectus (cut)



Levator palpebrae

Superior oblique

Superior rectus

Inferior oblique

Inferior rectus

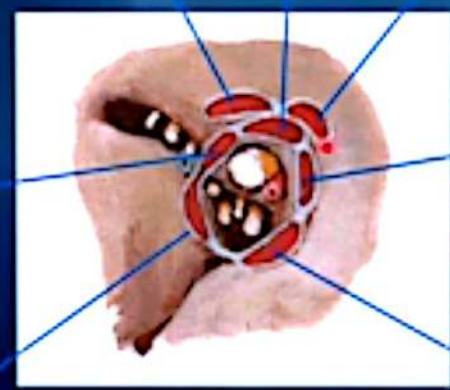
Medial rectus

LP
SR
SO

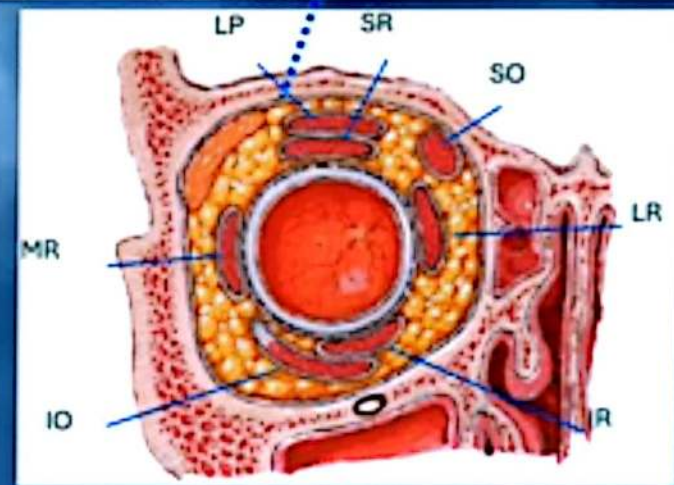
MR

IO

IR



Frontal views



LP

SR

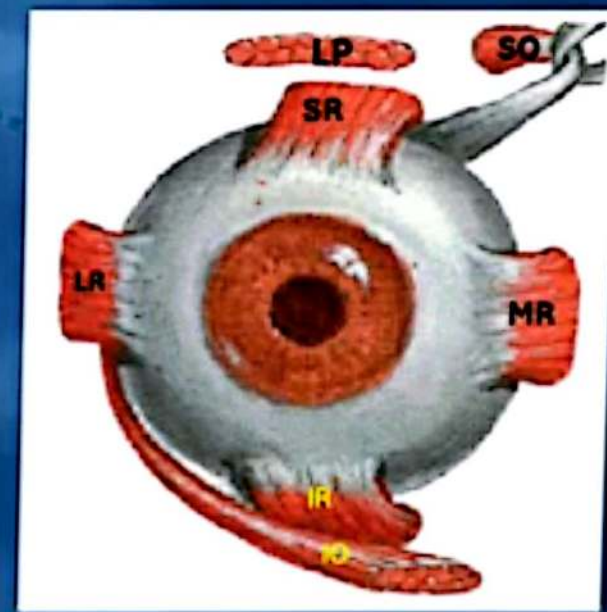
SO

MR

IO

LR

IR



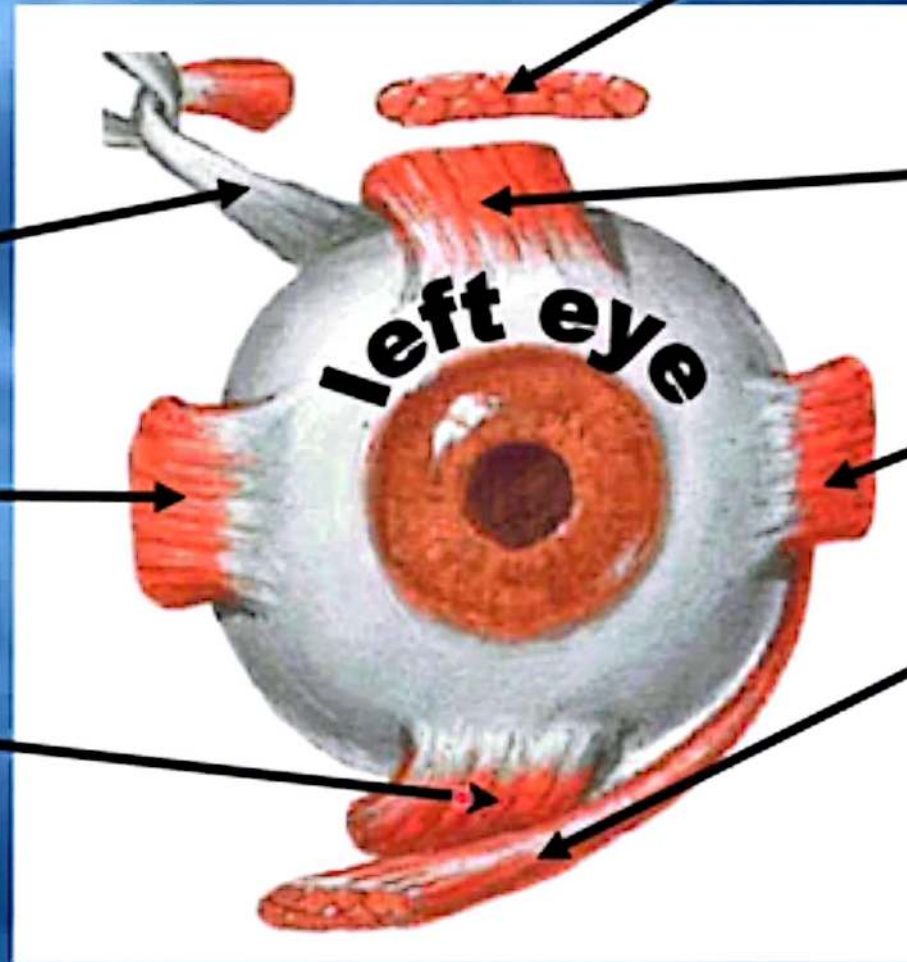
Extraocular Muscles

$(LR_6 SO_4)_3$

Superior oblique 4

Medial rectus 3

Inferior rectus 3



Levator palpebrae
superioris 3

Superior rectus 3

Lateral rectus 6

Inferior oblique 3

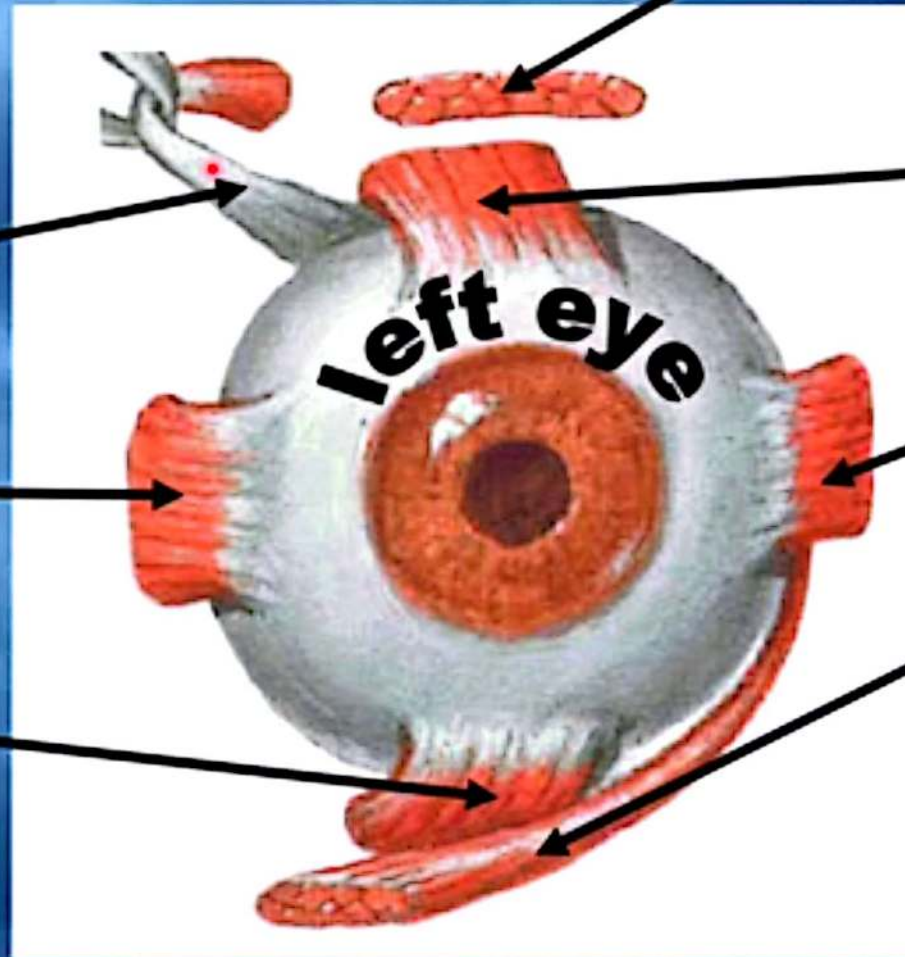
Extraocular Muscles

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Superior oblique 4

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superioris 3

Superior rectus 3

Lateral rectus 6

Inferior oblique 3

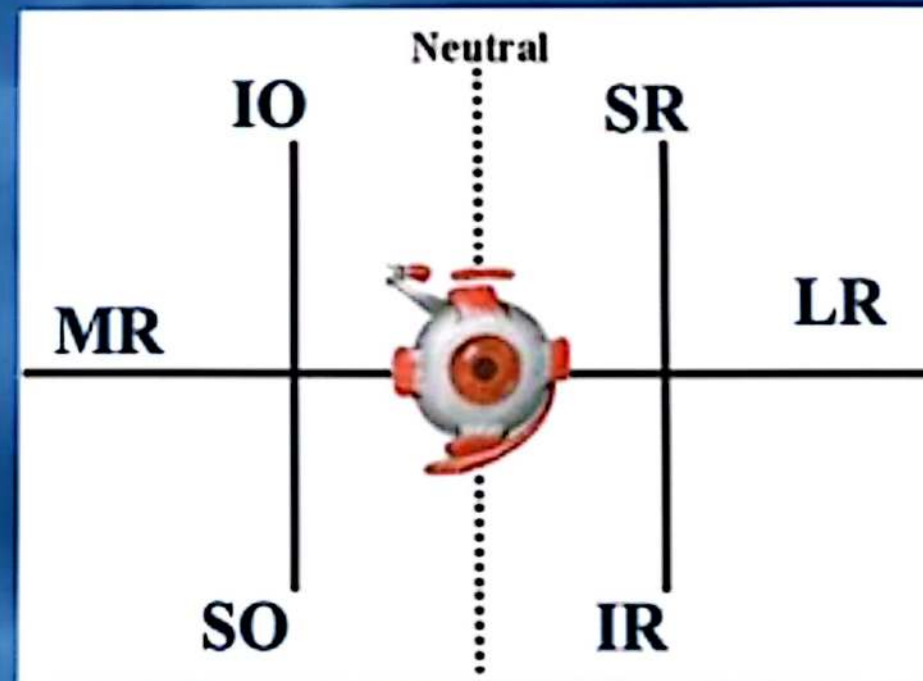
Note: Intraocular muscles are inside the eye and are smooth muscle

Muscle Testing VS Muscle Function

Muscle tested		Direction to move eye when testing muscle
Superior rectus		Look laterally and upward
Inferior rectus		Look laterally and downward
Lateral rectus		Look laterally
Medial rectus		Look medially
Inferior oblique		Look medially and upward
Superior oblique		Look medially and downward

- **Lateral rectus:** moves eye laterally.
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- **Superior rectus:** elevates or rolls eye upward.
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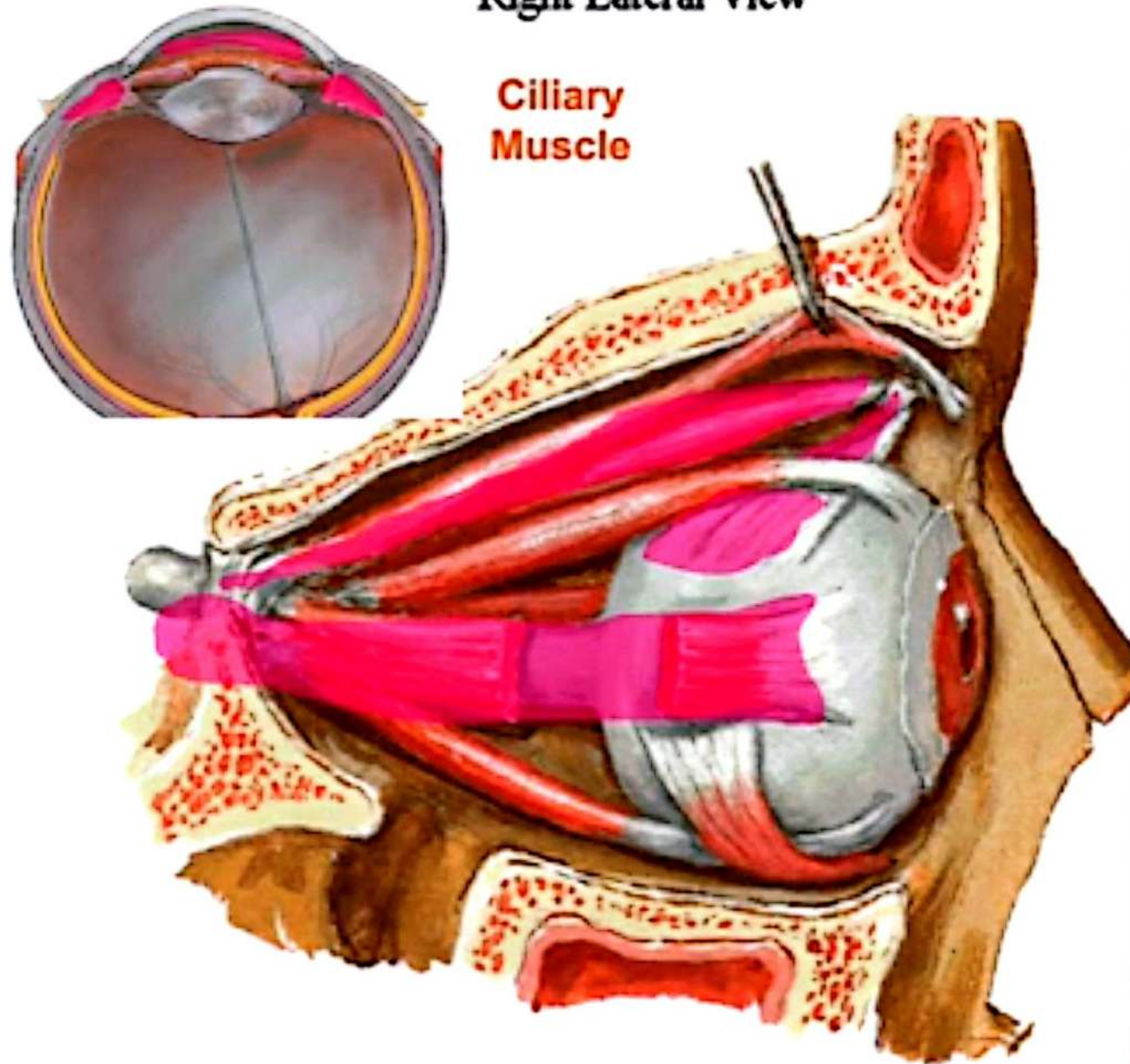
The “H” Test



**Constrictor Pupillae
Muscle**

Extrinsic Eye Muscles Right Lateral View

**Ciliary
Muscle**



EXTRINSIC EYE MUSCLES

**Motor - ALL Extrinsic
Eye Muscles- Except
SO4 & LR6**

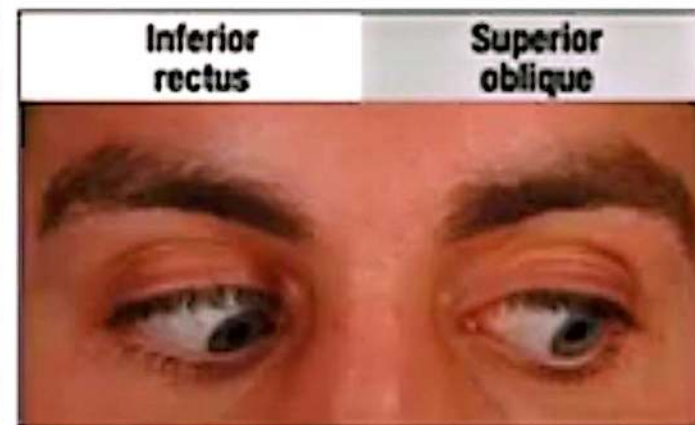
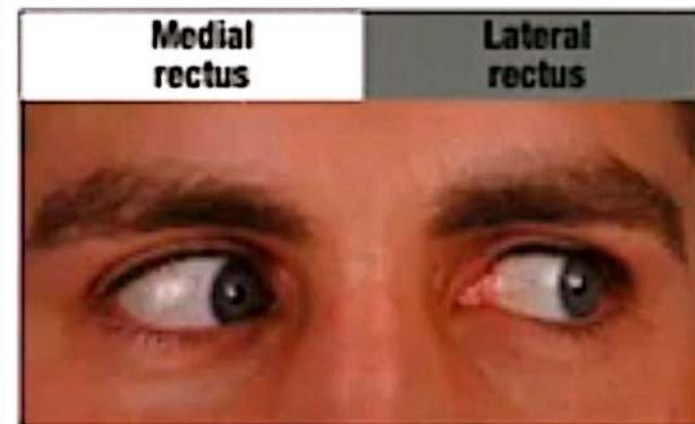
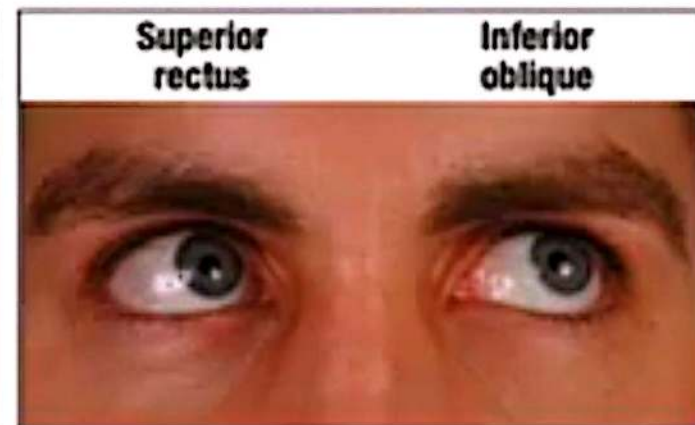
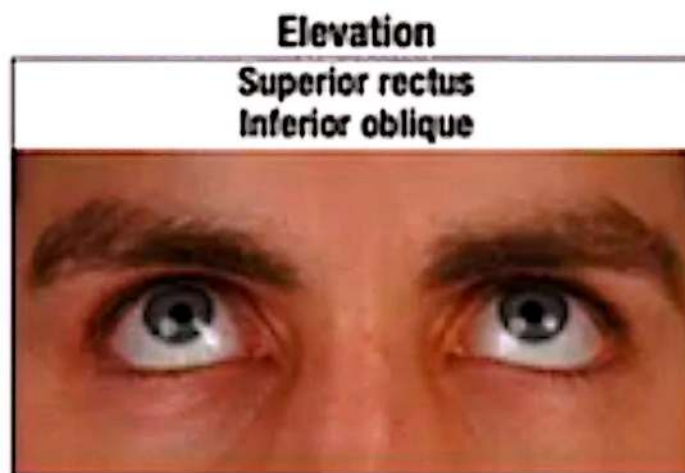
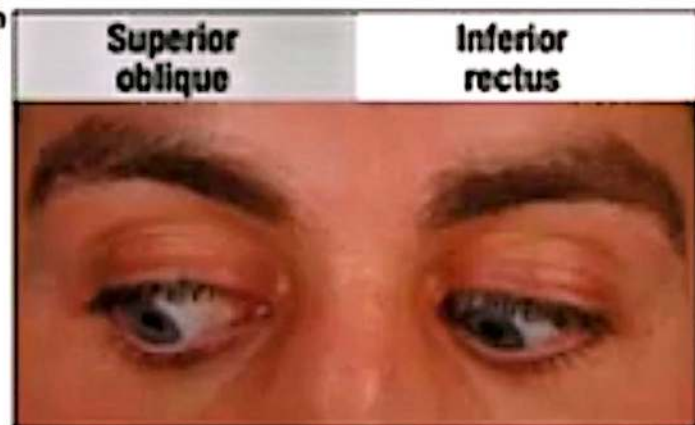
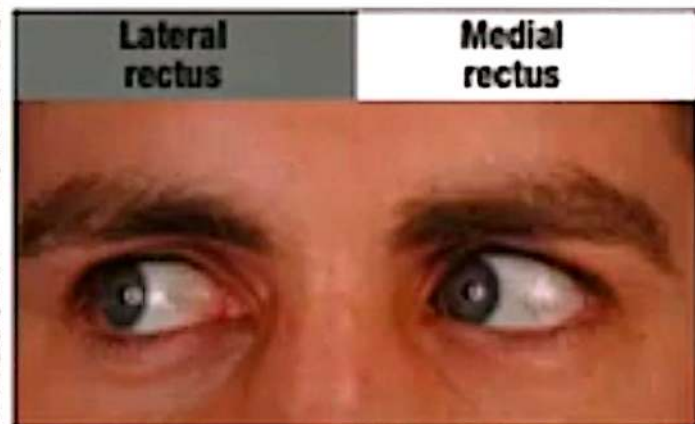
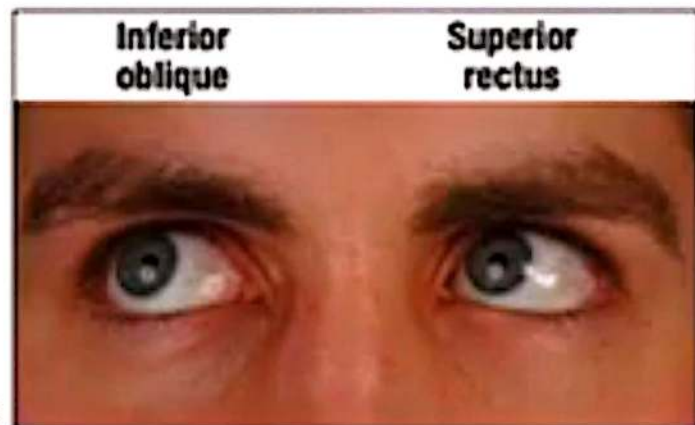


**Intrinsic Eye
Muscles (III)**

**Constrictor Pupillae
Muscle**

Ciliary Muscle

Right Abduction - Left Adduction

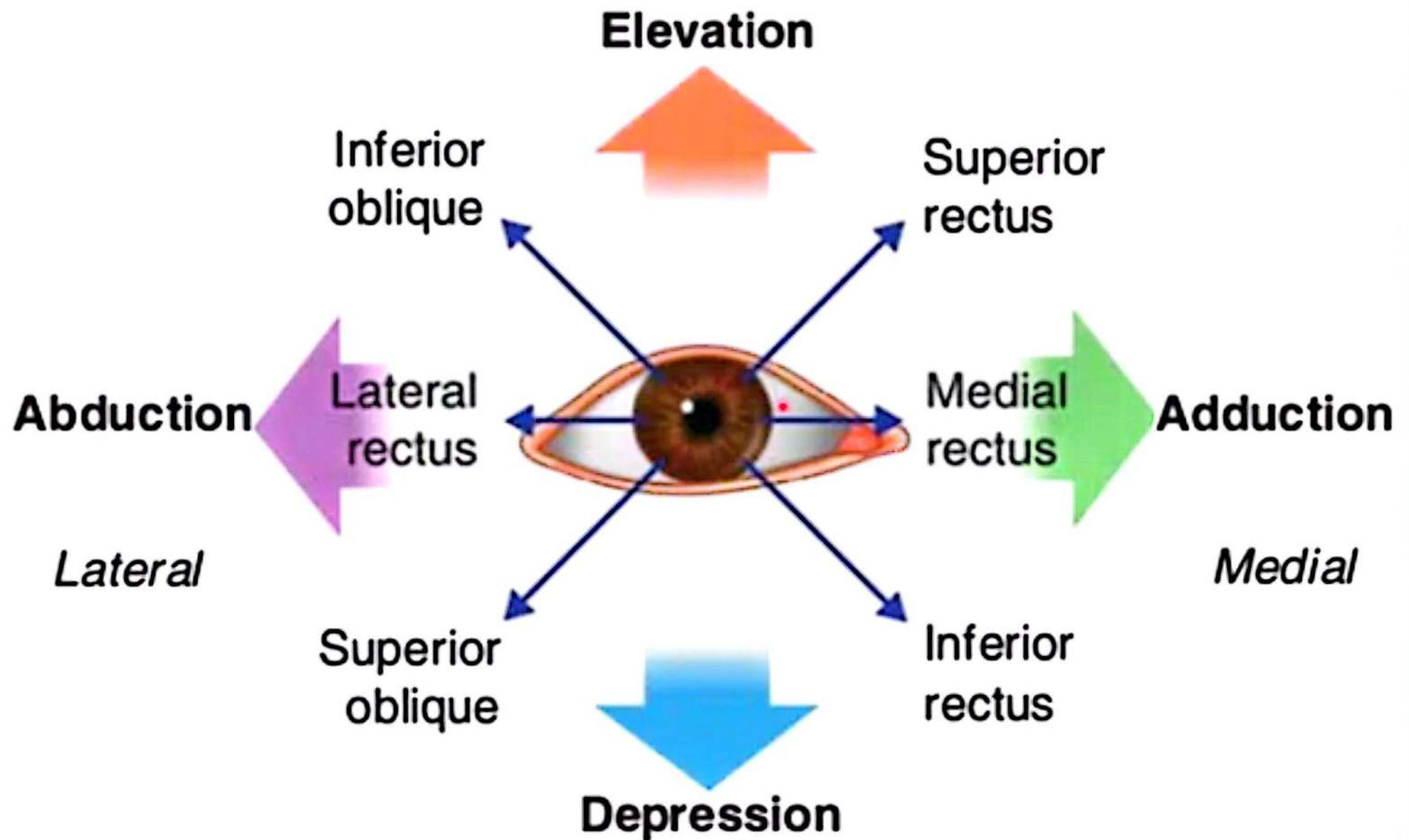


Left Abduction - Right Adduction

□ Oculomotor nerve (CN III)

□ Trochlear nerve (CN IV)

■ Abducent nerve (CN VI)



The background of the slide is a photograph of a bright blue sky filled with soft, white, fluffy clouds. The clouds are scattered across the frame, with some appearing more dense than others. The overall tone is bright and airy.

Will be continue ...

Lecture Points

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 - ✓ **The Roof.**
 - ✓ **The Floor.**
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 - ✓ **Lateral Wall.**
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- **Extrinsic Muscles.**

- **Structures (layer) of Eye.**
- **The outer fibrous tunic (the fibrous coat).**
 - ✓ **sclera.**
 - ✓ **cornea.**
- **The middle vascular tunic (the vascular pigmented coat).**
 - ✓ **The choroid.**
 - ✓ **The ciliary body.**
 - ✓ **The iris and Pupil.**
 - **Lens**
- **The inner nervous tunic.**
 - ✓ **The Retina.**
 - ✓ **Optic Disc**

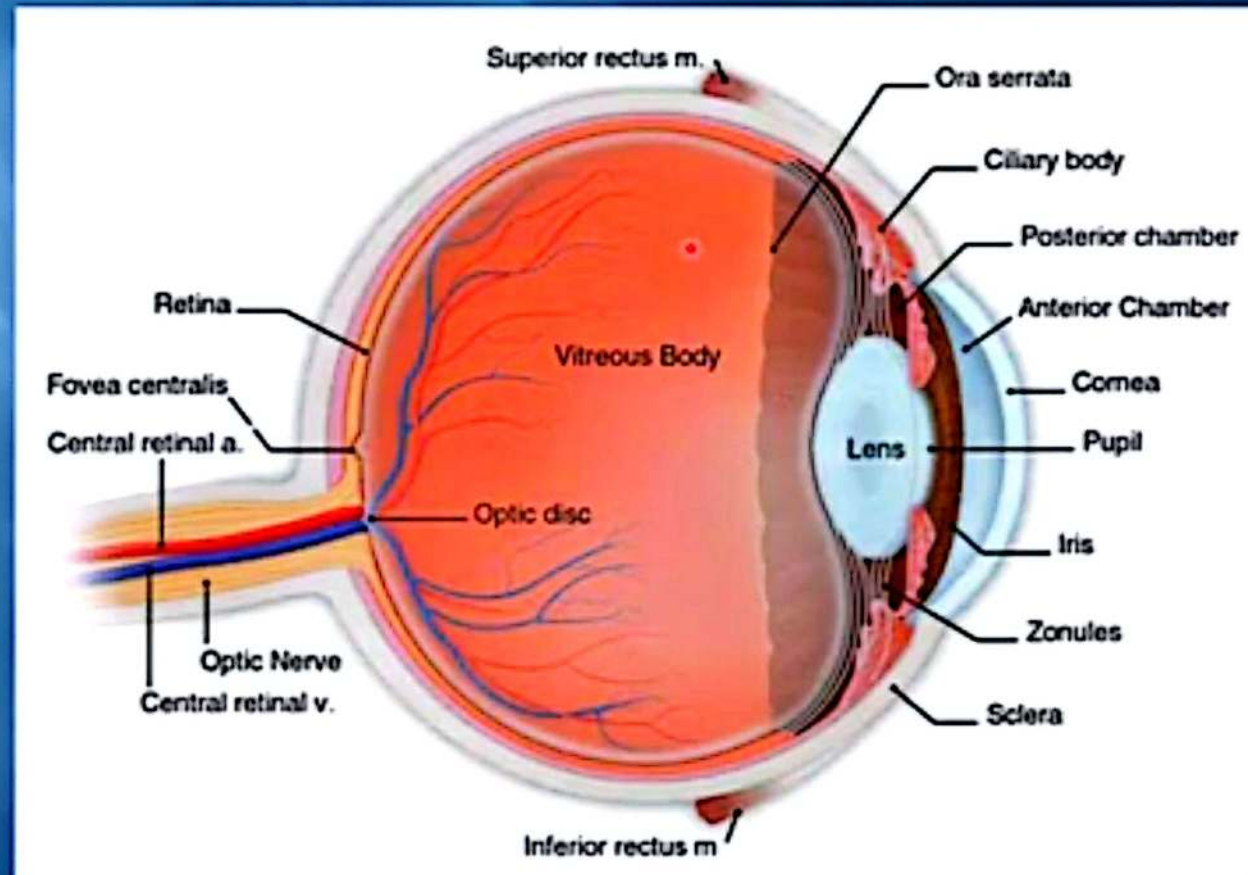
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(The refractive media):**
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 - ✓ **When you look at any object**

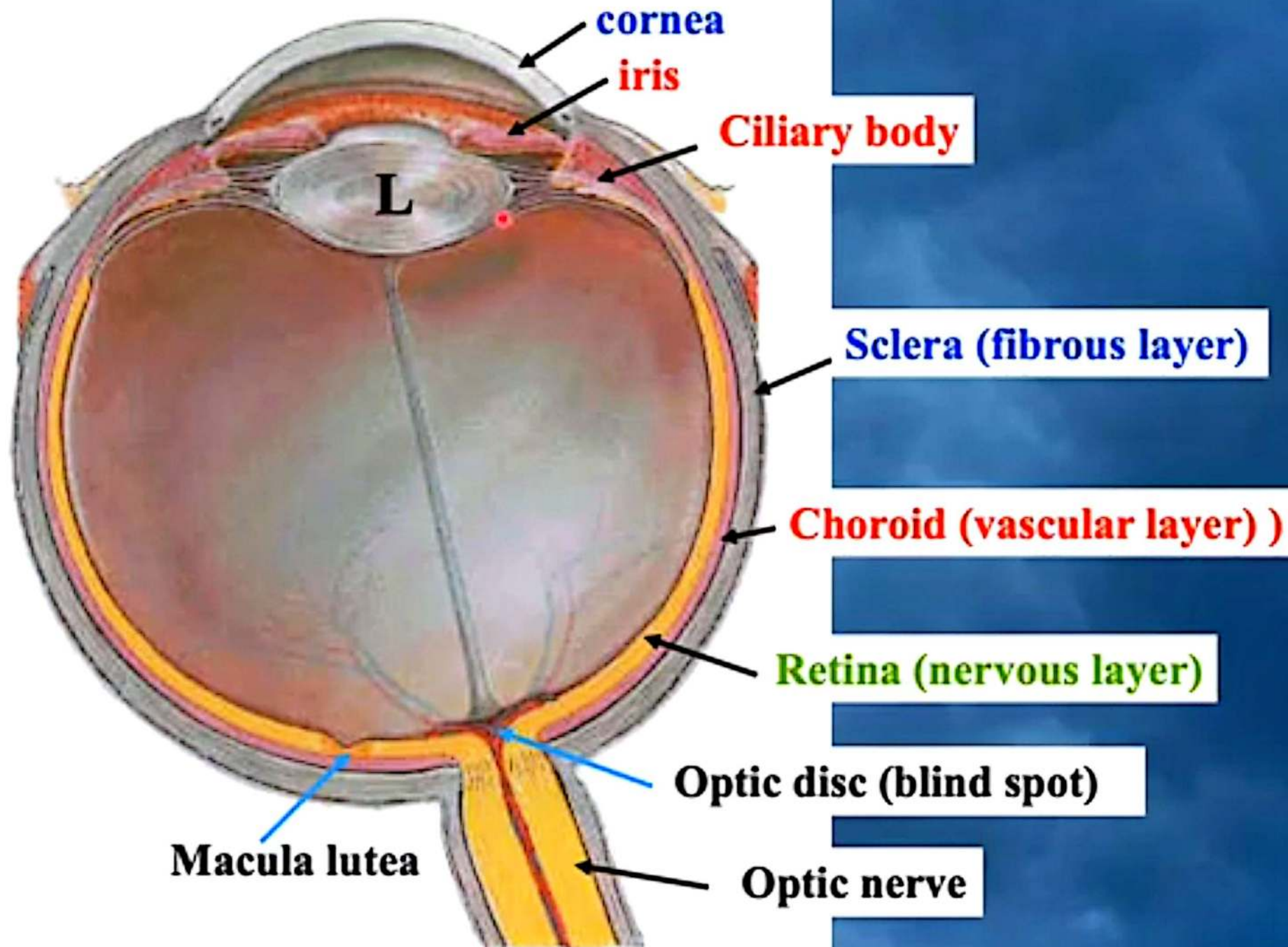
Accessory Structures

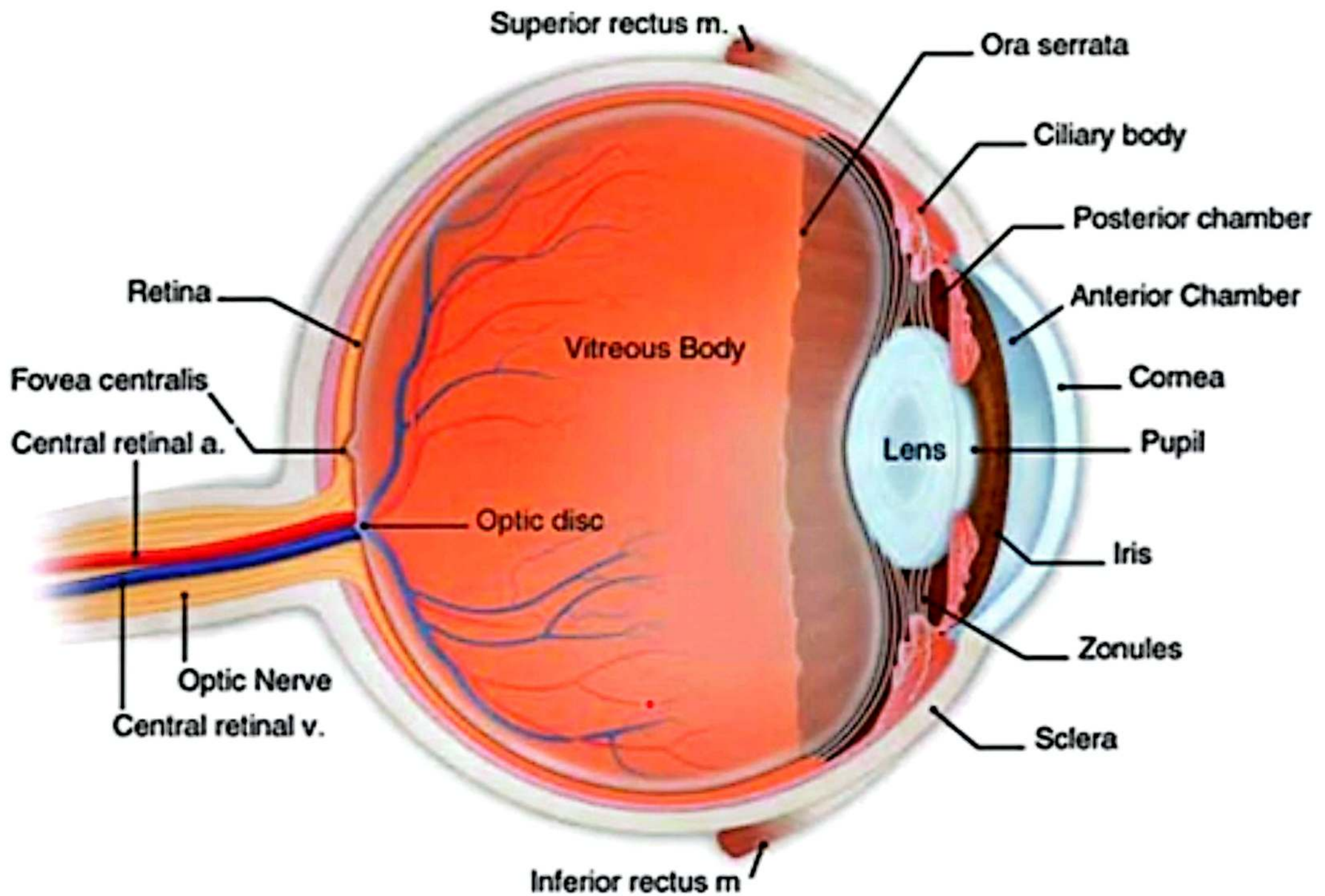
- ✓ **The Eyelids**
- ✓ **Lacrimal Apparatus**

- The eye is a spherical structure about 2.5 cm in diameter.
- Its wall consists of three distinct layers, (or tunics).
 - The outer fibrous tunic (the fibrous coat)
 - The middle vascular tunic (the vascular pigmented coat).
 - The inner nervous tunic.
- Inside the eye are structures that divide the eye into fluid-filled compartments.

Structures (layer) of Eye

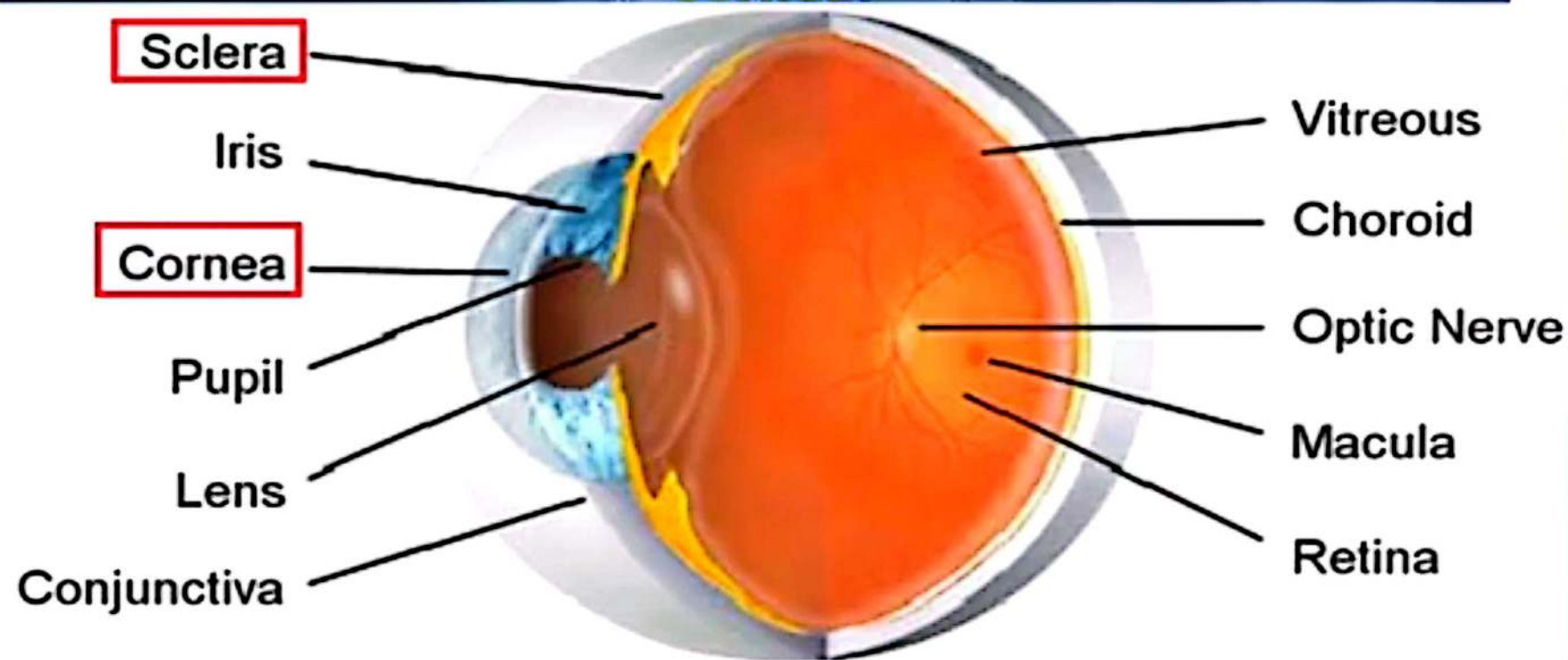
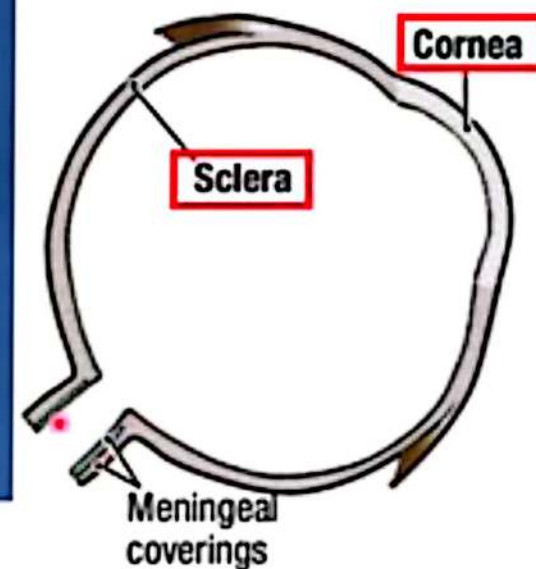






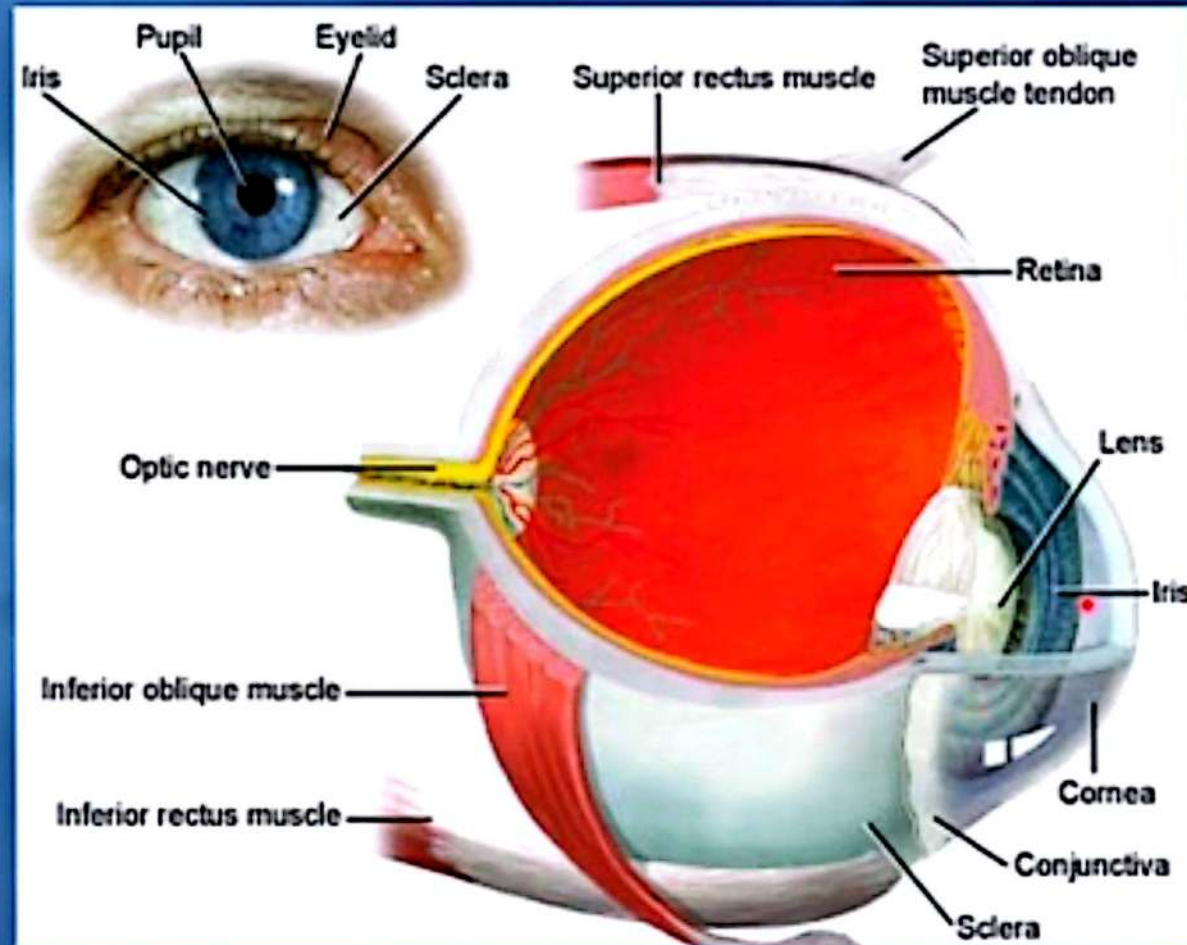
Fibrous Tunic

- It is thick outermost layer of the eyeball.
- It contains two regions:
 - Posterior opaque part **sclera**.
 - Anterior transparent part **cornea**.



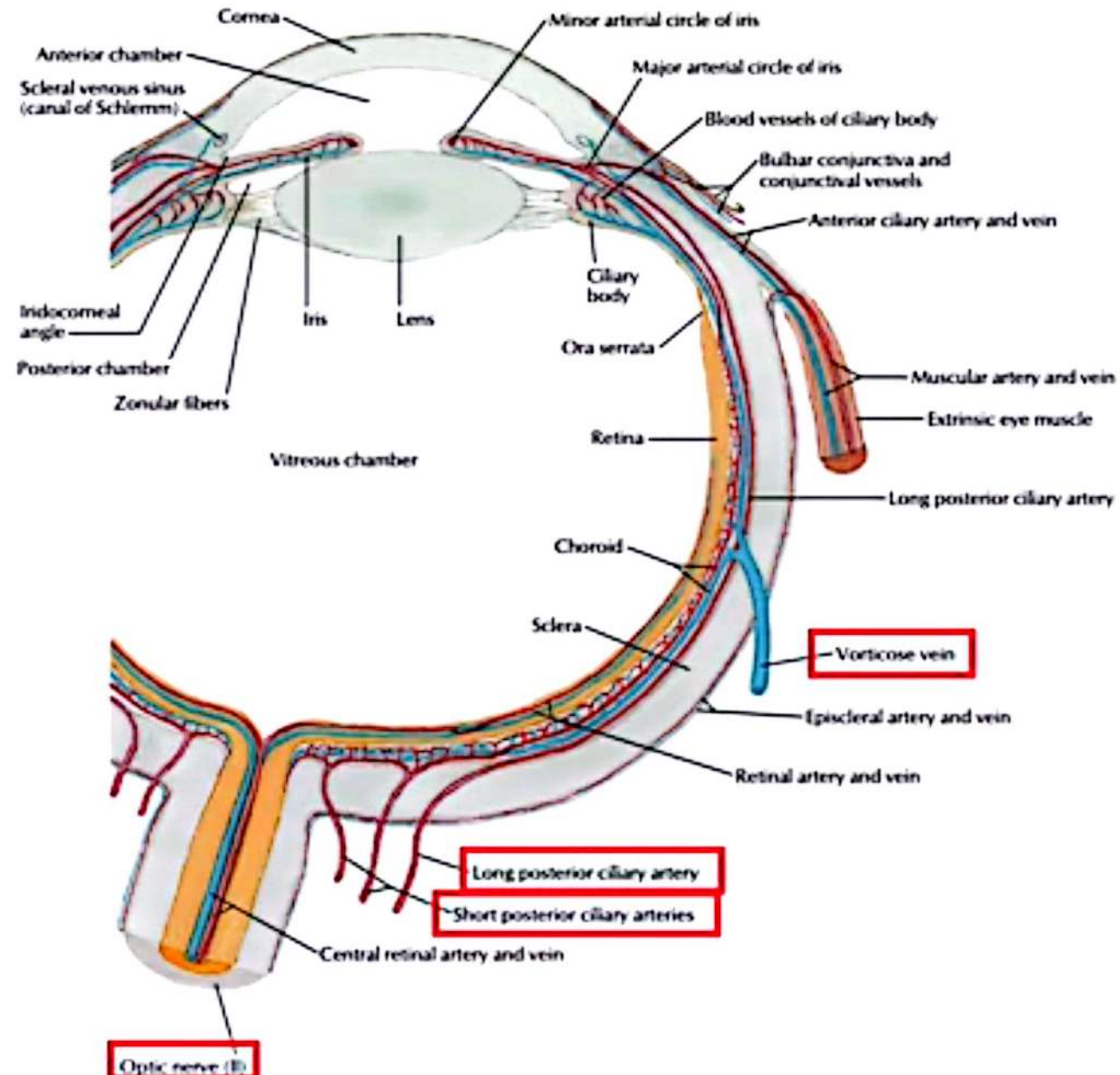
- Forms most of the **fibrous layer**.
- It is composed of **white fibrous connective tissue** and is often called the “**white of the eye**.”
- It forms a **thick, tough protective layer** that provides **shape** to the **eyeball** and **protects** its inner parts.
- It contains an **abundant** supply of **blood vessels**.
- The **posterior surface** of the sclera is **penetrated** by the **optic nerve**.

Sclera



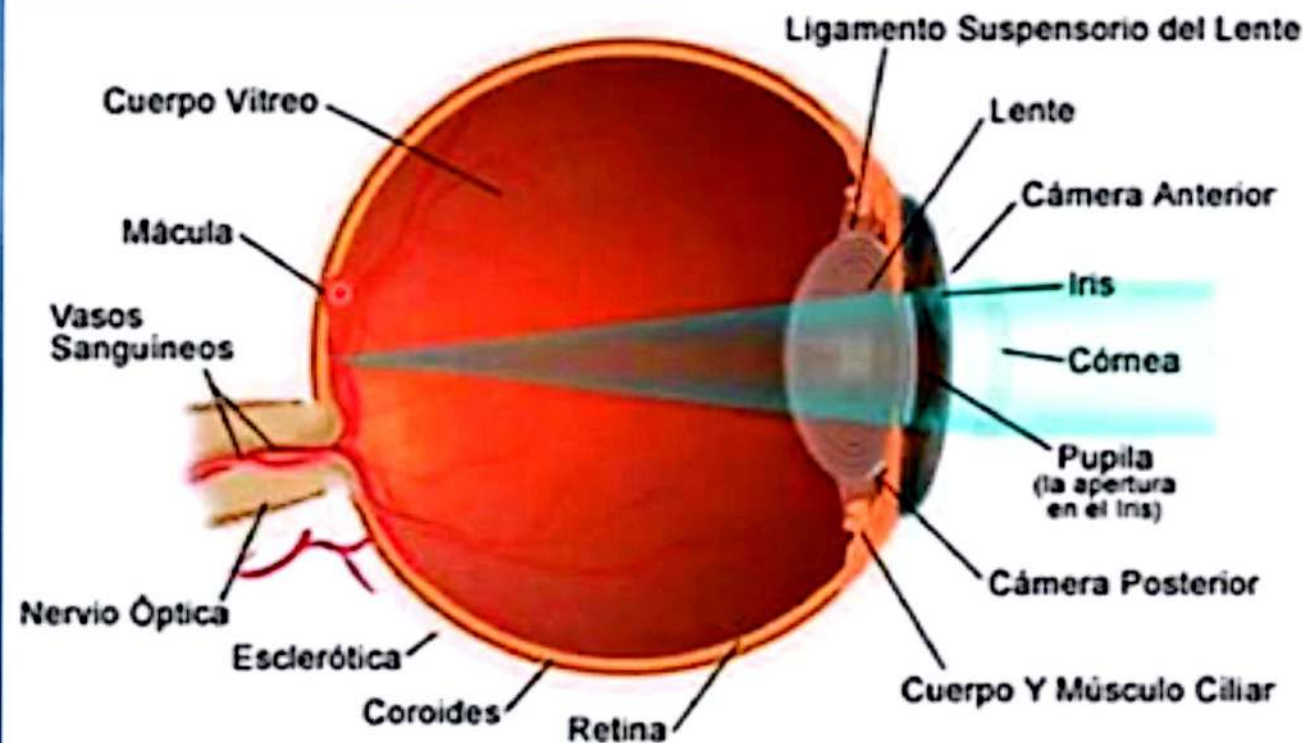
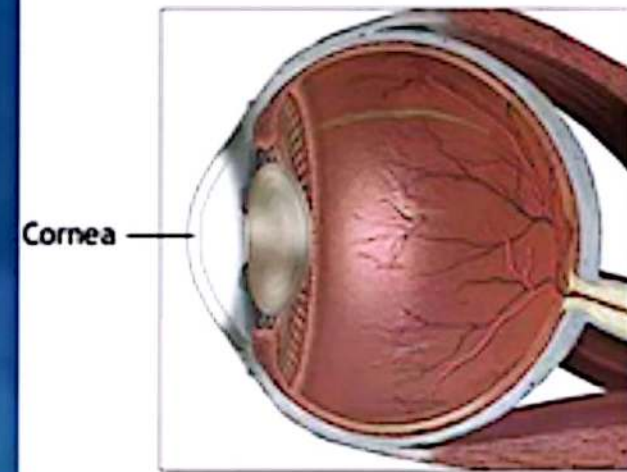
Sclera

- The **lamina cribrosa** is the area of the sclera that is **pierced** by the **nerve fibers** of the **optic nerve**.
- The sclera is also pierced by the **ciliary arteries** and **nerves** and their associated **veins**, the **venae vorticosae**.
- The sclera is **directly continuous in front** with the **cornea** at the **corneoscleral junction**, or **limbus**.



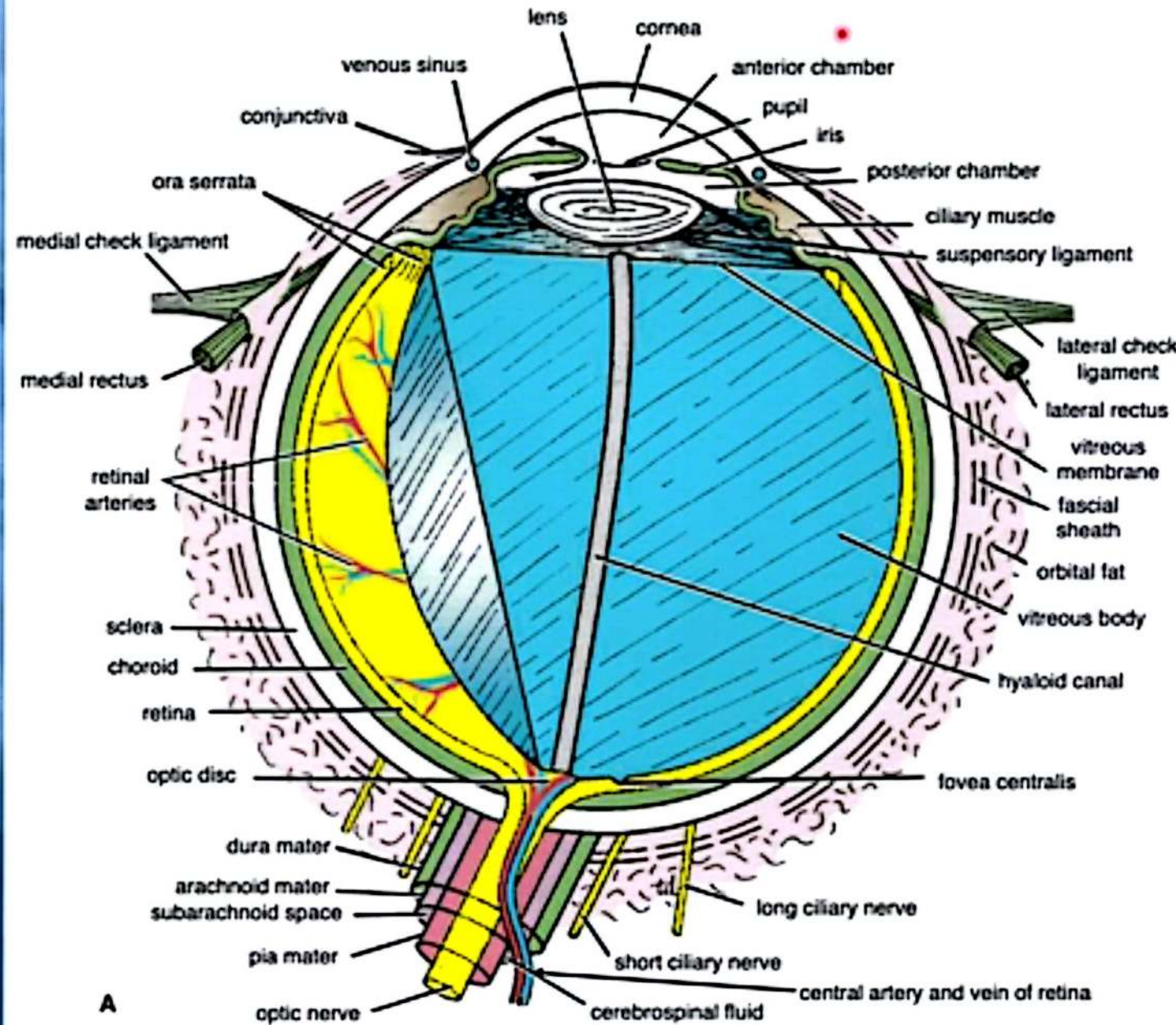
- It is the **anterior, transparent part of the fibrous tunic** that **bulges outward slightly**.
- It is the “**window**” of the eye, as **light must pass through it** before entering the internal structures and cavities.
- It functions as important **refractive medium**.
- It is **transparent** due to **lack of blood vessels** and **regular arrangement of protein fibers**.

Cornea



Cornea

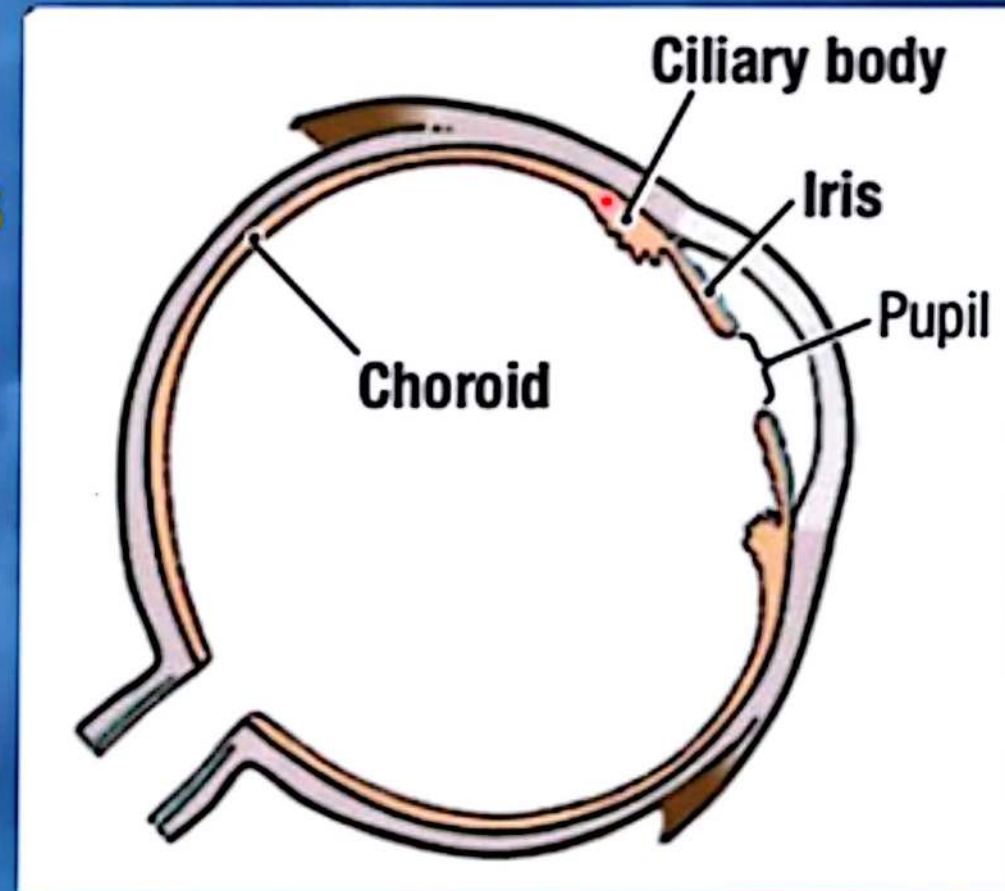
- It is responsible for the **refraction of the light** entering the eye.
- It is in **contact posteriorly** with the **aqueous humor**.
- **Blood Supply:**
 - ✓ It is **avascular** and **devoid of lymphatic drainage**.
 - ✓ It is **nourished** by diffusion from the **aqueous humor** and from the **capillaries at its edge**.
- **Nerve Supply:**
 - ✓ It is by **long ciliary nerves** from the **ophthalmic division of the trigeminal nerve**.



Vascular Tunic

➤ The vascular pigmented coat consists from behind forward of:

- The choroid.
- The ciliary body.
- The iris.



➤ A **thin, dark-brown membrane** that lines most of the internal surface of the sclera.

➤ The **brown pigment absorbs light.**

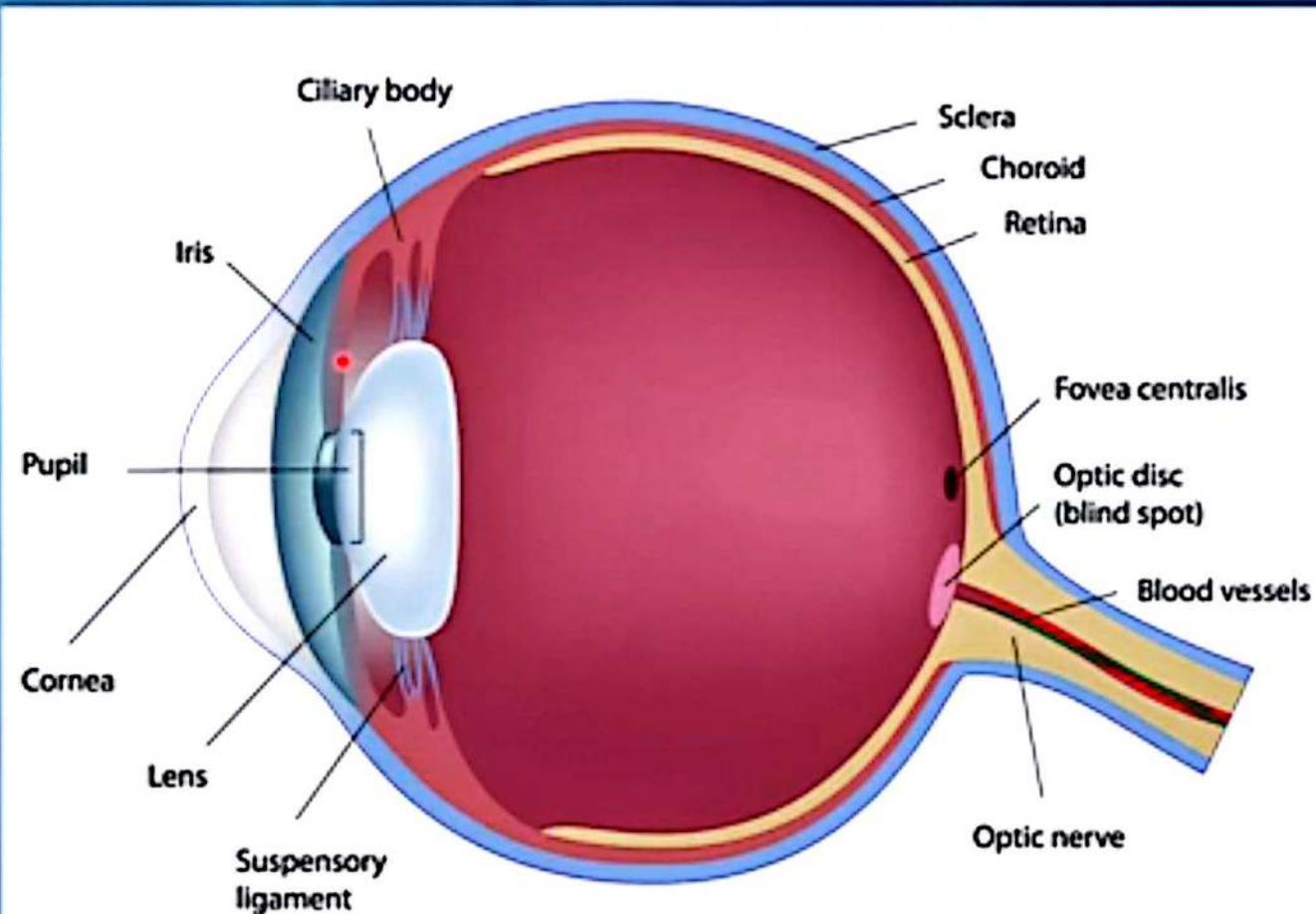
• It is composed of an **outer pigmented layer** and an **inner, highly vascular layer.**

➤ Anterior to the choroid, the vascular tunic is modified to form two structures composed of **smooth muscle:**

➤ **The ciliary body.**

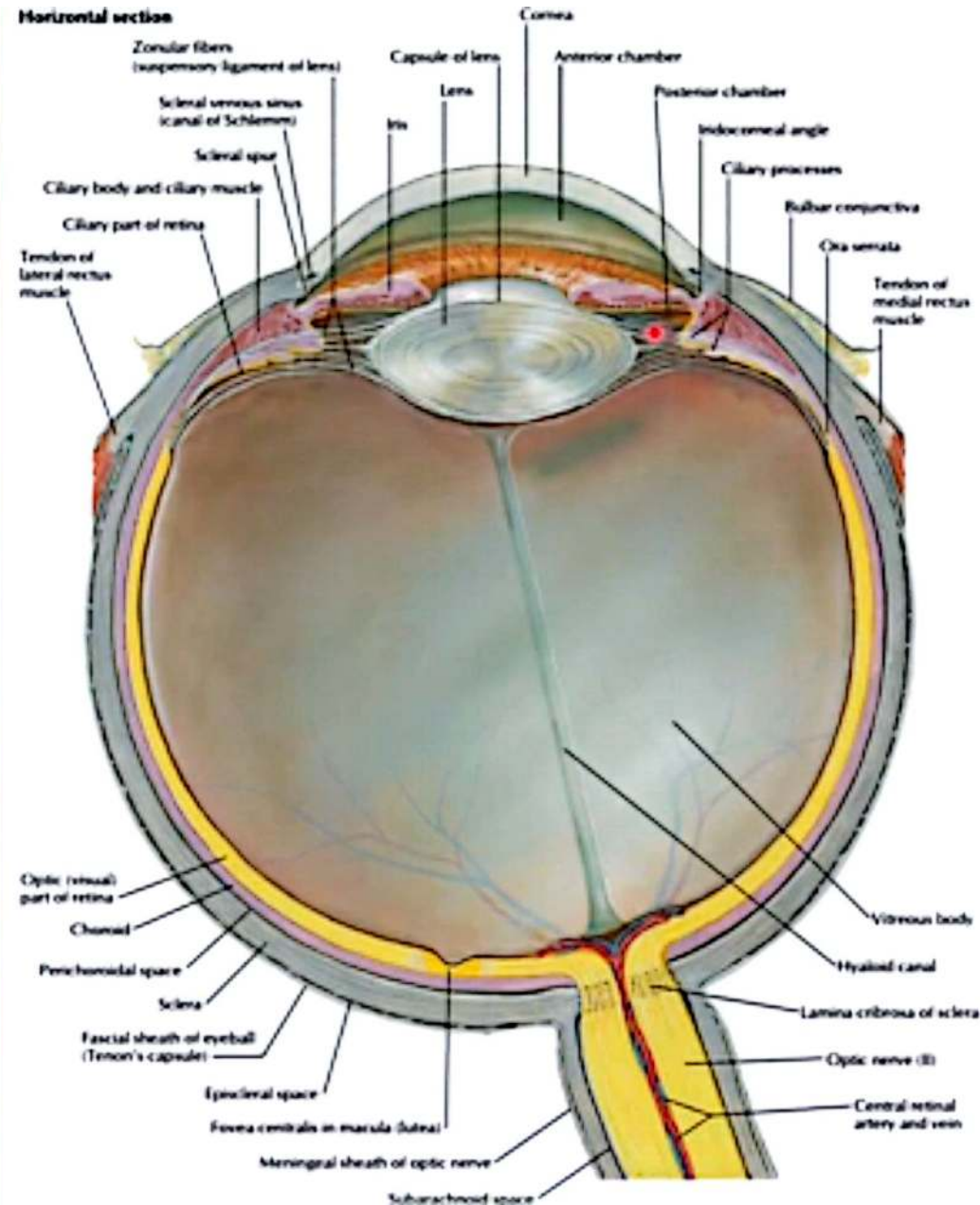
➤ **The iris.**

Vascular Tunic Choroid



The Ciliary Body

- It is continuous **posteriorly** with the **choroid**.
- It is **anteriorly** lies behind the peripheral margin of the **iris**.
- It is composed of:
 - ✓ The ciliary ring.
 - ✓ The ciliary processes.
 - ✓ The ciliary muscle.

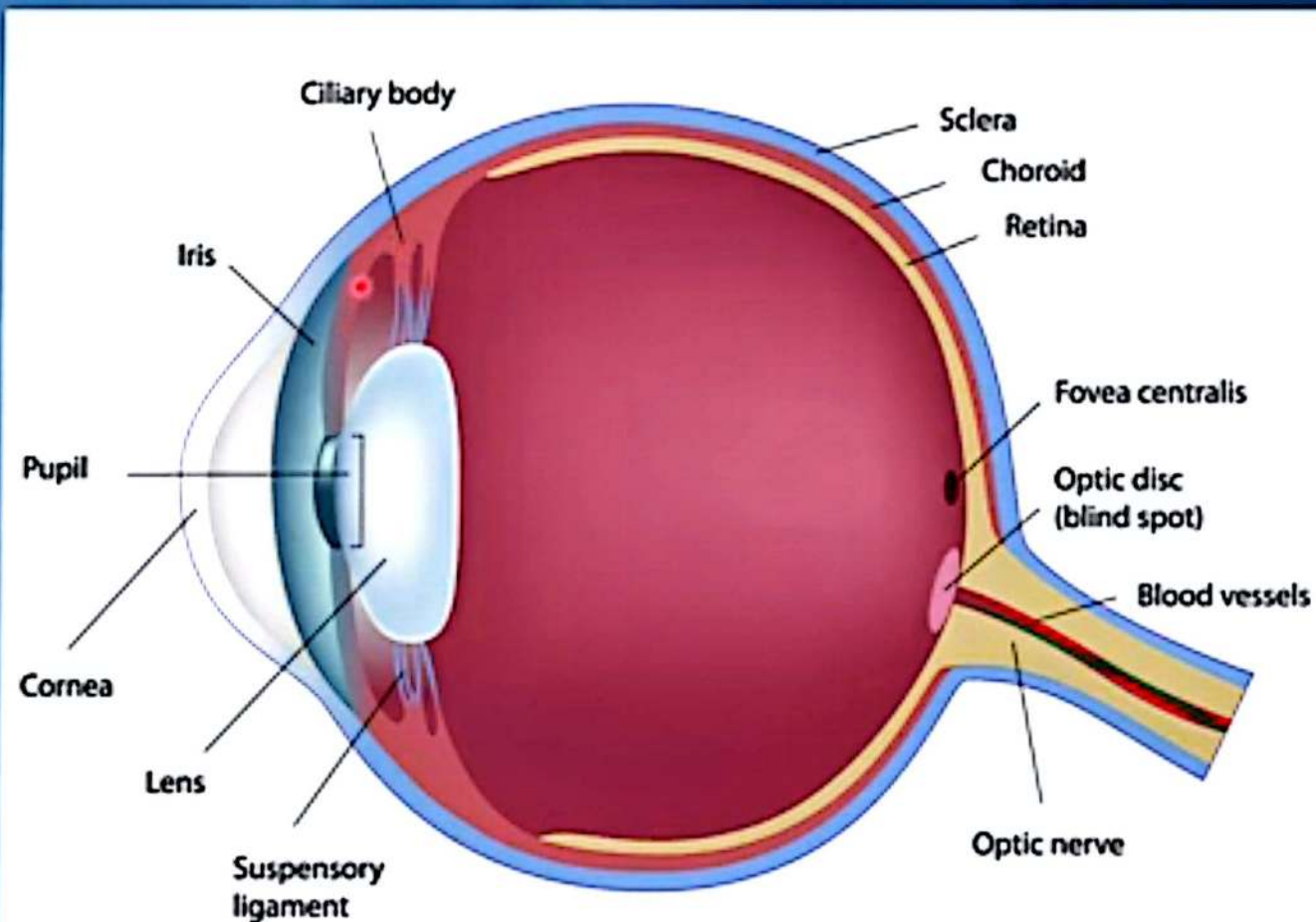


➤ The **ciliary body** continues from the **choroid** to become the **thickest part** of the **vascular tunic**.

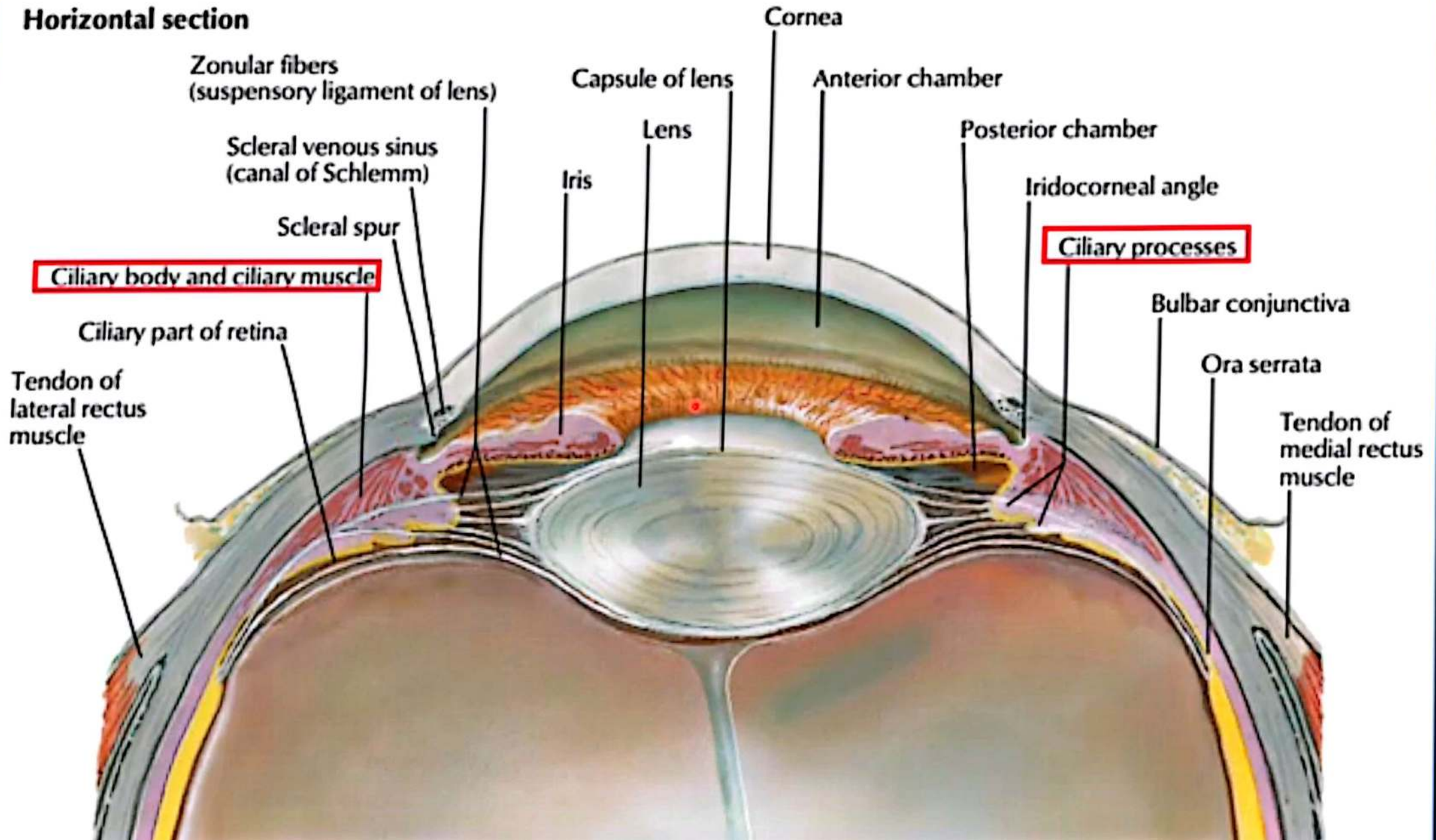
➤ It consists of **smooth muscle fibers**, which connect to the **lens** by way of **suspensory ligaments**.

➤ **Attached** to the inner margin of the **ciliary body** is the **iris**.

Ciliary Body



Horizontal section



➤The ciliary ring:

- It is the **posterior part of the body**.
- Its surface has **shallow grooves**, the **ciliary striae**.

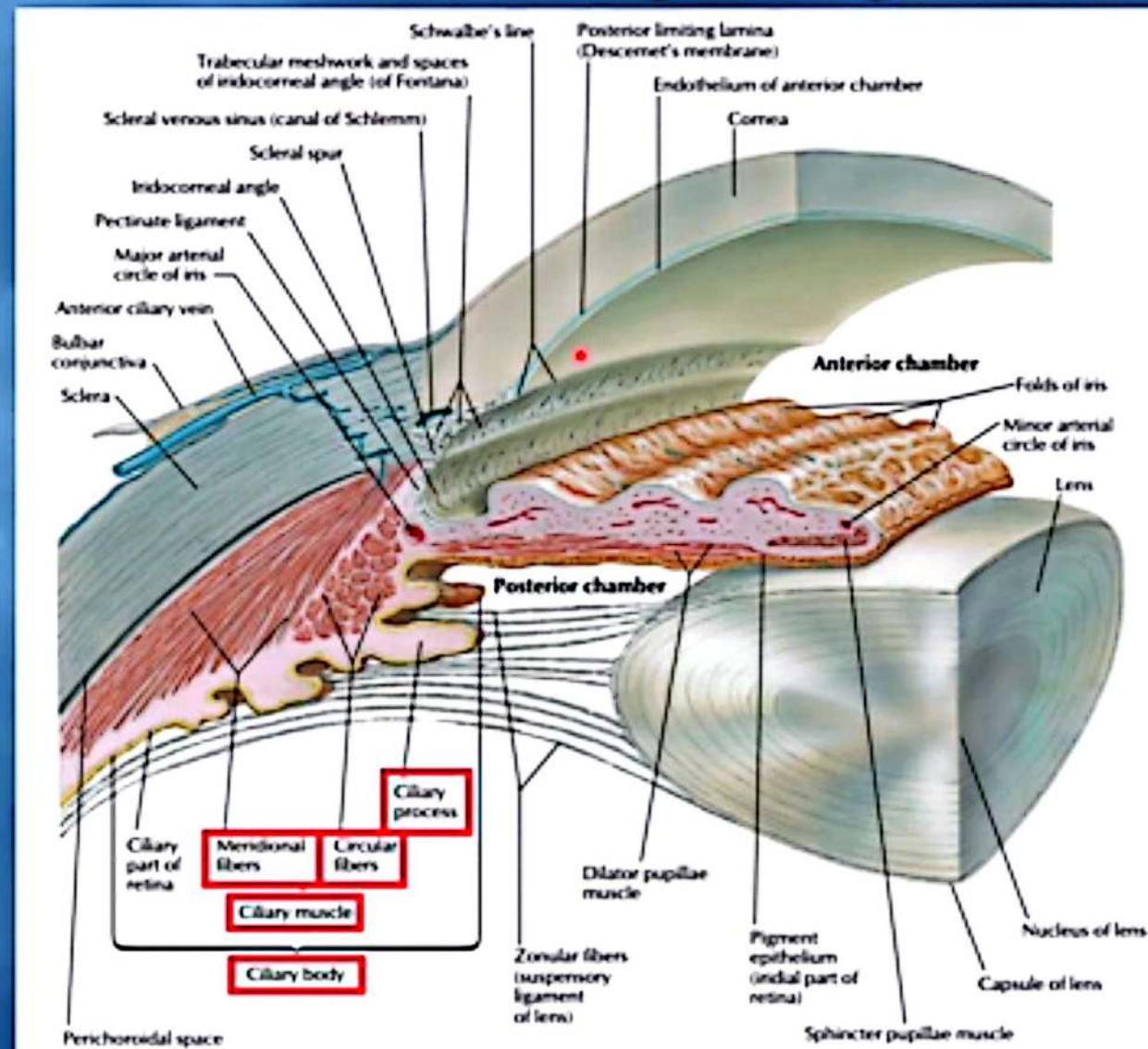
➤The ciliary processes:

- They are radially **arranged folds**, or ridges, to the posterior surfaces of which are connected the **suspensory ligaments of the lens**.

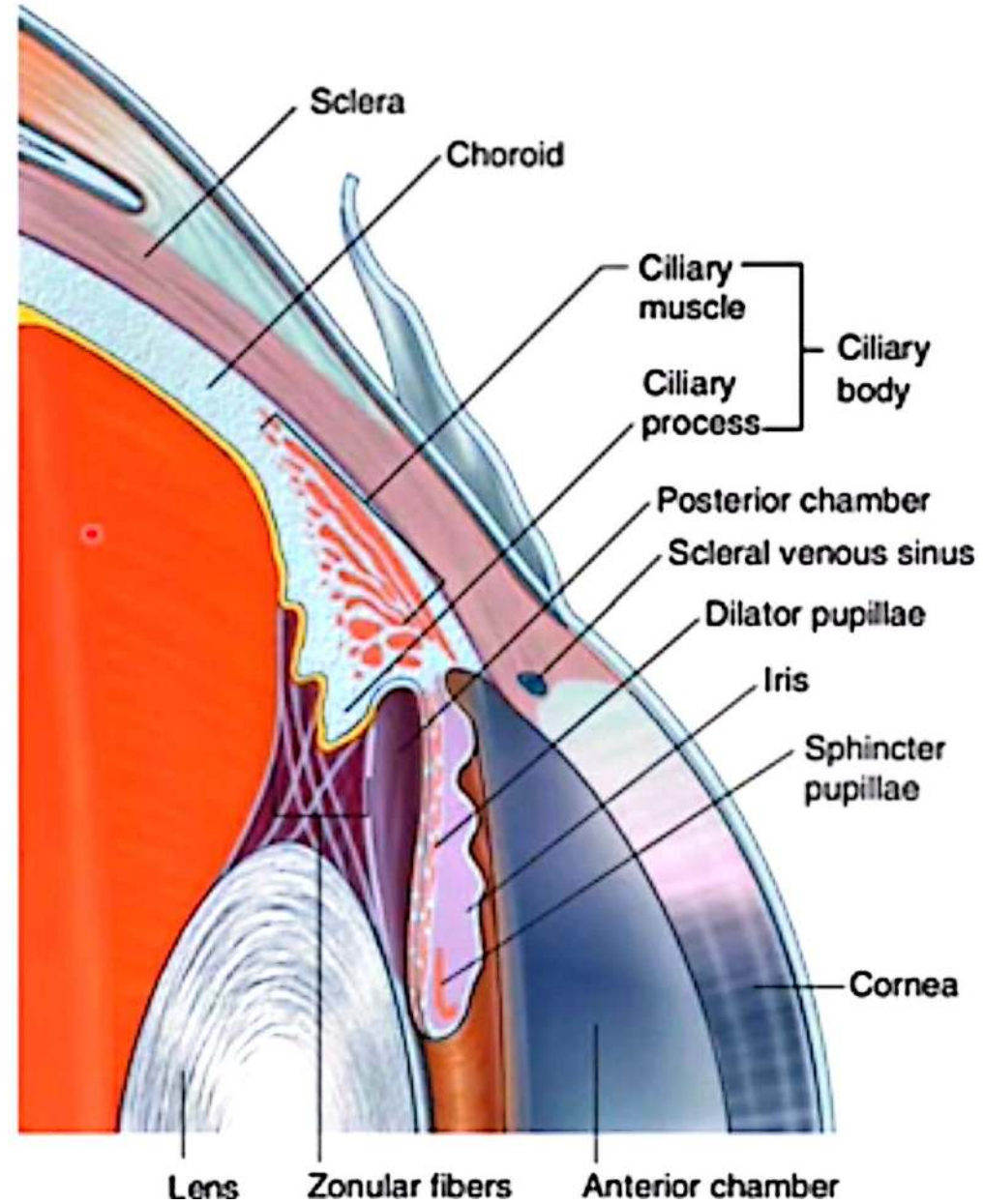
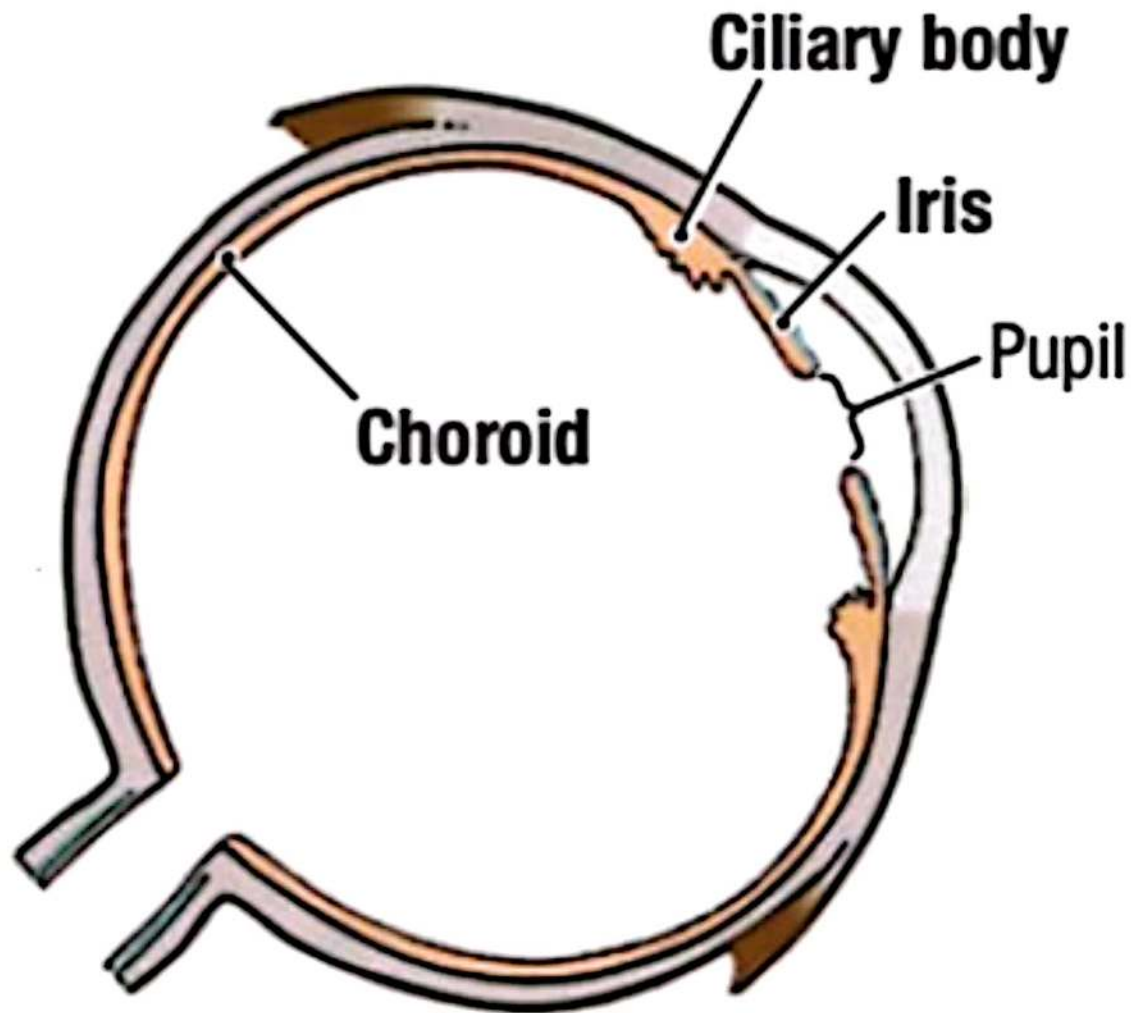
➤The ciliary muscle:

- It is composed of **meridional** and **circular fibers** of **smooth muscle**.
- The meridional fibers run backward from the region of the **corneoscleral junction** to the **ciliary processes**.
- The **circular fibers** are **fewer** in number and lie **internal** to the **meridional fibers**.

The Ciliary Body



The Ciliary Body



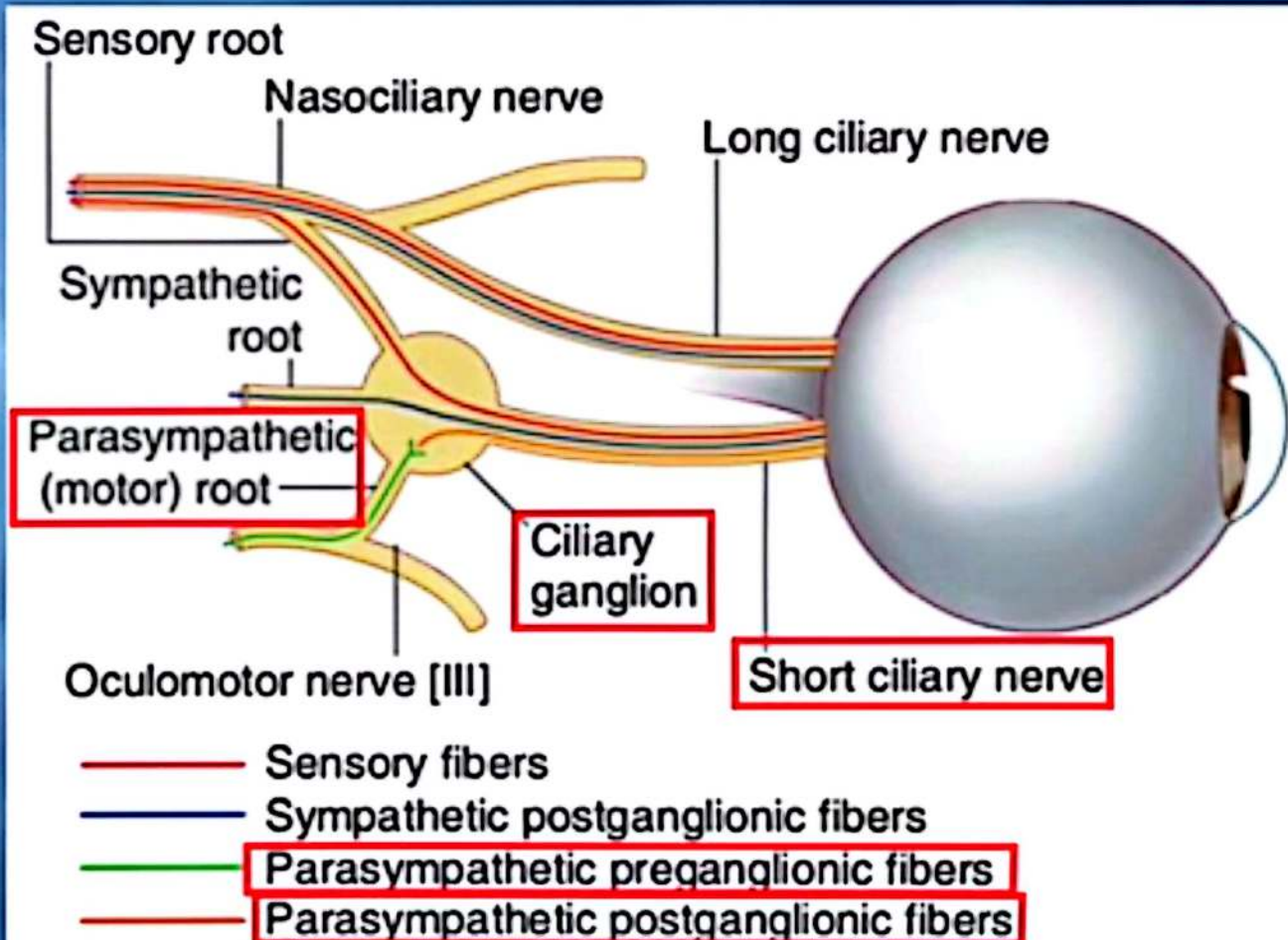
➤ Nerve supply:

- ✓ The ciliary muscle is supplied by the **parasympathetic fibers** from the **oculomotor nerve**.
- ✓ After synapsing in the **ciliary ganglion**, the **postganglionic fibers** pass forward to the eyeball in the **short ciliary nerves**.

➤ Action:

- ✓ **Contraction** of the **ciliary muscle**, especially the **meridional fibers**, pulls the **ciliary body forward**.
- ✓ This **relieves the tension** in the **suspensory ligament**, and the **elastic lens** becomes **more convex**.
- ✓ This **increases the refractive power** of the lens.

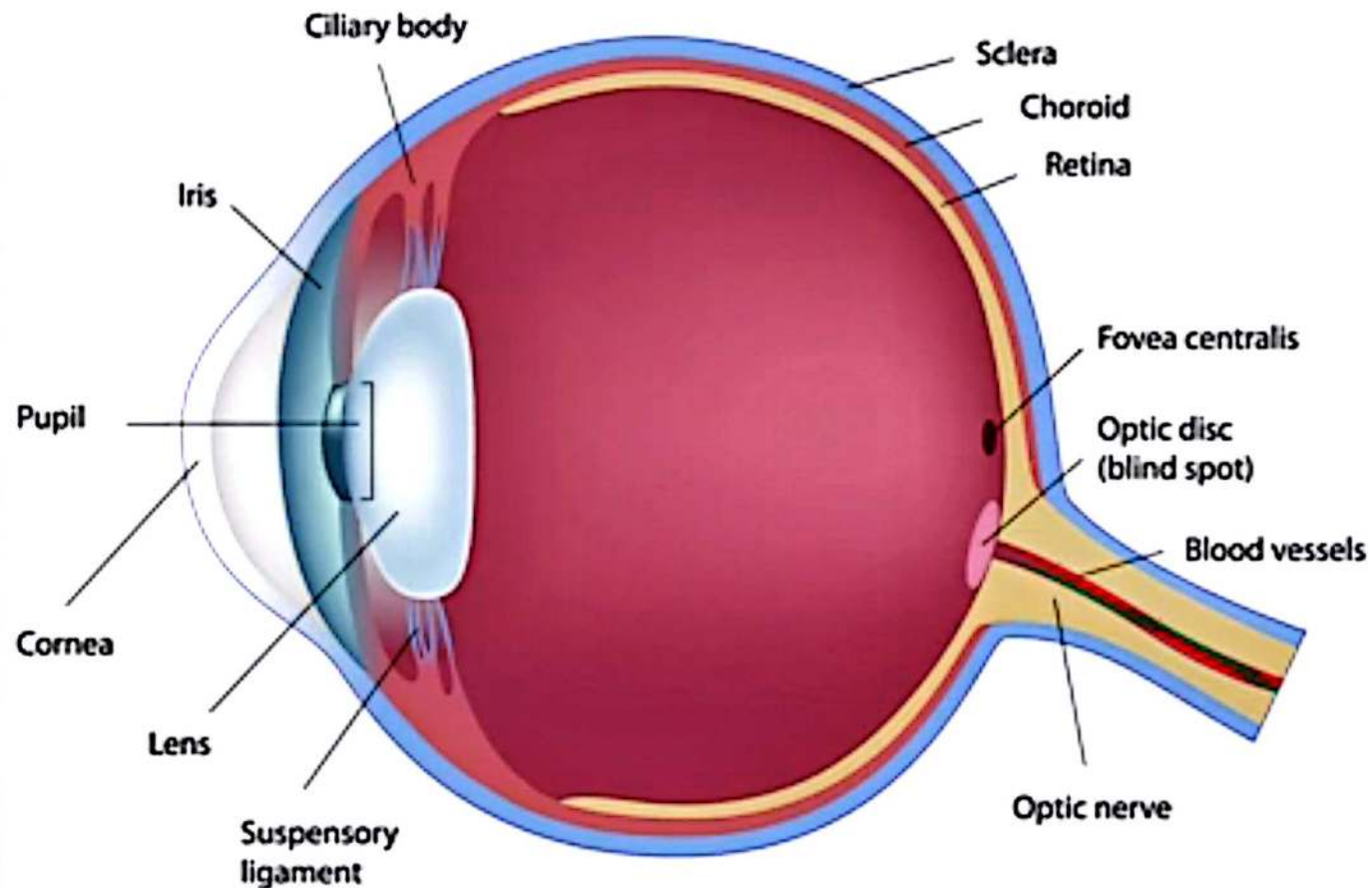
The Ciliary Body



Iris

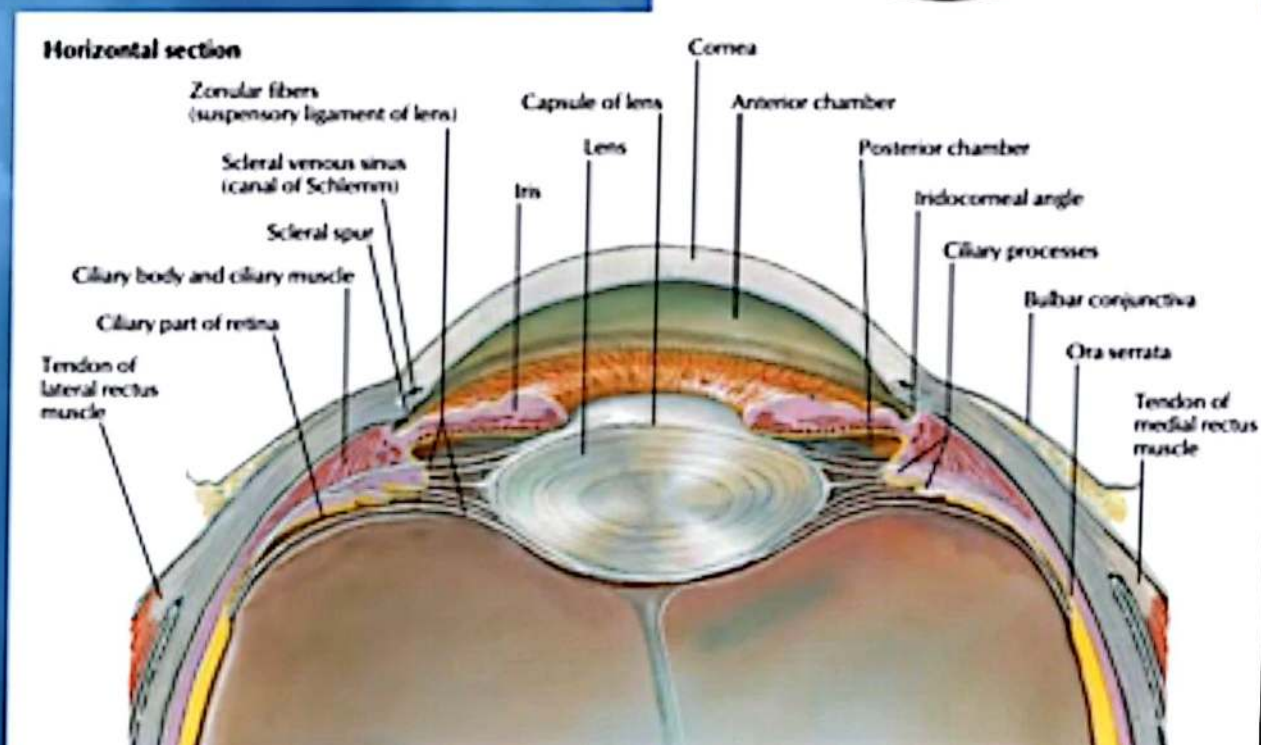
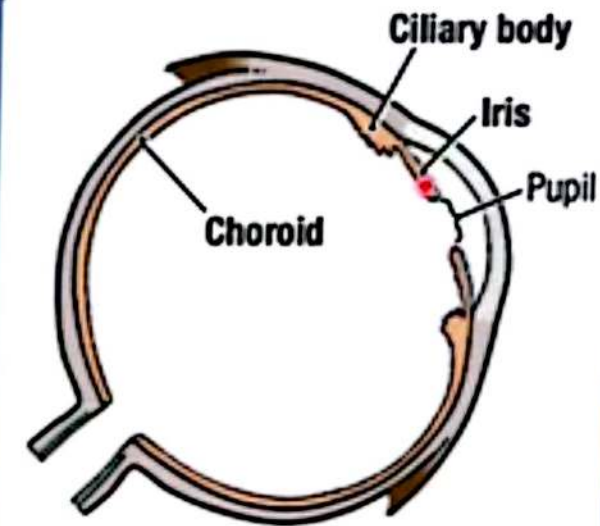
Vascular Tunic

- It is **beautifully colored part** of the eye that **can be seen** from the exterior.
- Suspended between the **cornea** and the **lens**.
- The **pupil** appears as a **black spot** in the **center** of iris.
- **Contraction of muscle fibers changes the diameter of the pupil** to regulate the amount of light entering the inner eye cavity.



The Iris and Pupil

- The iris is a **thin, contractile, pigmented diaphragm** with a **central aperture** (the **pupil**).
- It is suspended in the **aqueous humor** between the **cornea** and the **lens**.
- The **periphery of the iris** is attached to the **anterior surface** of the **ciliary body**.
- It divides the space between the **lens** and the **cornea** into an **anterior** and a **posterior chamber**.



The Iris and Pupil

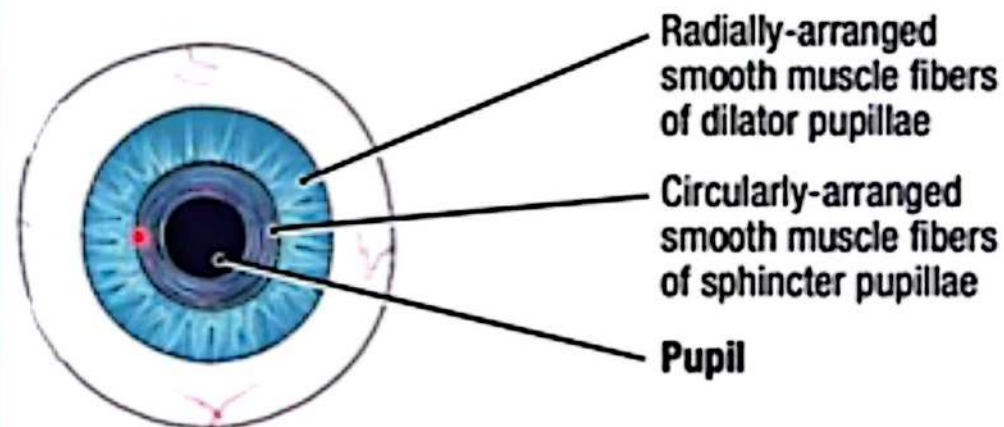
➤ **The muscle fibers** of the iris are **involuntary** and consist of **circular** and **radiating fibers**.

1) The circular fibers:

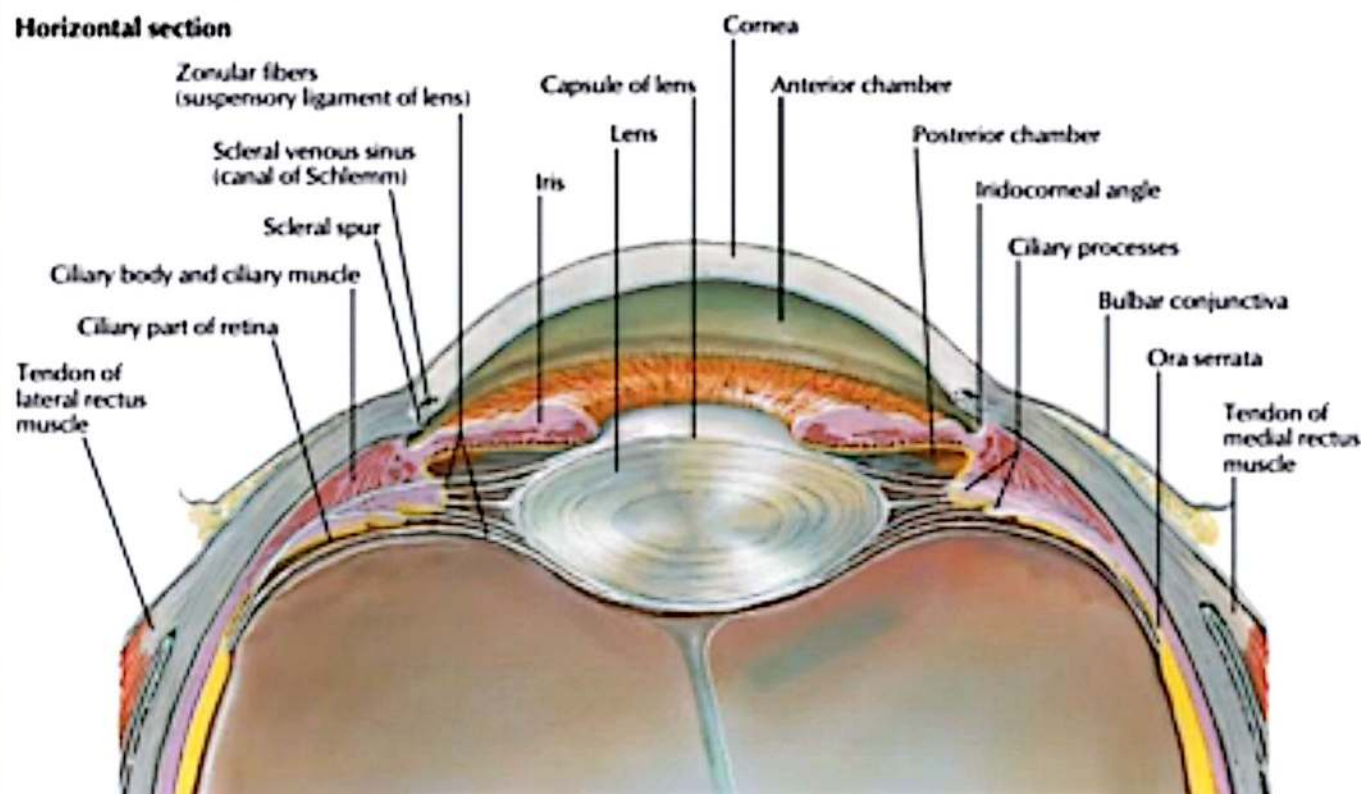
- ✓ They form the **sphincter pupillae**.
- ✓ They are arranged **around the margin of the pupil**.

2) The radial fibers:

- ✓ It form the **dilator pupillae**.
- ✓ It consist of a **thin sheet of radial fibers**.
- ✓ They are lie close to the **posterior surface**.



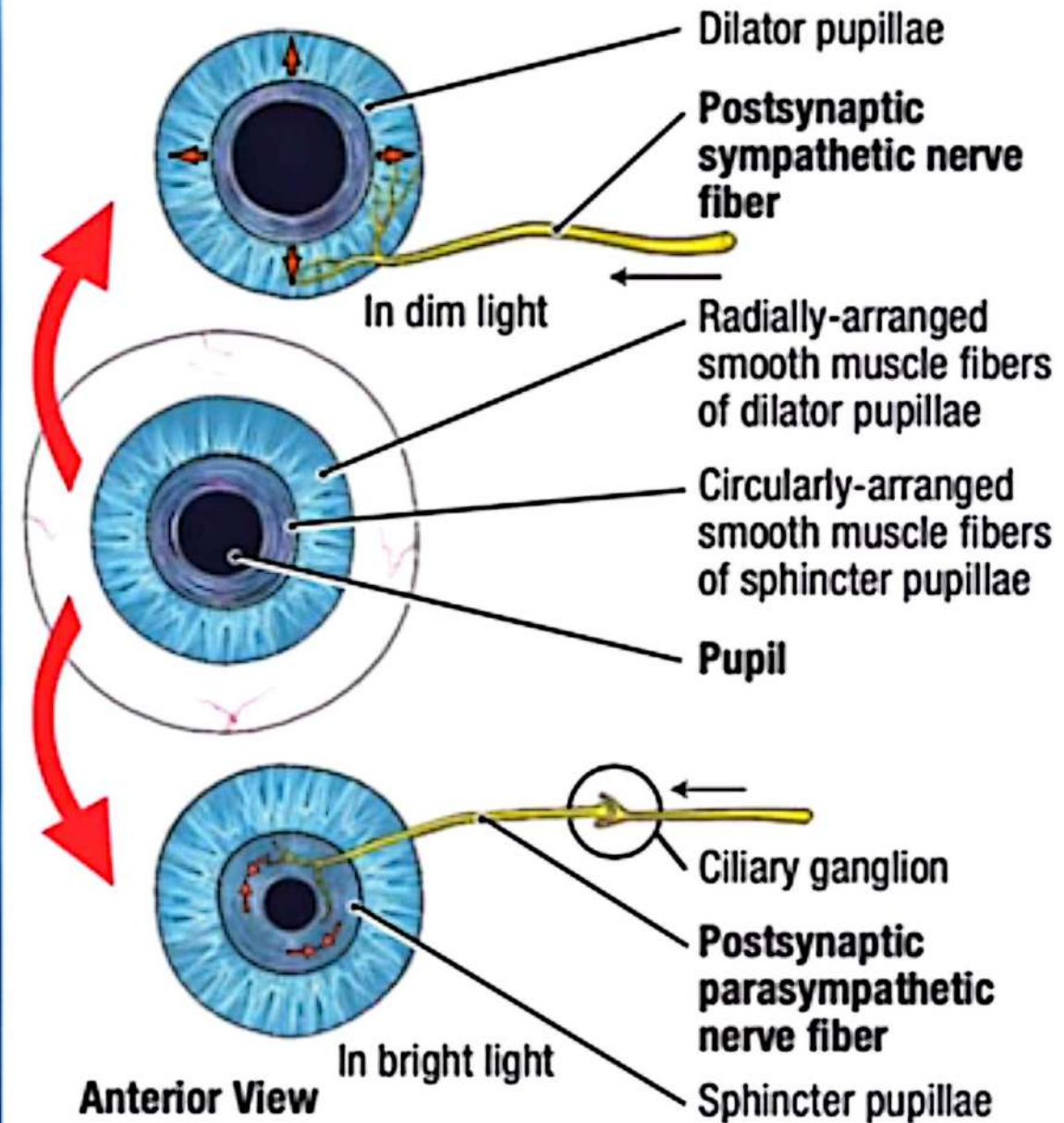
Horizontal section



The Iris and Pupil

➤ Nerve supply:

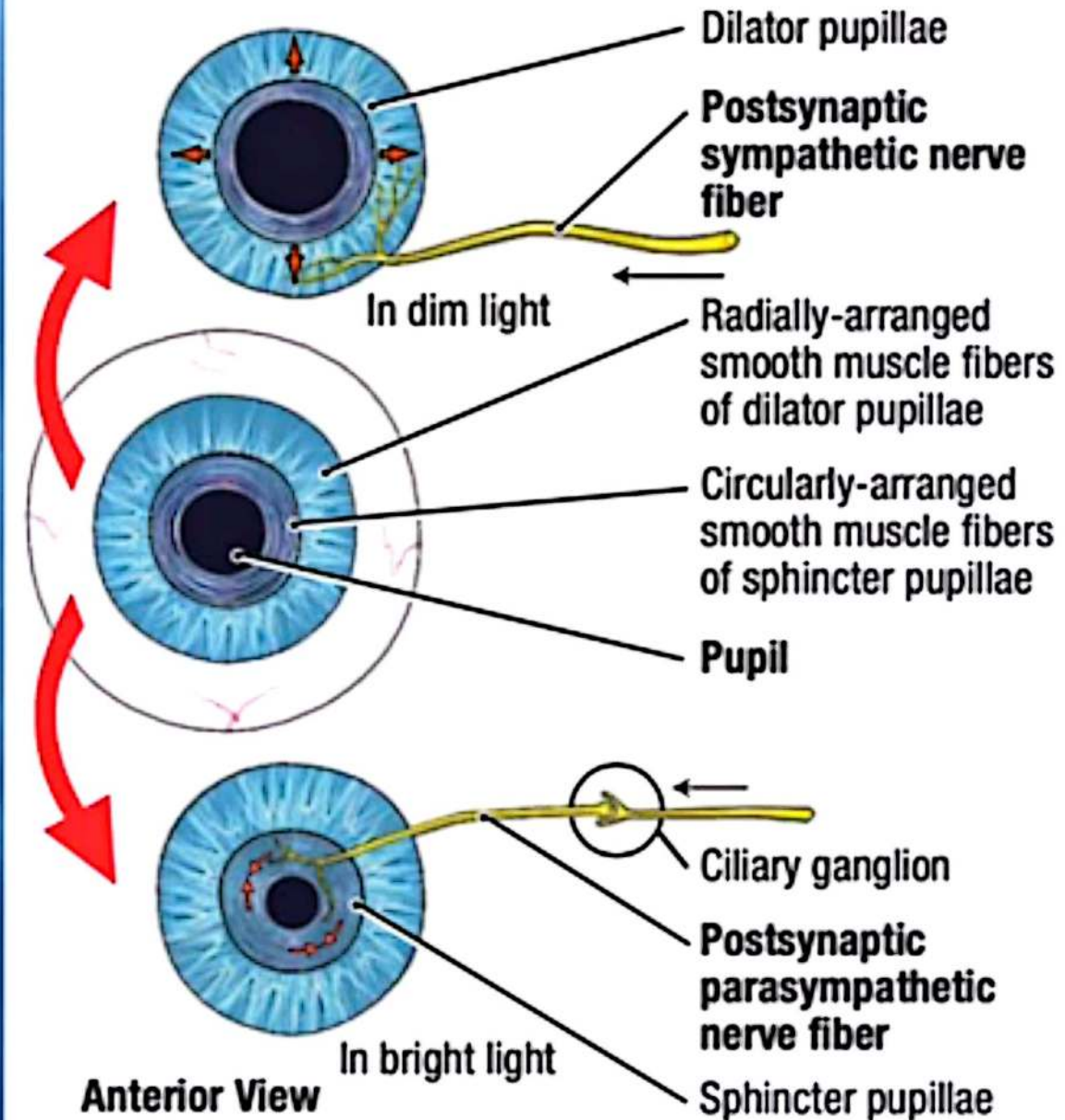
- ✓ The **sphincter pupillae** is supplied by **parasympathetic fibers** from the **oculomotor nerve**.
- After synapsing in the **ciliary ganglion**, the **postganglionic fibers** pass forward to the eyeball in the **short ciliary nerves**.
- The **dilator pupillae** is supplied by **sympathetic fibers**, which pass forward to the eyeball in the **long ciliary nerves**.



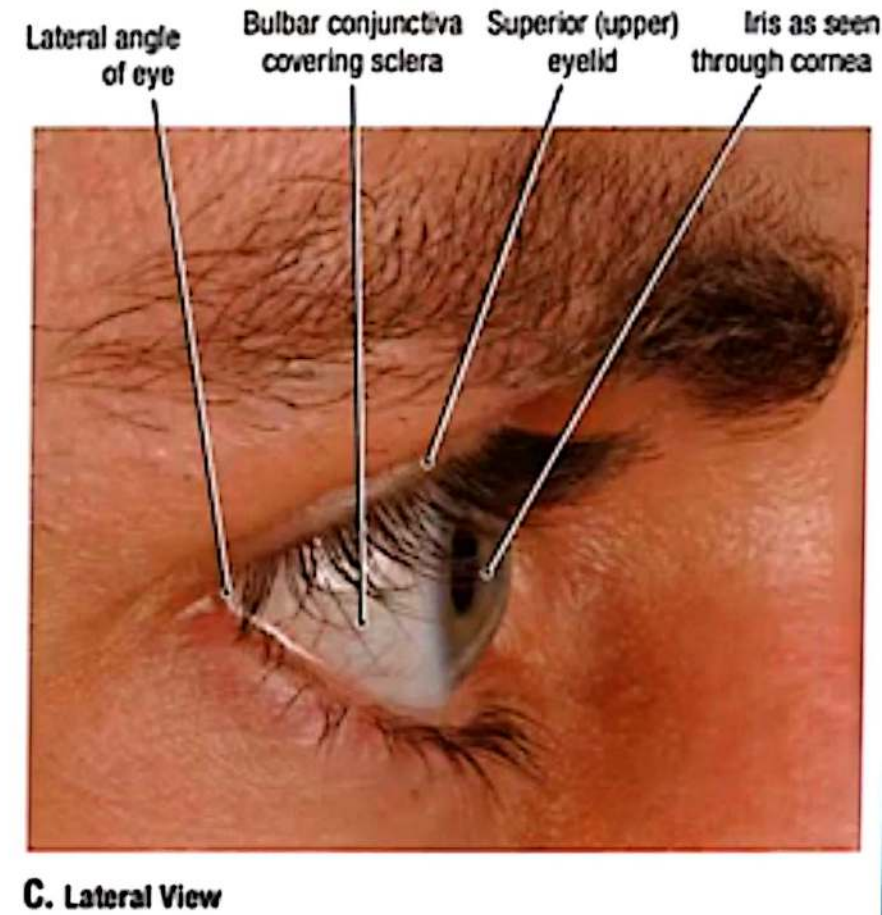
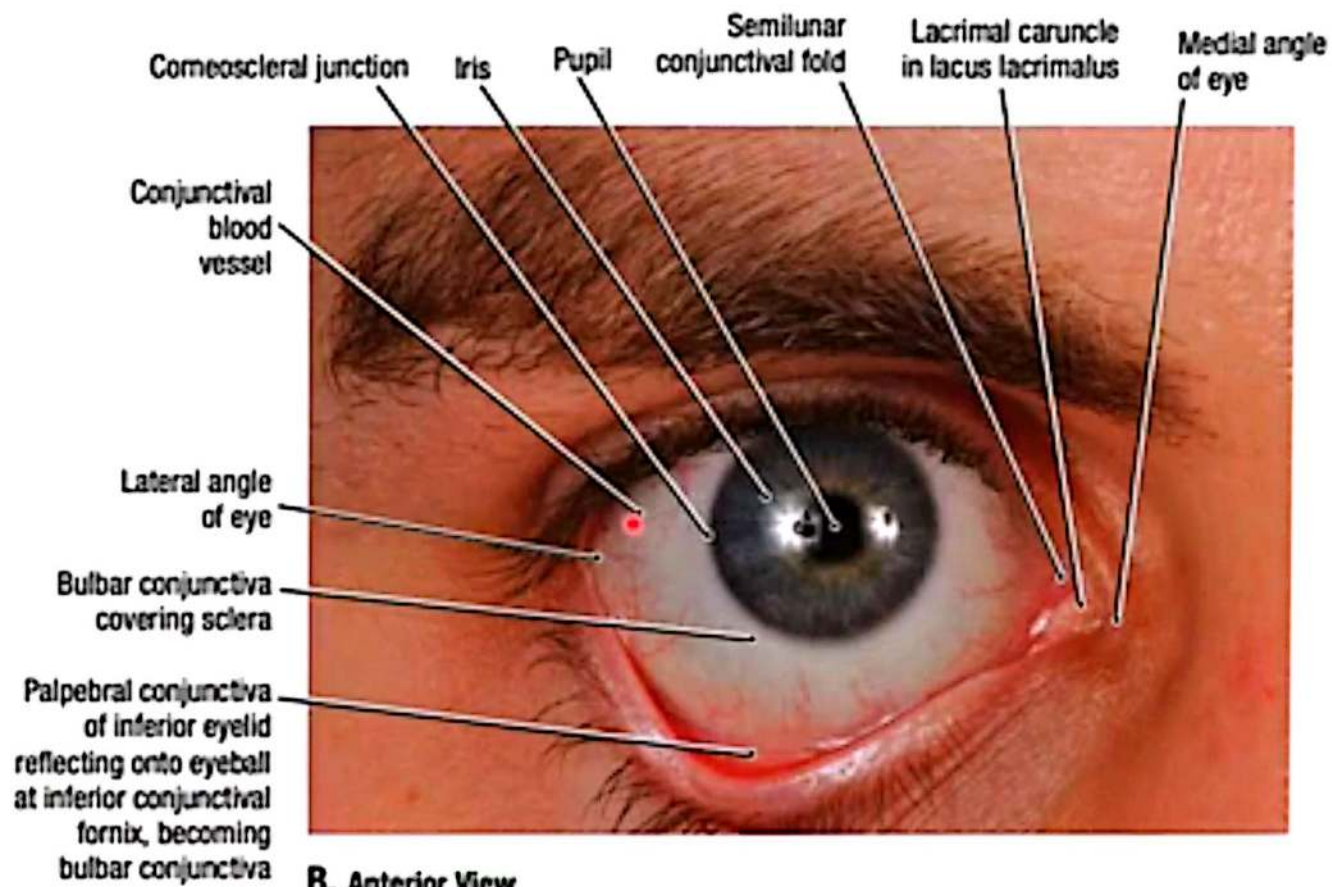
The Iris and Pupil

➤ Action:

- ✓ The **sphincter pupillae constricts the pupil** in the presence of **bright light** and during **accommodation**.
- ✓ The **dilator pupillae dilates the pupil** in the presence of light of low intensity or in the presence of **excessive sympathetic activity** such as occurs in fright.



The Iris and Pupil

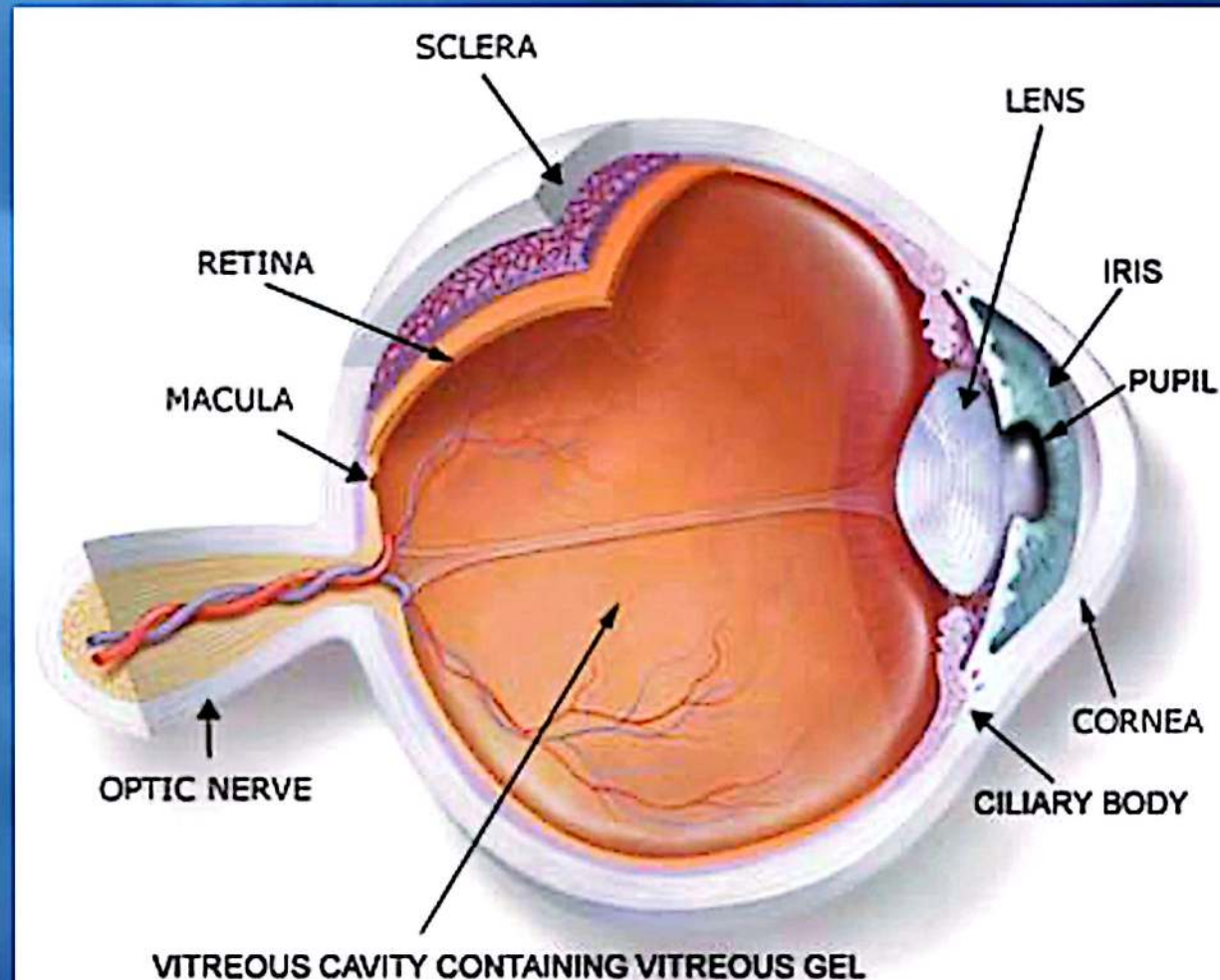


The background of the slide is a photograph of a bright blue sky filled with soft, white, and light blue clouds. The clouds are scattered across the frame, with some appearing more dense and others more wispy. The overall tone is bright and airy.

Will be continue ...

Nervous layers (Nervous Tunic)

- The nervous tunic consists of the **retina**, which is a **thin, fragile** layer of neurons that forms the **inner lining of the eyeball's posterior wall**.
- The retina contains its own supply of blood vessels.
- Function of the retina is the **detection of light** and **subsequent transport of nerve impulses to the optic nerve**.



Nervous layers (Nervous Tunic) (The Retina)

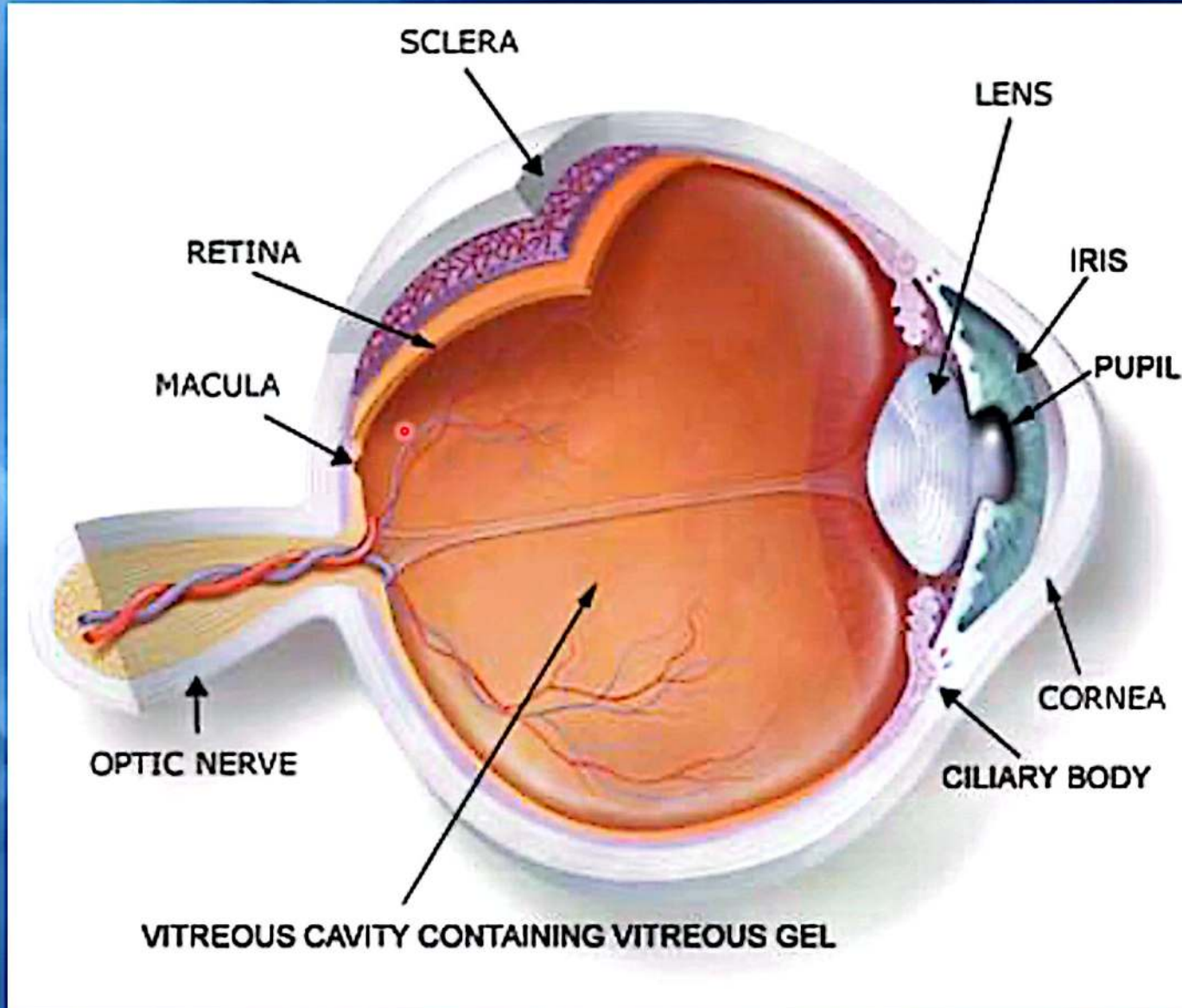
➤ The retina consists of:

1) Outer pigmented layer:

✓ Its outer surface is in contact with the **choroid**.

2) Inner nervous layer:

✓ Its inner surface is in contact with the **vitreous body**.



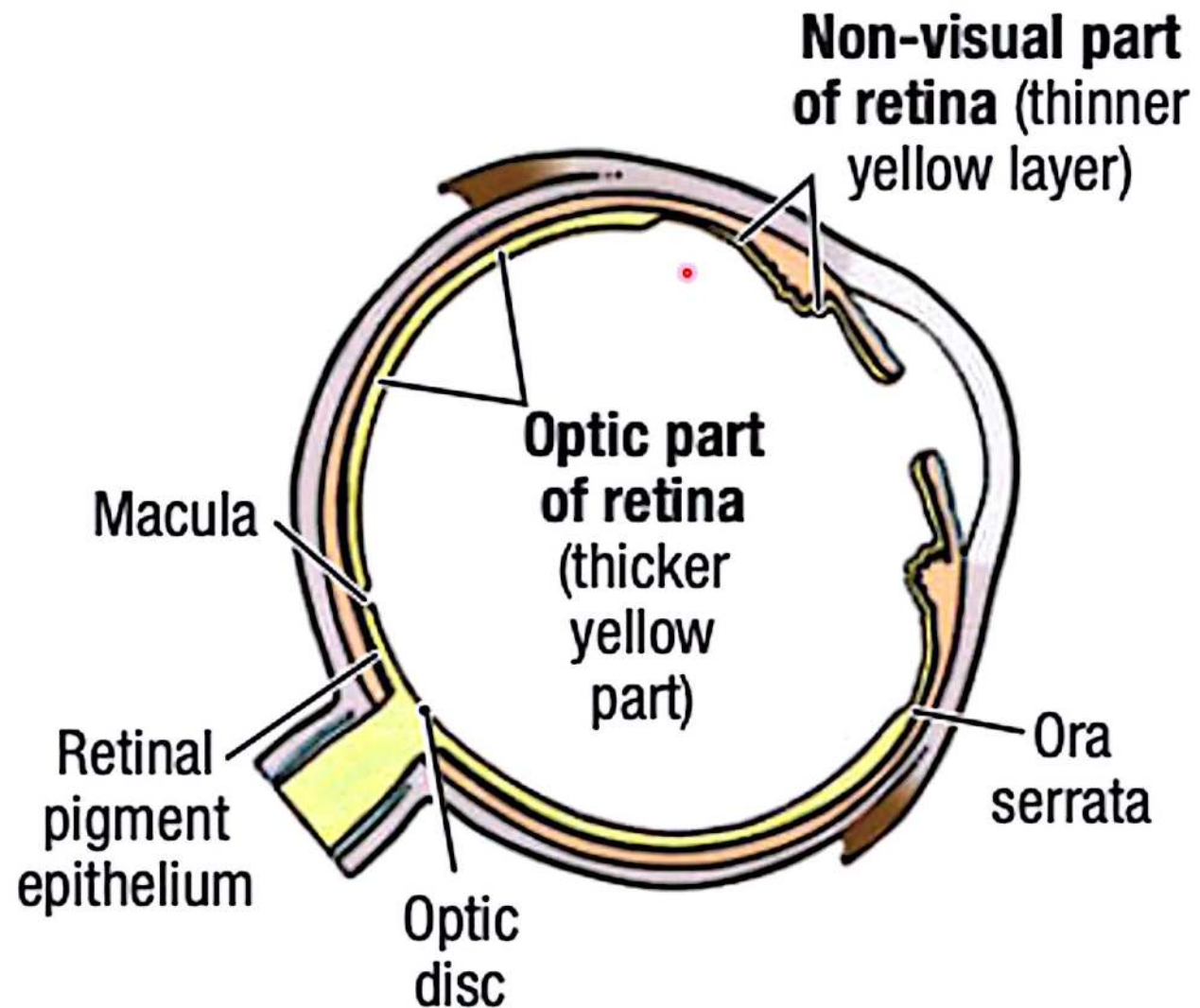
Nervous layers (Nervous Tunic) (The Retina)

- **The posterior three quarters of the retina:**

- It is the **receptor organ**.
- Its anterior edge forms a wavy ring, the **ora serrata**, and the nervous tissues end here.

- **The anterior part of the retina:**

- It is **nonreceptive**.
- It consists merely of **pigment cells**, with a deeper **layer of columnar epithelium**.
- This anterior part of the **retina** covers the **ciliary processes** and the **back of the iris**.

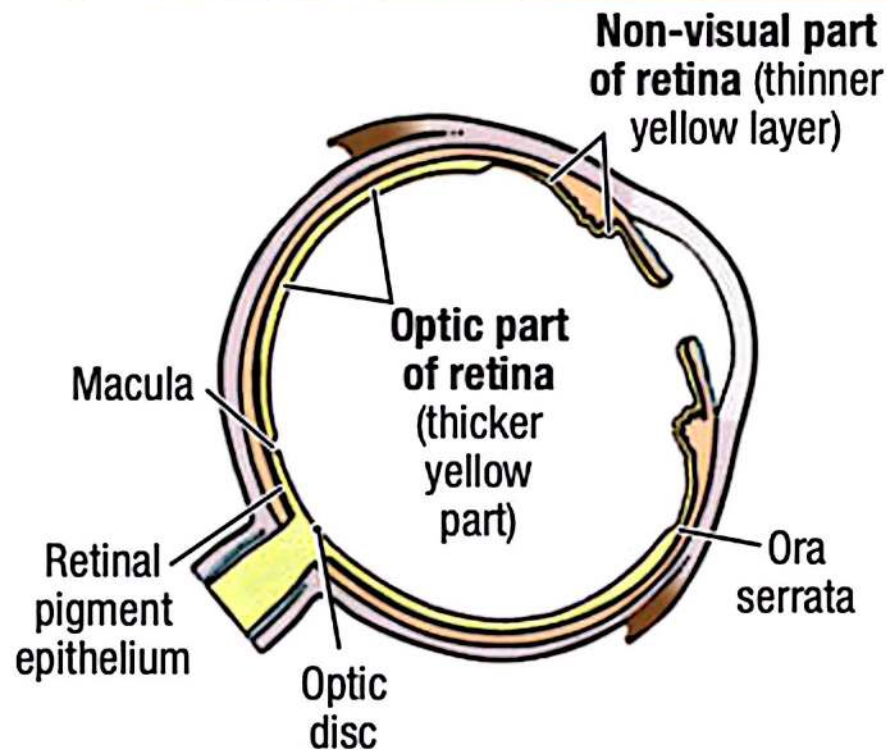


The macula lutea:

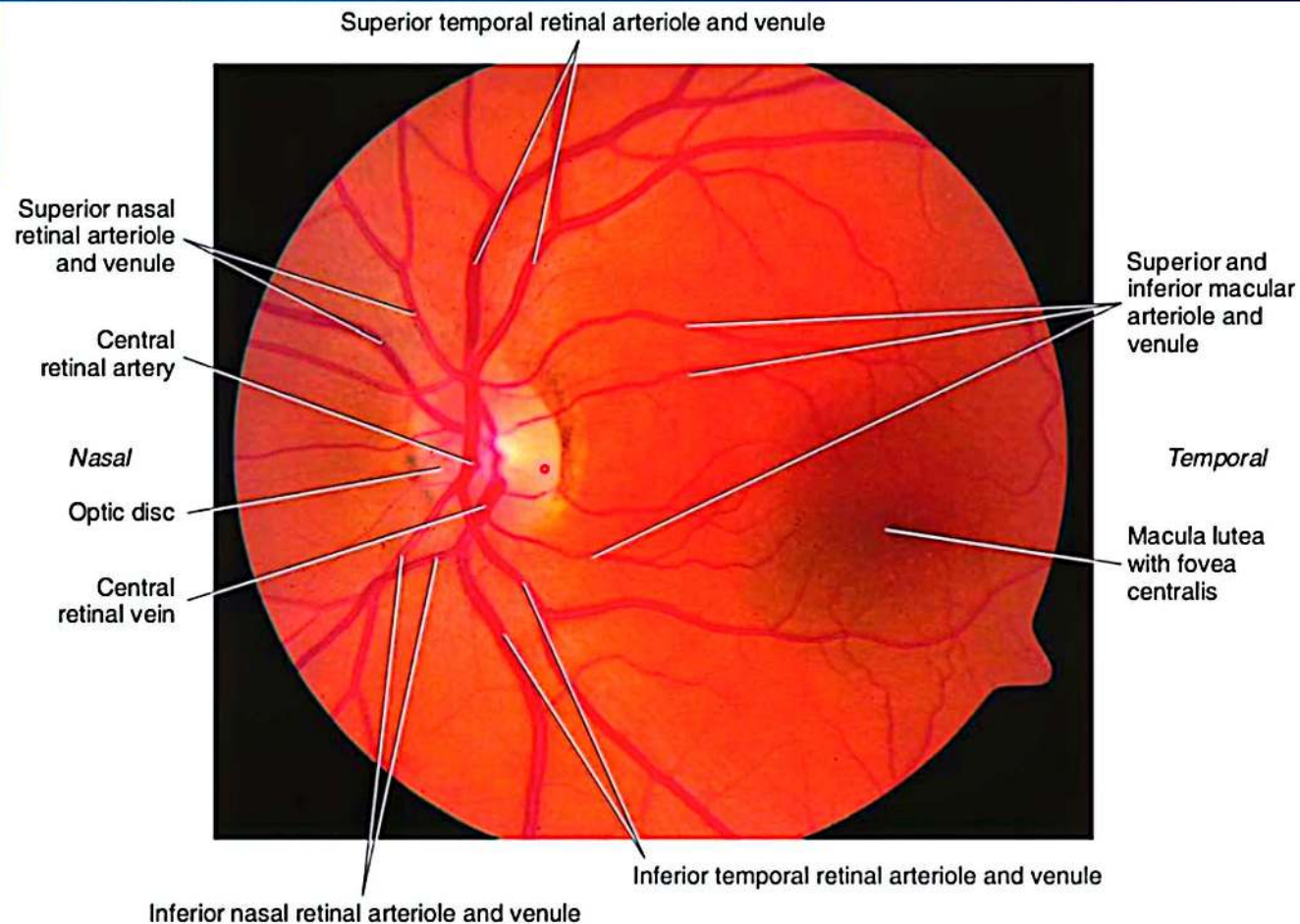
- It is oval, yellowish area at the center of the posterior part of the retina for the most distinct vision.

The fovea centralis:

- It has a central depression.

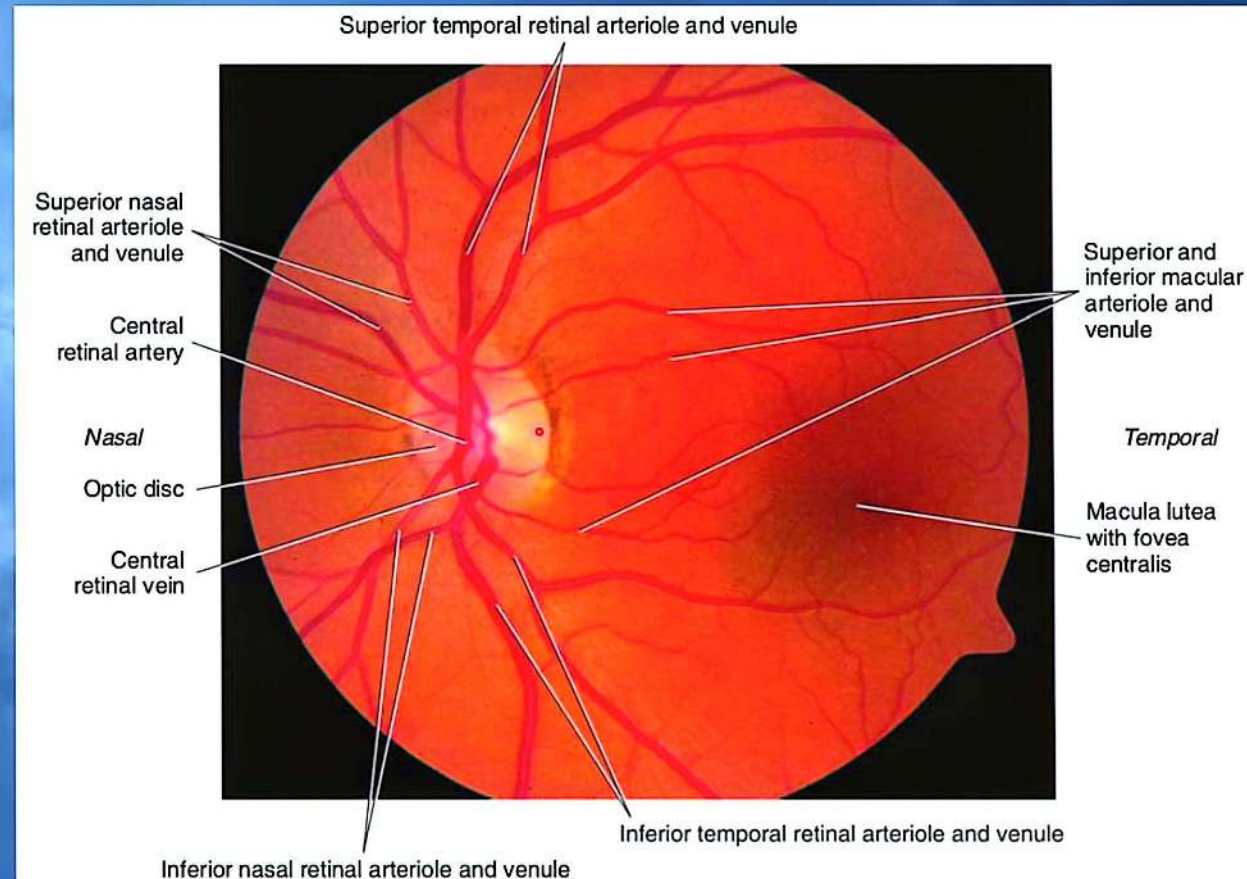


The Retina



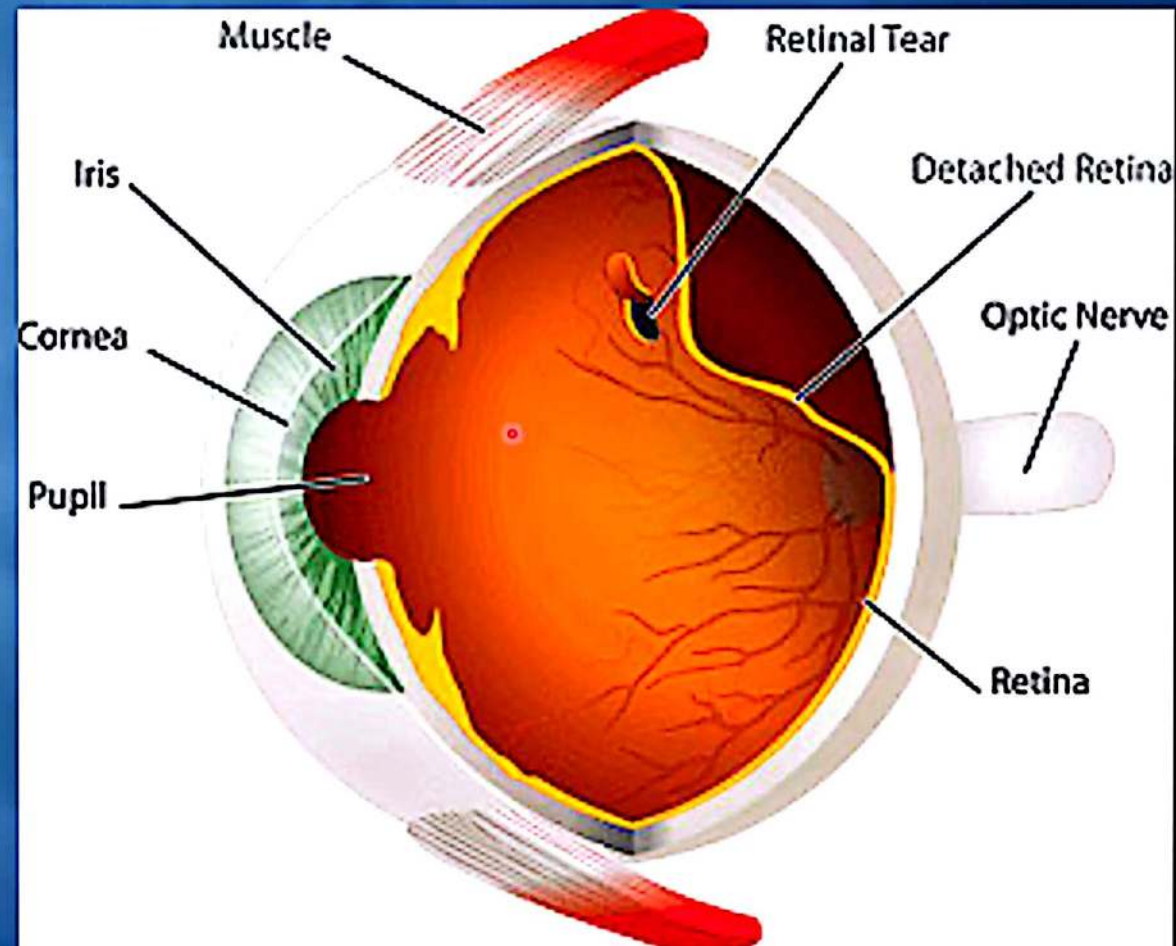
- The **optic nerve** leaves the retina about **3 mm to the medial side** of the **macula lutea** by the **optic disc**.
- The optic disc is slightly **depressed** at its **center**, where it is **pierced by the central artery** of the **retina**.
- At the optic disc is a complete absence of **rods** and **cones** so that it is **insensitive to light** and is referred to as the "**blind spot**." On **ophthalmoscopic examination**, the optic disc is seen to be **pale pink** in color, much paler than the surrounding **retina**.

The optic disc



- The retina consists of many distinct **layers of neurons**.
- The layer nearest the choroid contains the **neurons that are specialized to respond to light, the photoreceptor cells**.

Retina



Photoreceptor Cells

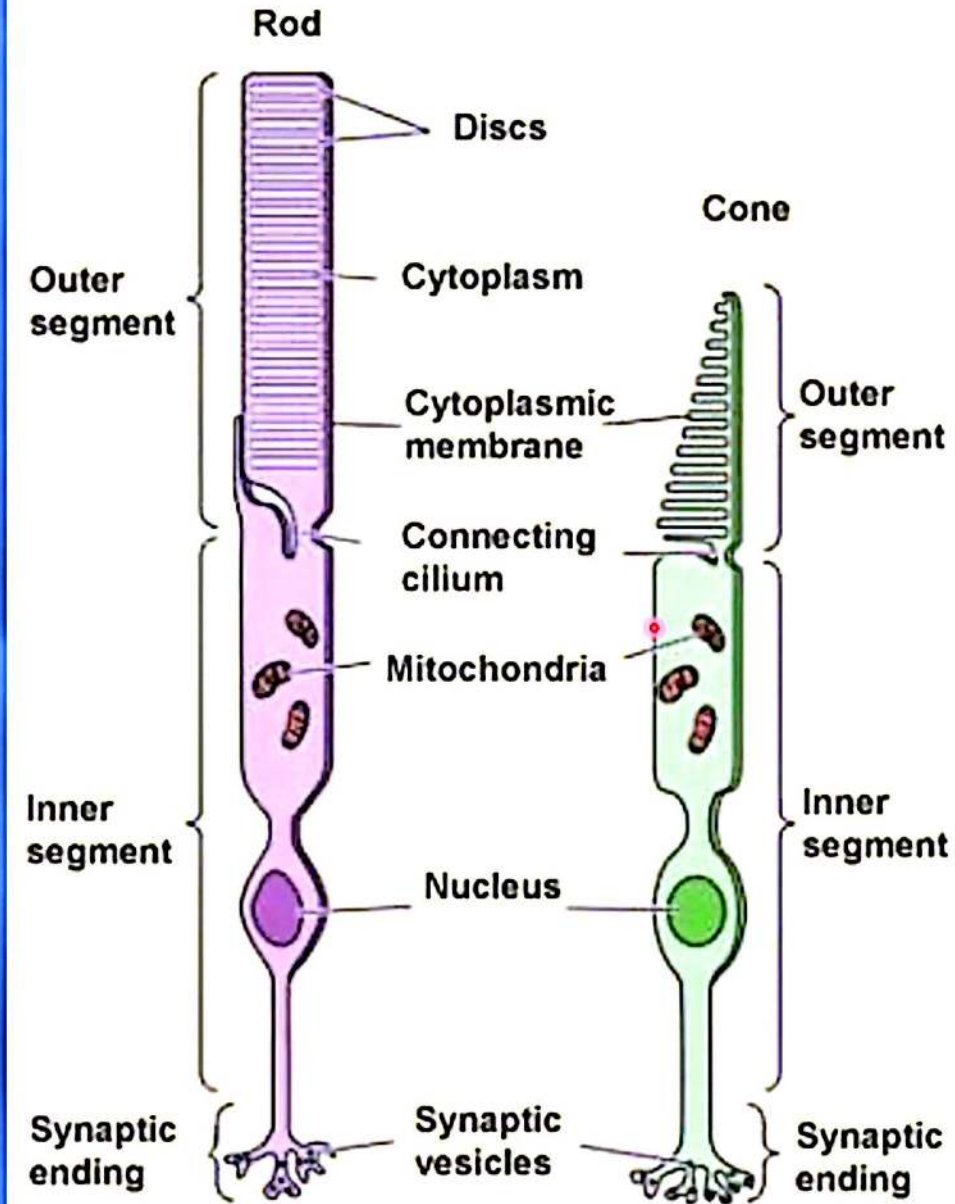
➤ Two types:

❖ Rod cells:

- ✓ They are containing an **elongated**, cylindrical dendrite.
- ✓ They are **not color-sensitive**, but are sensitive to very small levels of light (**night vision**).

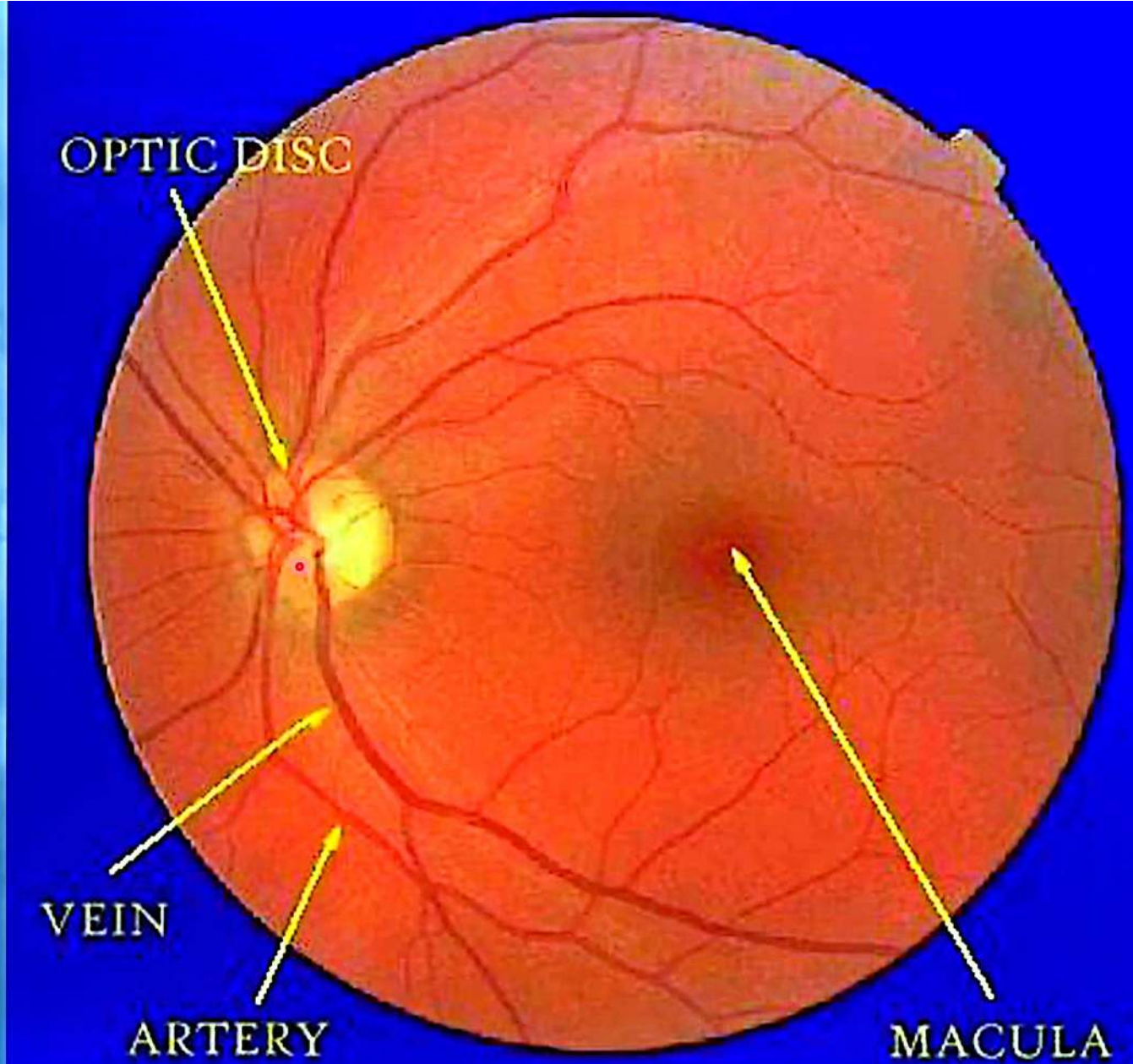
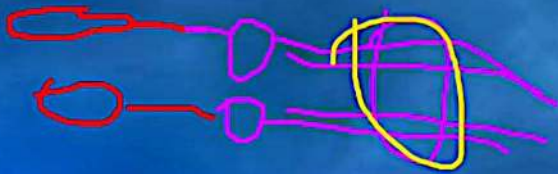
❖ Cone cells:

- ✓ whose dendrites are tapered to a point like cones.
- They are **color-sensitive** and require **lighter** but provide a sharper image.
- **Both cells** respond to light by **initiating a signal**, which is transmitted to the **cells** forming the **middle layer of the retina**, the **bipolar neurons**.



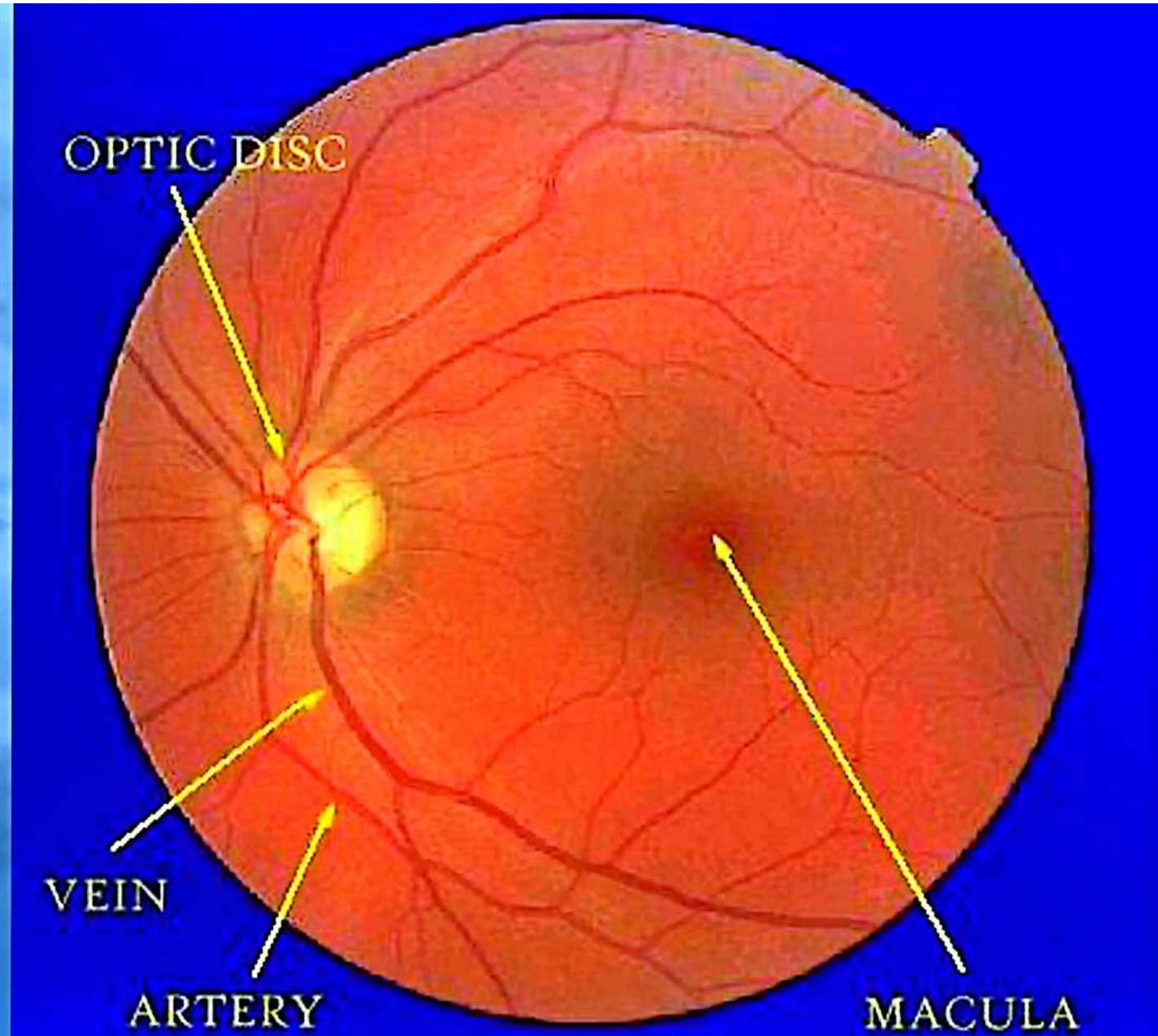
Optic Disc

- Point near the center of the retina, where **axons** from the **ganglion cells** converge to form the **optic nerve**.



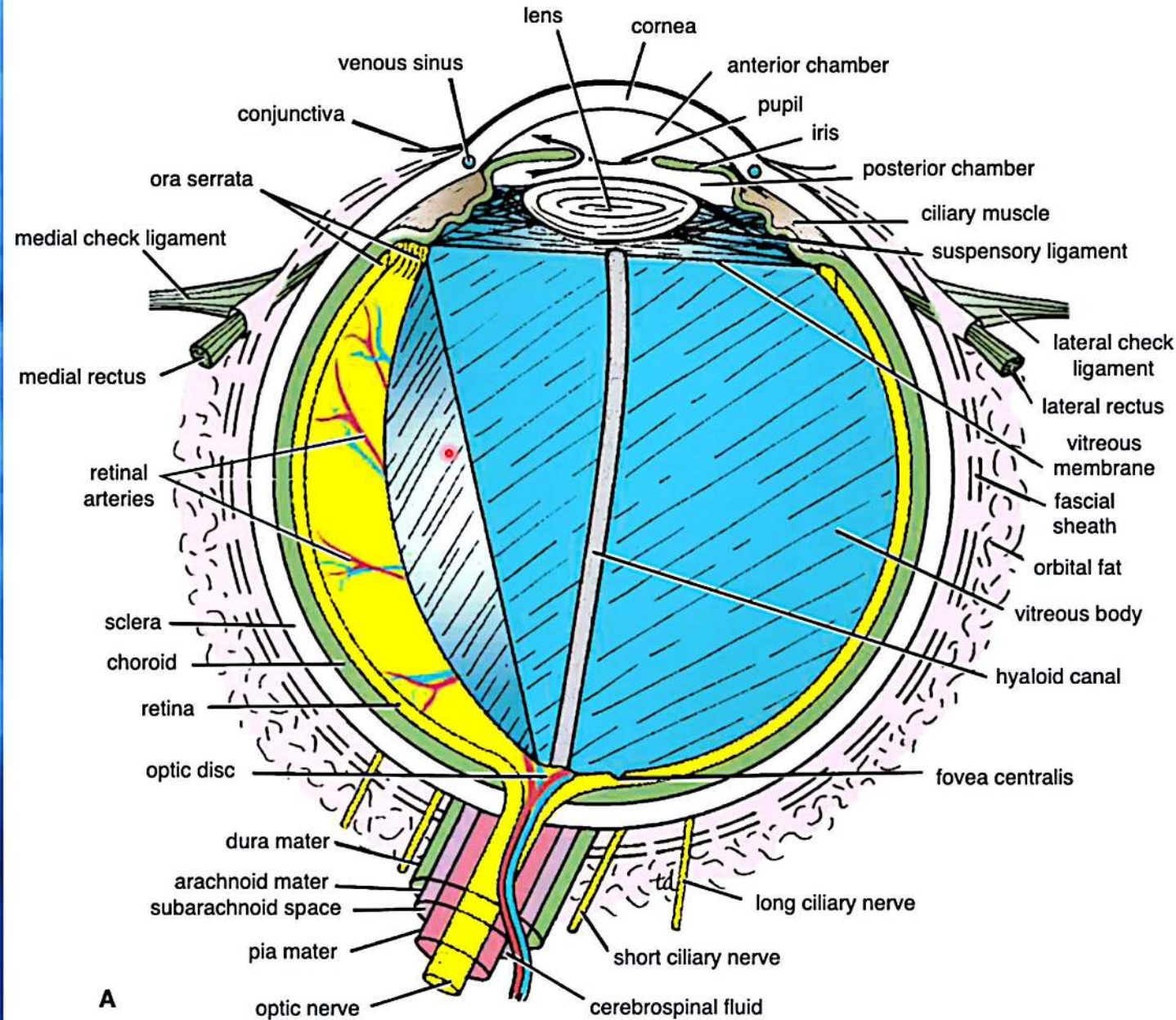
Optic Disc

- Point near the center of the retina, where **axons** from the **ganglion cells** converge to form the **optic nerve**.
- Called the “blind spot”



Contents of the Eyeball

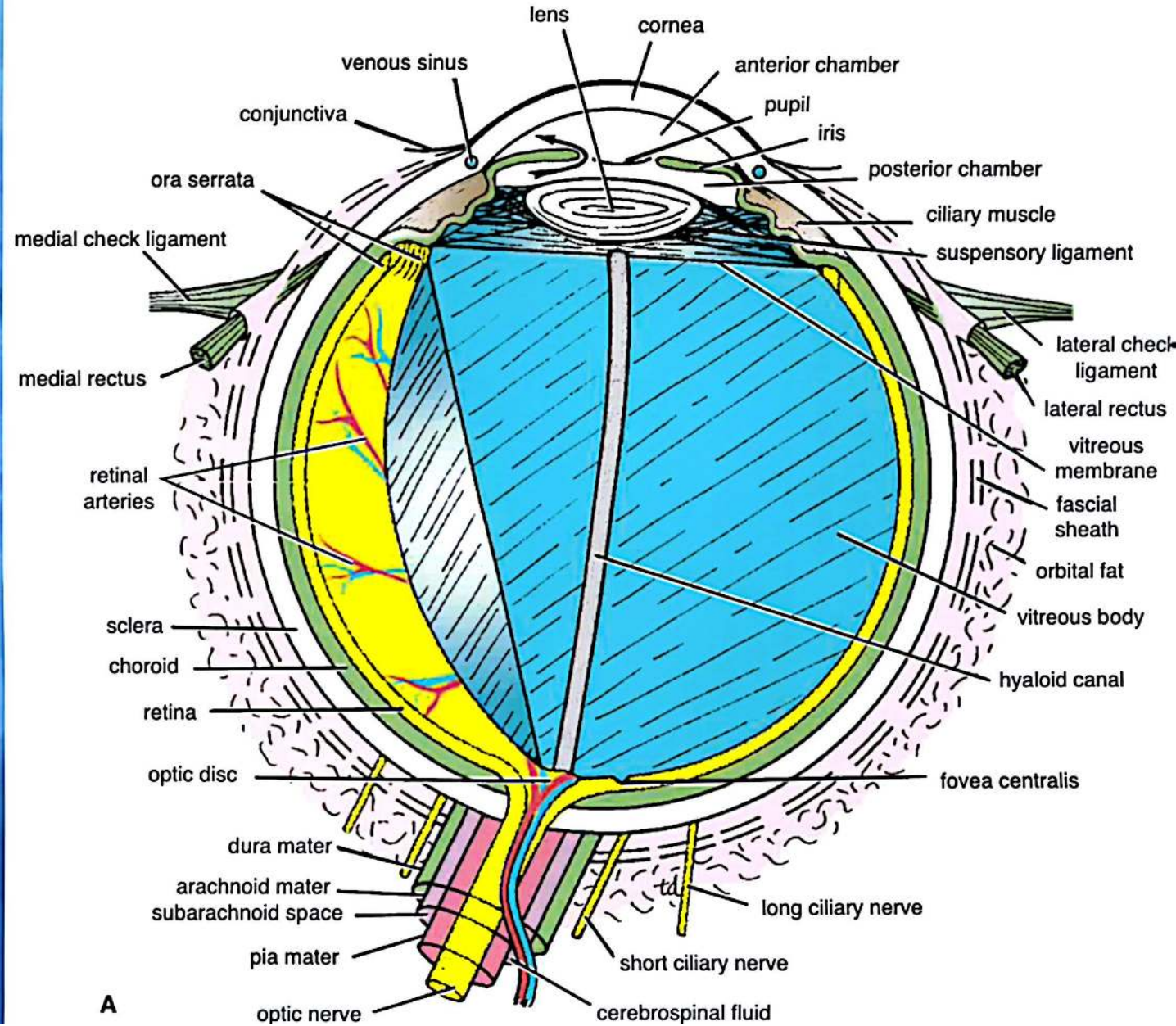
- The contents of the eyeball consist of (**The refractive media**):
 - The aqueous humor.
 - The vitreous body.
 - The lens.



Contents of the Eyeball

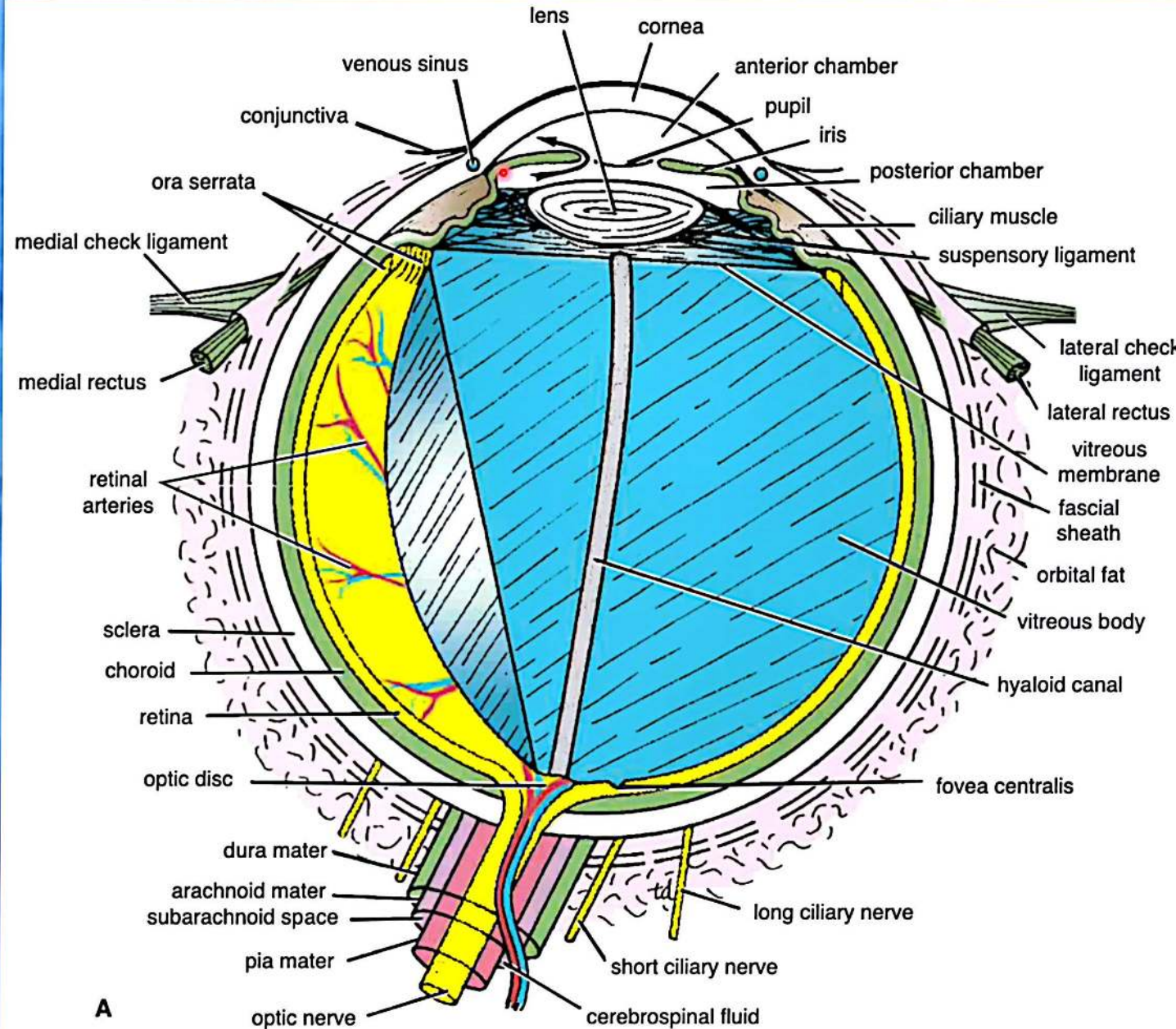
➤ Aqueous Humor:

- ✓ It is a clear fluid that fills the **anterior** and **posterior chambers** of the eyeball.
- ✓ It is **secreted** from the **ciliary processes** than enters the **posterior chamber**.
- ✓ It then **flows into** the **anterior chamber** through the **pupil** and is drained away through the spaces at the **iridocorneal angle** into the **canal of Schlemm**.



Contents of the Eyeball

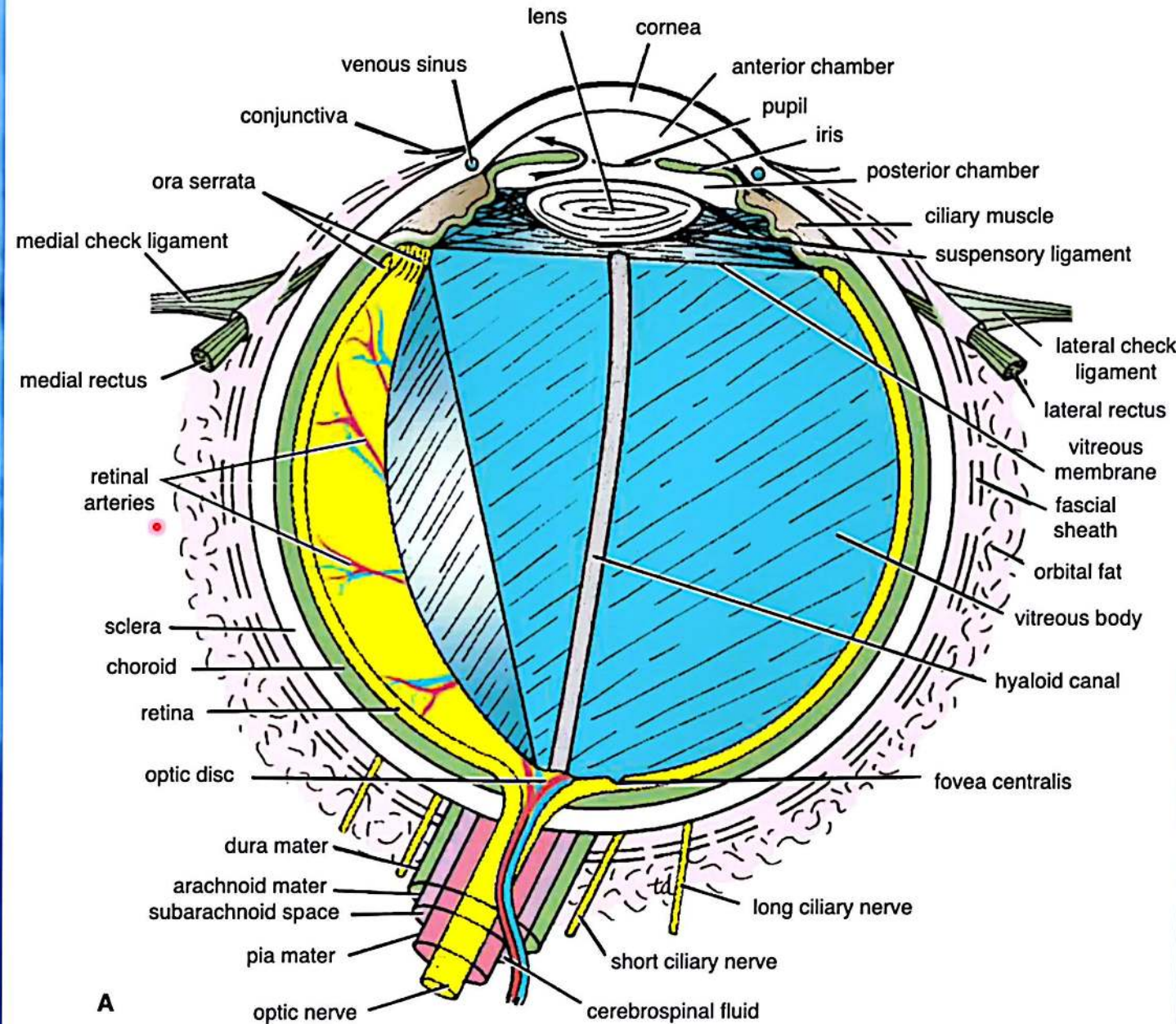
- **Obstruction** to the draining of the **aqueous humor** results in a rise in **intraocular pressure** called **glaucoma**.
- This can produce **degenerative changes** in the **retina**, with consequent **blindness**.



A

Contents of the Eyeball

- The function of the aqueous humor is to:
- ✓ support the wall of the eyeball by exerting internal pressure and thus maintaining its optical shape.
 - ✓ It is nourishes the cornea and the lens and removes the products of metabolism.
 - ✓ These functions are important because the cornea and the lens do not possess a blood supply.



Vitreous Body:

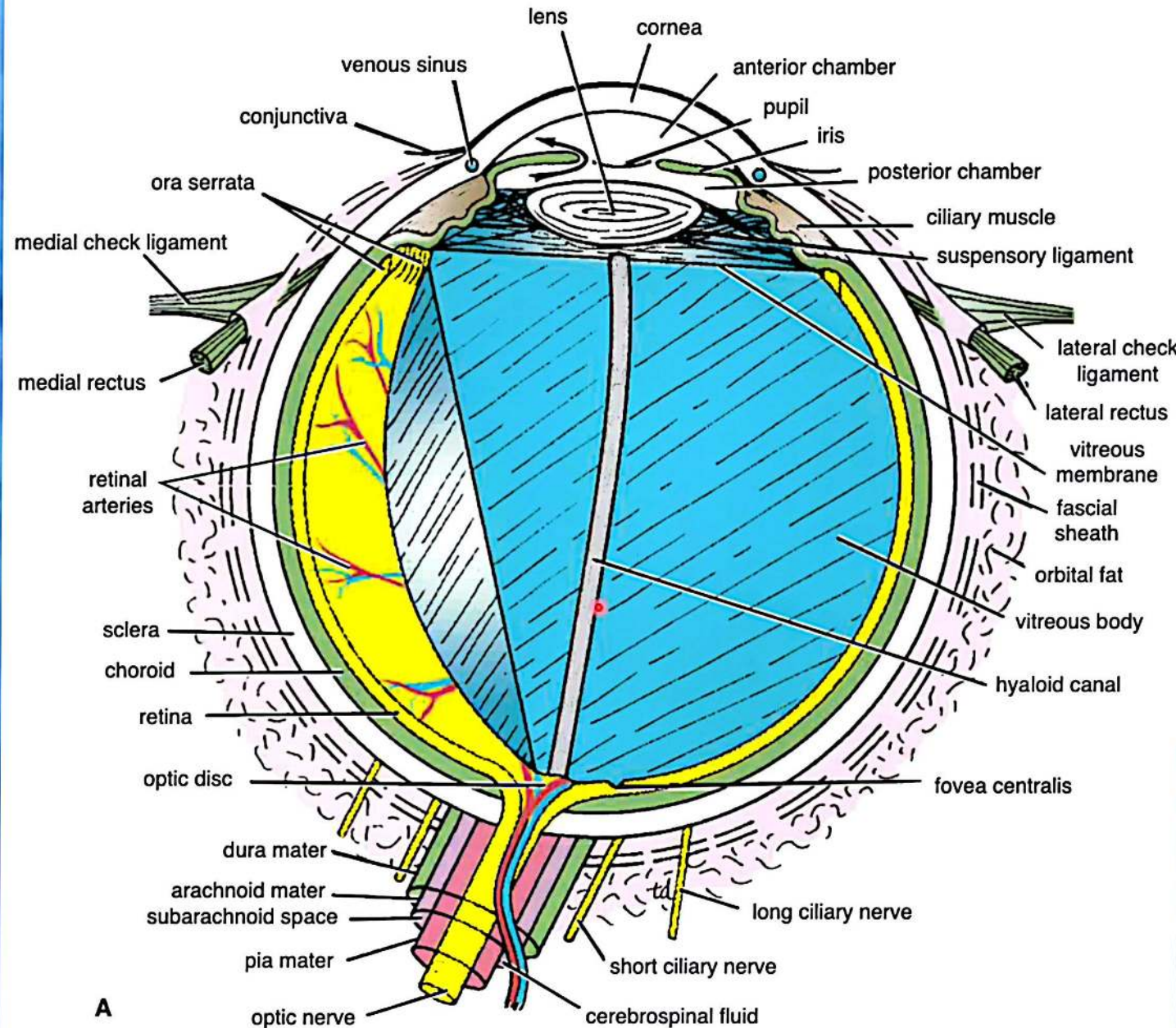
- ✓ It fills the eyeball behind the lens.
- ✓ It is a transparent gel.

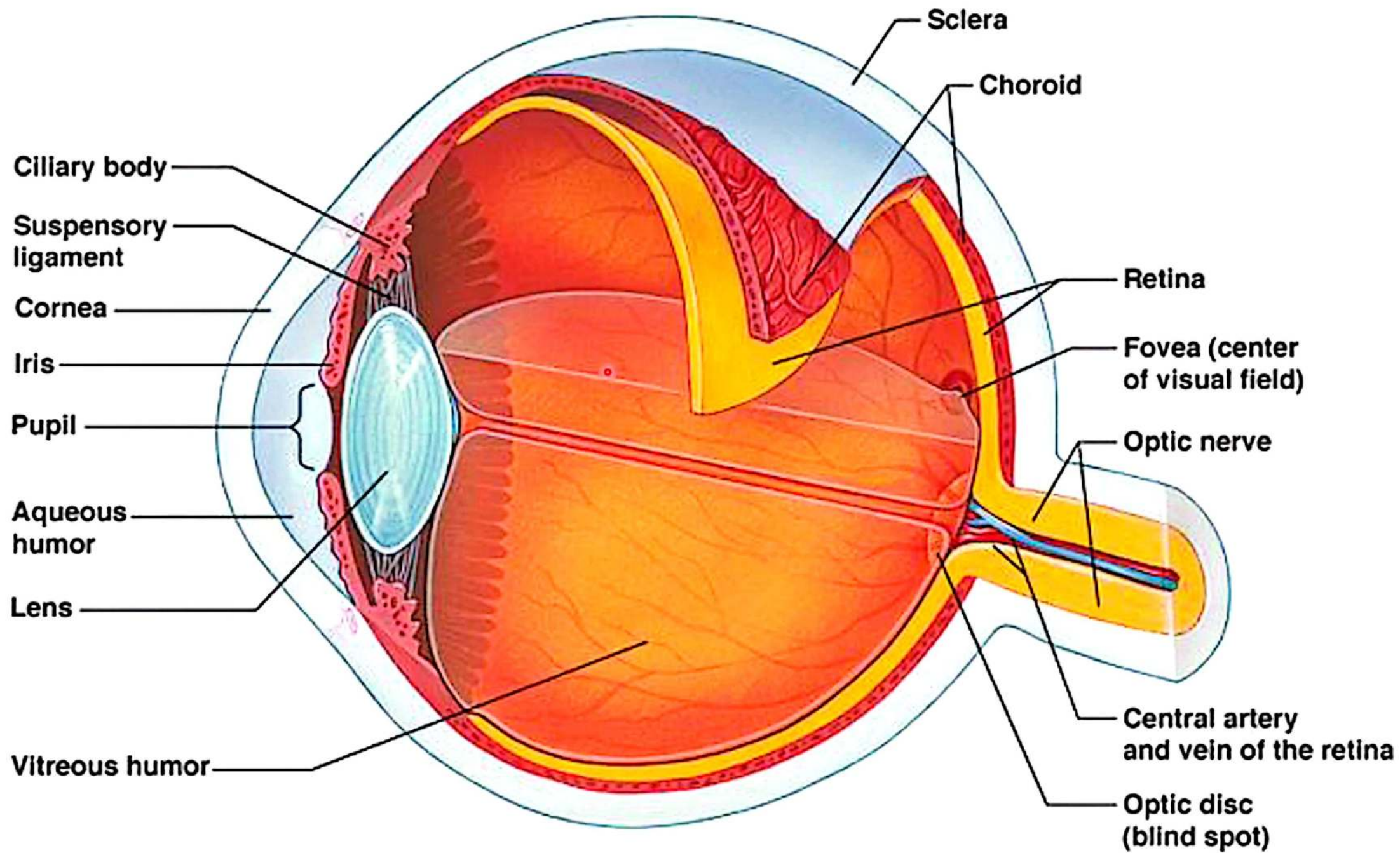
➤ The hyaloid canal:

- ✓ It is a narrow channel that runs through the **vitreous body** from the **optic disc** to the posterior surface of the **lens**.
- ✓ (in the **fetus**) it is filled by the **hyaloid artery**, which disappears before birth.

➤ The function of the vitreous body:

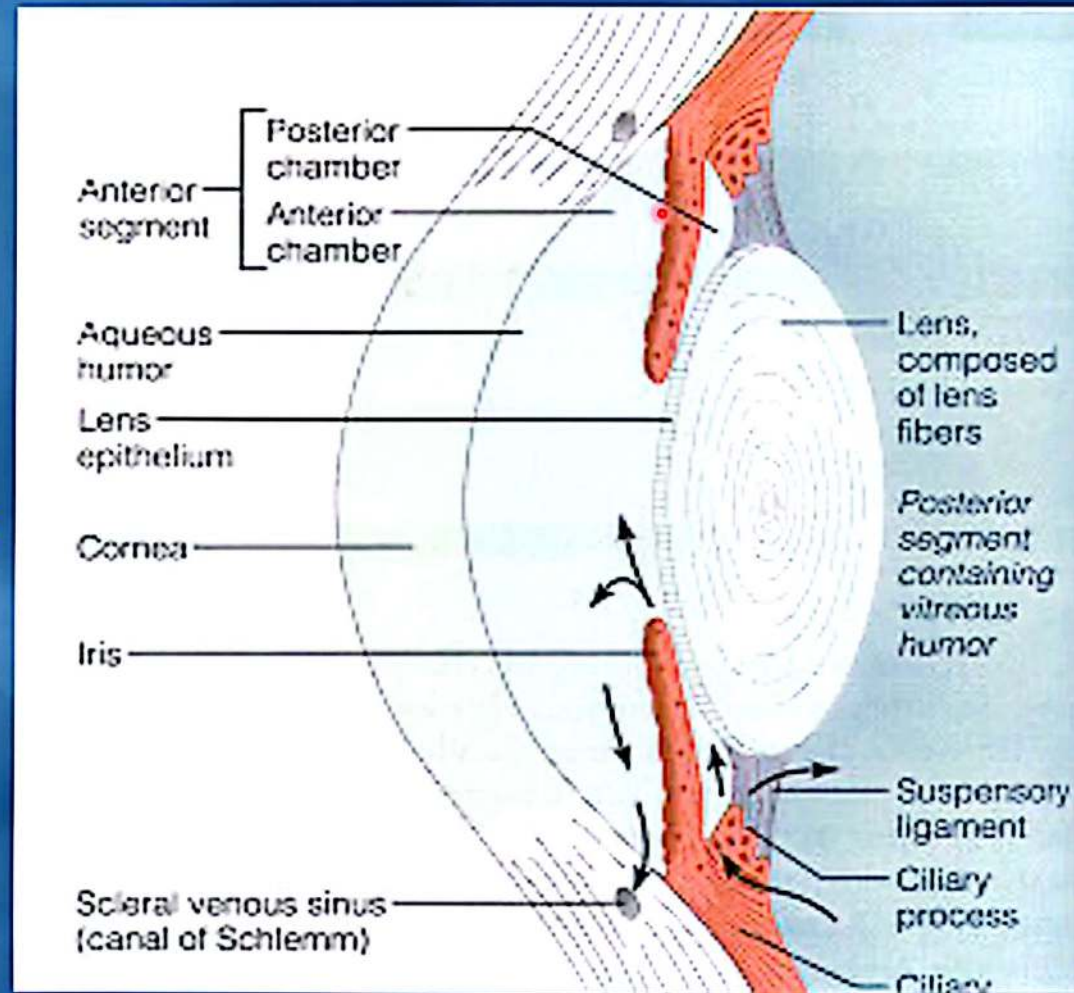
- It is to contribute slightly to the **magnifying power** of the eye.
- It **supports the posterior surface** of the **lens** and
- It **assists** in holding the **neural part** of the retina against the **pigmented part** of the **retina**.





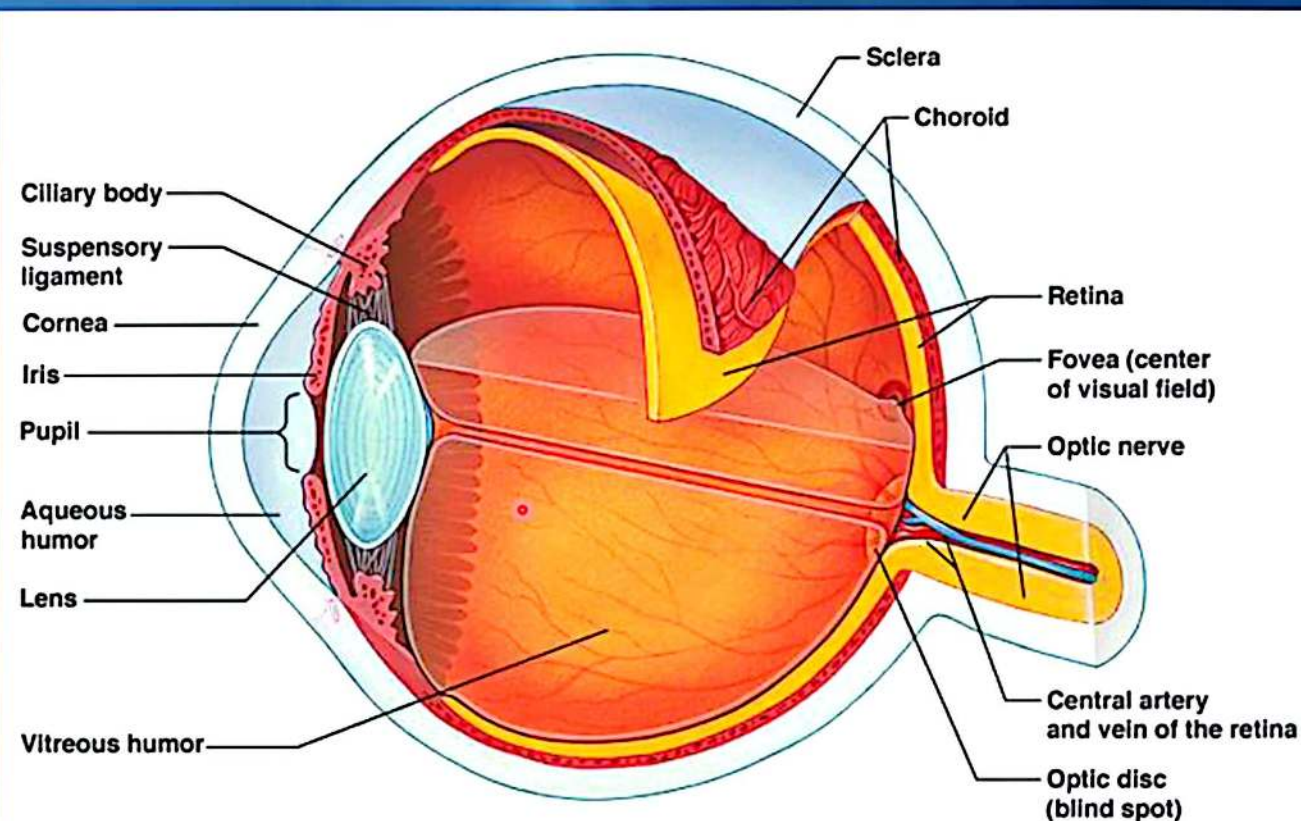
Main Compartments

- The anterior compartment is divided by the iris into two chambers:
 - The anterior chamber between the cornea and the iris.
 - The posterior chamber between the iris and the lens.
- The anterior and posterior chambers also play a major role in maintaining its normal shape, through the mechanism of balancing the production and drainage of aqueous humor, the fluid which fills both of them.



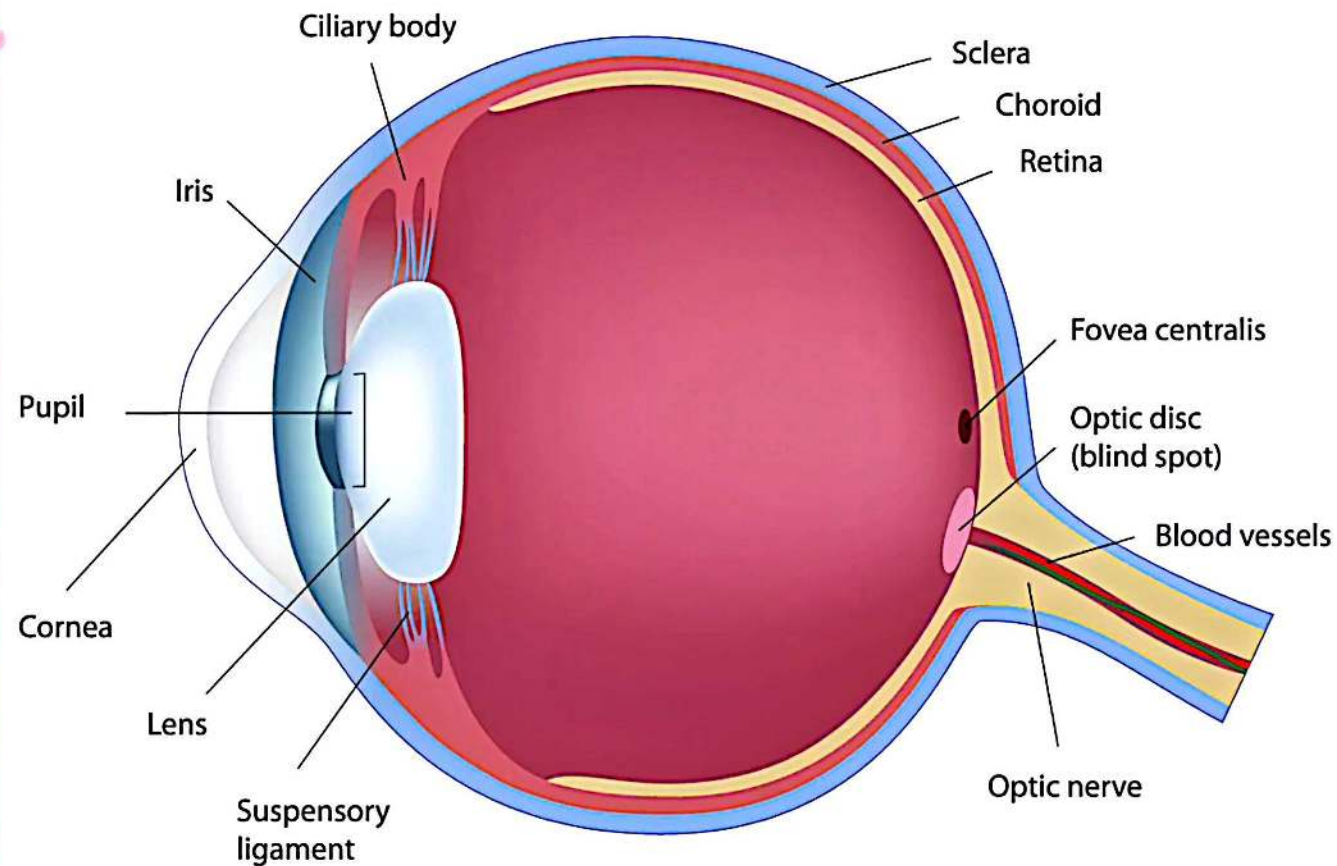
- It is the **clear gel** that **fills** the **space** between the **lens** and the **retina** of the **eyeball of humans**.
- It is often referred to as the **vitreal body** or simply "the vitreal".
- The vitreal is the **transparent, colourless, gelatinous mass** that fills the space between the lens of the eye and the retina lining the back of the eye.
- It is **present** at **birth** and **does not change** much over the course of aging.
- Its composition is similar to the cornea, but **contains** very few cells (mostly **phagocytes** which **remove unwanted cellular debris** in the **visual field**).

Vitreous Humor



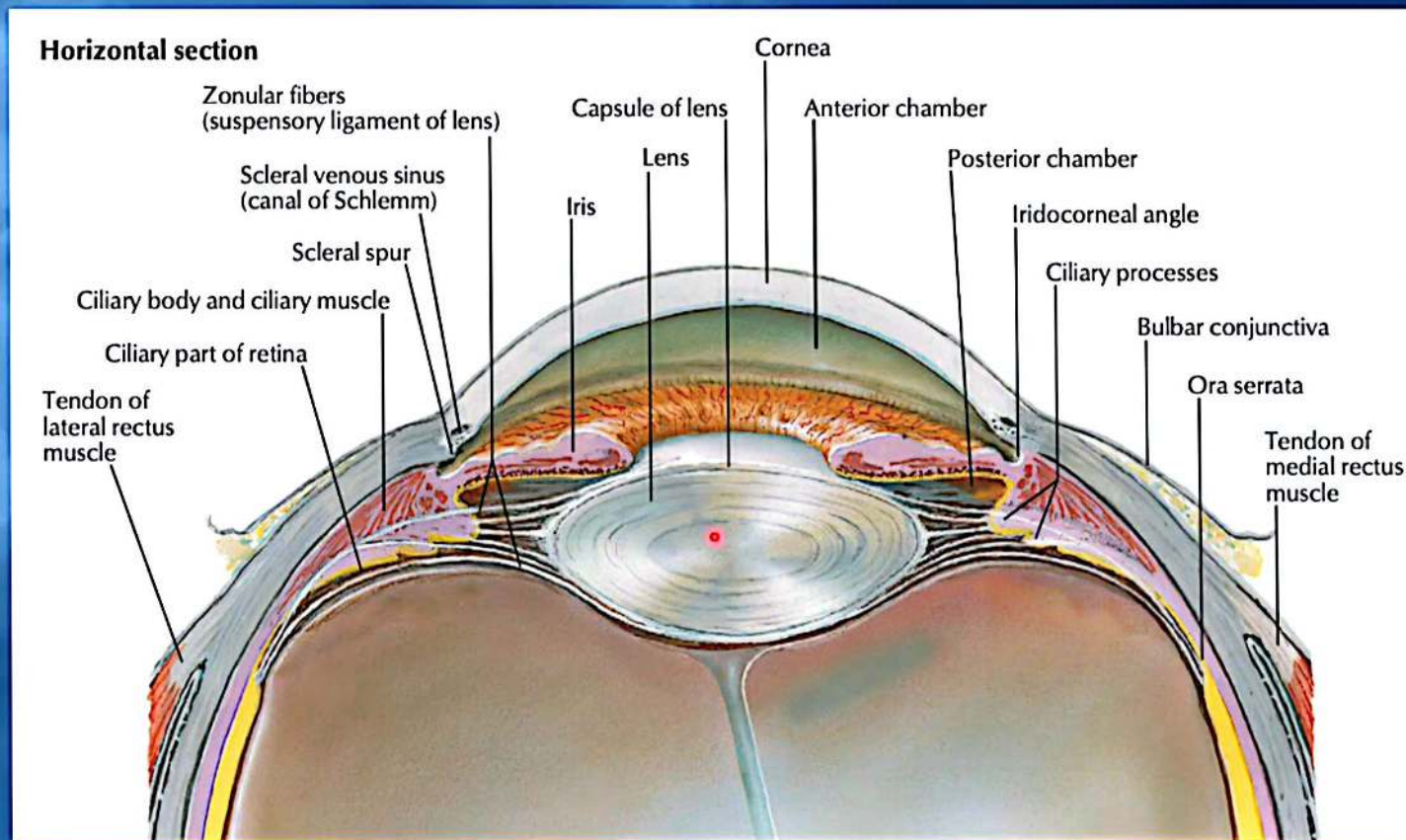
Lens

- It is a **transparent, biconvex** structure enclosed in a **transparent capsule**.
- It is situated **behind the iris** and
- It is in front of the **vitreous body** and
- It is **encircled** by the **ciliary processes**.
- It consists of an **elastic capsule**.
- It is formed from the cuboidal epithelium at the equator of the lens.



Lens

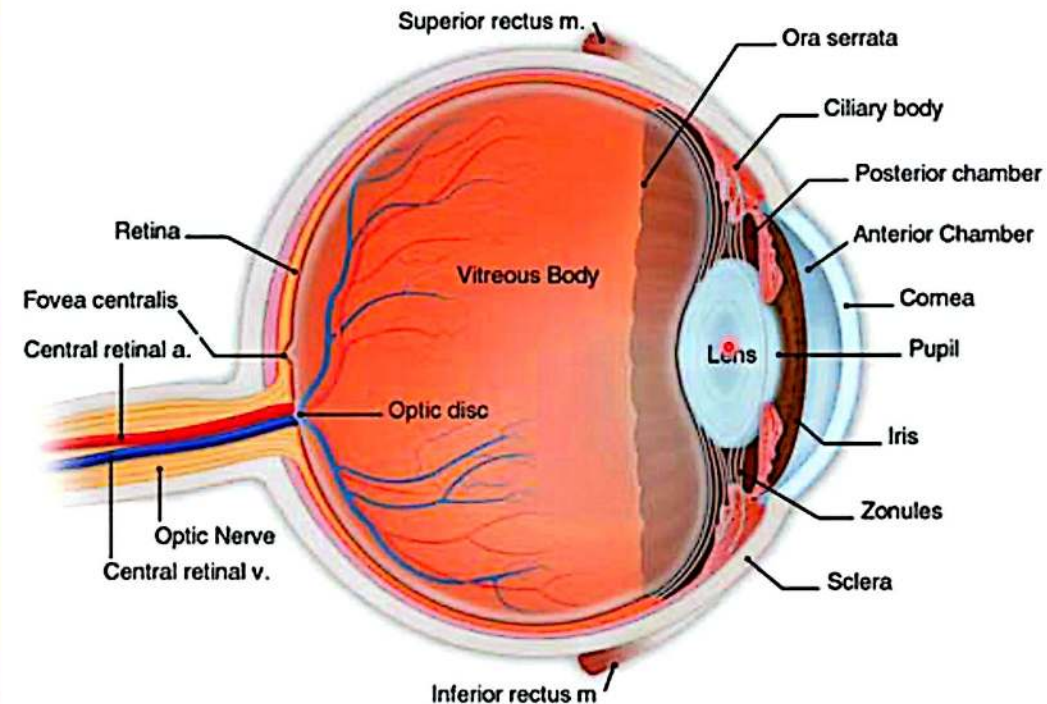
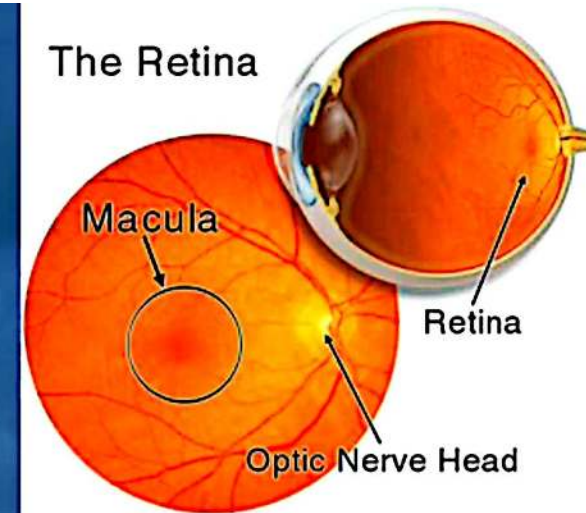
- The **lens fibers** make up the bulk of the lens (**disc shape**).
- It is attached to **the ciliary processes** of the **ciliary body** by the **suspensory ligament**.
- The pull of the **radiating fibers of the suspensory ligament** tends to keep the **elastic lens flattened** so that the **eye can be focused** on **distant objects**.



When You Look at any Object

- Light waves from that object enter the eye first through the **cornea**, which is the clear dome at the front of the eye.
- Light waves progress through the **pupil**, the circular opening in the center of the colored iris.
- Immediately behind the **iris** (and pupil) is the crystalline **lens**, and light passes through that also.

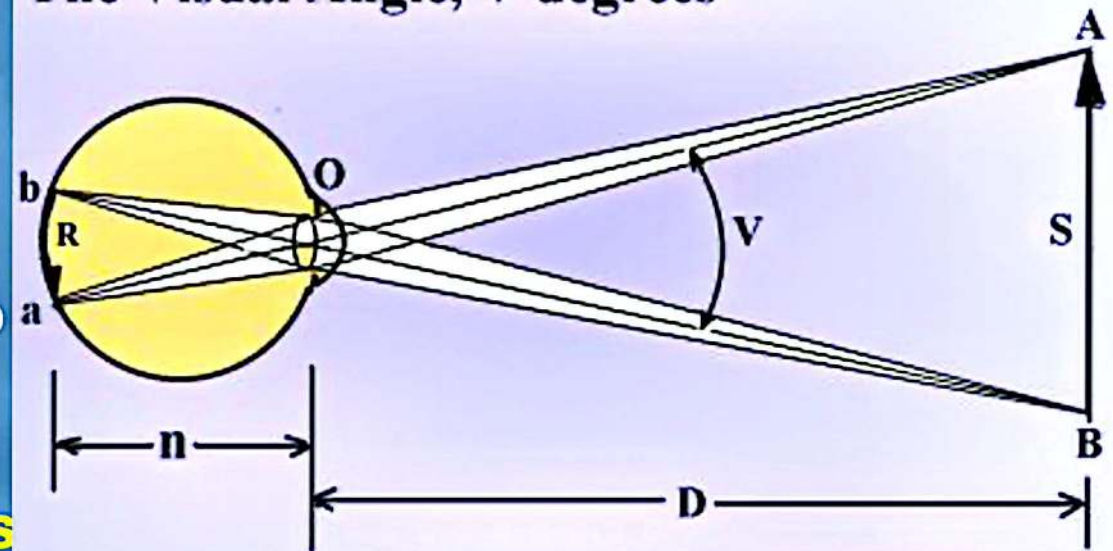
The Retina



When You Look at any Object

- If you think of your **eye as a camera**, the **retina** would be like **the film inside a camera** that registers the tiny **photons of light** with produces the picture.
- **Light waves are bent (converged)** first by the cornea, then even more so by the **crystalline lens**, to a **nodal point** which is **immediately** behind the lens
- At the **nodal point**, the **light waves** (image) become **reversed (turned backwards)** and **inverted (turned upside down)**.

The Visual Angle, V degrees

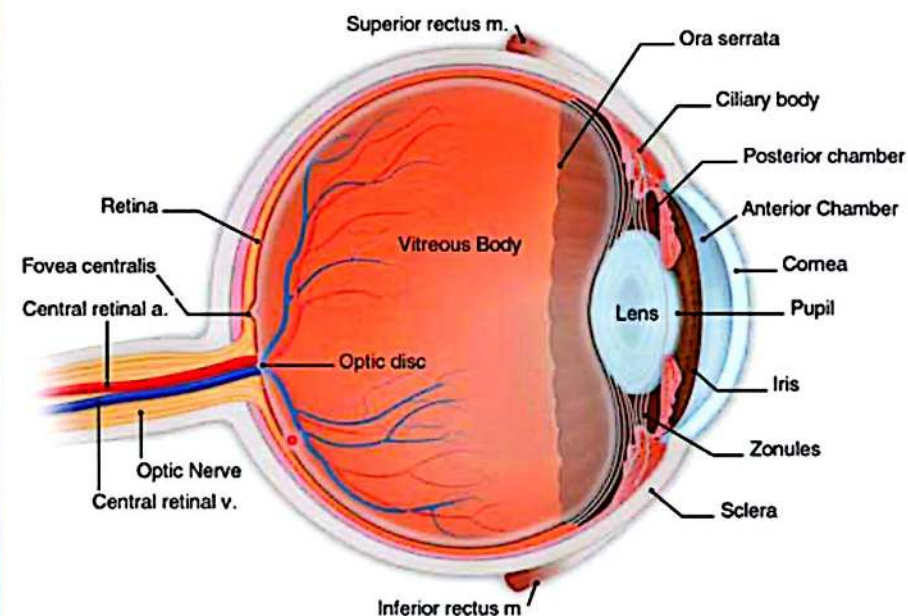
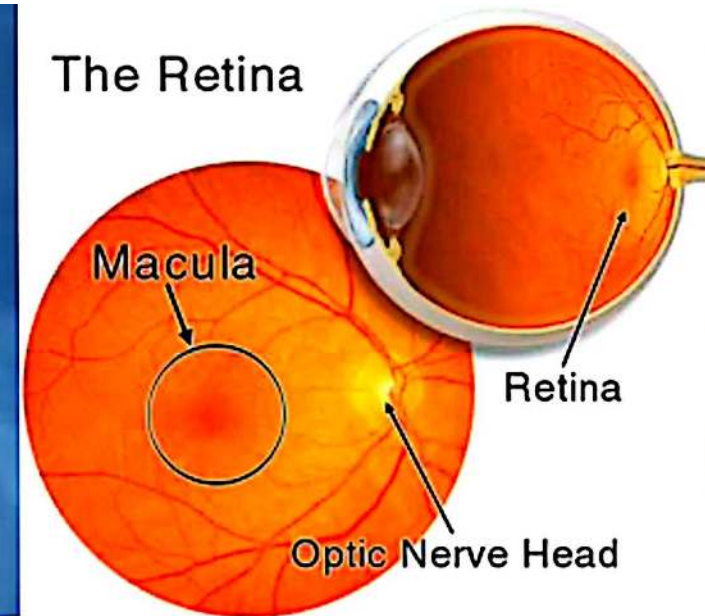


$$V = 2 \arctan(S/2D)$$

When You Look at any Object

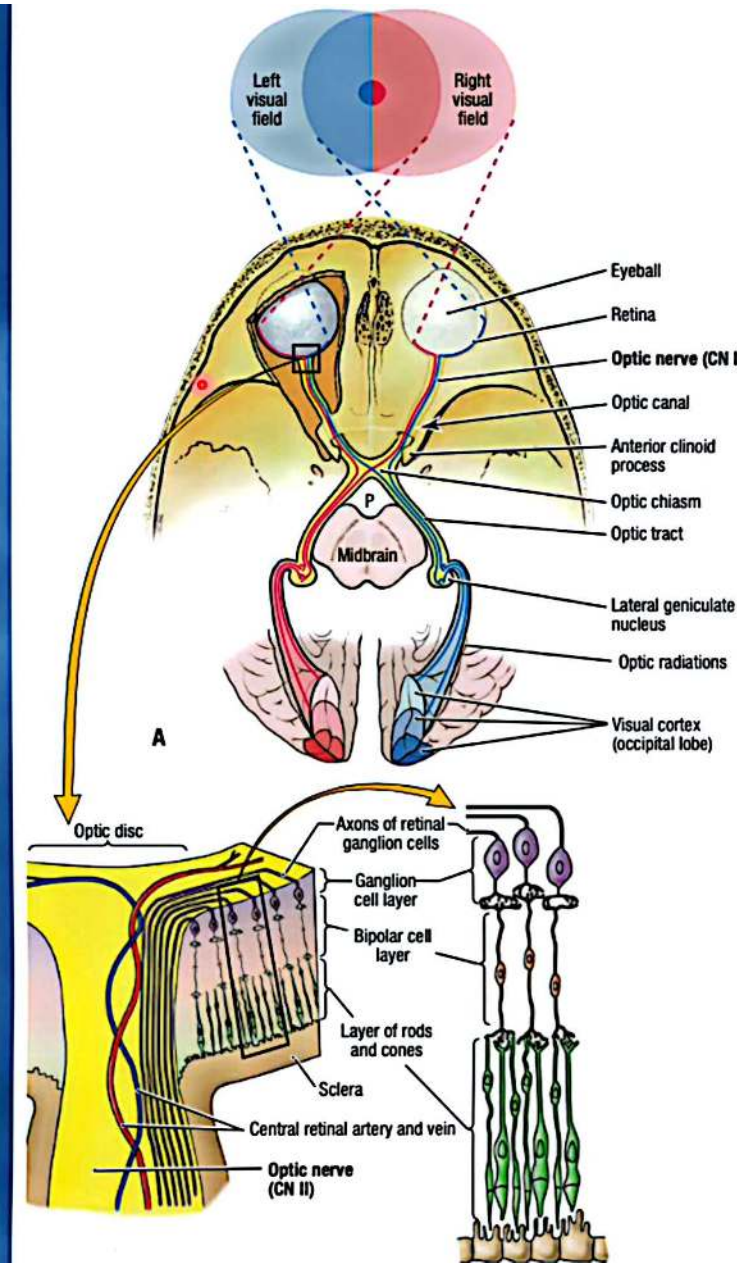
- Light waves continue through the **vitreous humor**, the **clear gel** that makes up about **80% of the eye's volume**, and then back to a clear focus on the **retina** behind the vitreous.
- The small, central area of the retina is the **macula**; it provides the best vision of any location in the retina.

The Retina



When you look at any object

- The light impulses are changed into electrical signals, then sent through the **optic nerve** along the visual pathway to the **occipital cortex**, or posterior (back), of the brain.
- This is where the **electrical signals** are seen by the brain as a visual image.
- When light entering the eye is bright enough, the **pupils** will get smaller (**constrict**) due to pupillary light response.





Will be continue ...